Literature Review

Individual Paper Summaries

25 subject areas; one paragraph summary for each retained paper.

Generated from the scholarly literature-review artifacts.

01 Hallucination Surveys

1. Maynez et al. (2020)

Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. “On Faithfulness and Factuality in Abstractive Summarization.” *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 1906–1919, 2020. DOI: 10.18653/v1/2020.acl-main.173 (https://doi.org/10.18653/v1/2020.acl-main.173); ACL Anthology (https://aclanthology.org/2020.acl-main.173/). Maynez et al. conduct a large-scale human evaluation of abstractive summarization systems and analyze the kinds of content that is not supported by the input document. They find substantial hallucinated content across generated summaries, while pretrained systems perform better than earlier systems on both automatic metrics and human judgments of faithfulness and factuality; textual-entailment measures correlate with human faithfulness judgments better than standard overlap metrics such as ROUGE. The study is limited by its focus on document summarization and by the difficulty of reducing human judgments of factuality and faithfulness to a single general criterion. For this project, its key contribution is the distinction between source-grounded faithfulness and broader factuality, which supports labeling claims as entailed by supplied premises, contradicted by them, or unsupported rather than treating every factual error as the same phenomenon.

2. Ji et al. (2023)

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Delong Chen, Wenliang Dai, Ho Shu Chan, Andrea Madotto, and Pascale Fung. “Survey of Hallucination in Natural Language Generation.” *ACM Computing Surveys* 55(12), 1–38, 2023. DOI: 10.1145/3571730 (https://doi.org/10.1145/3571730); arXiv:2202.03629 (https://arxiv.org/abs/2202.03629). This survey synthesizes definitions, evaluation metrics, mitigation methods, and task-specific findings across abstractive summarization, dialogue, generative question answering, data-to-text generation, machine translation, visual-language generation, and large language models. Its central organizing distinction is intrinsic hallucination, where an output contradicts the source, versus extrinsic hallucination, where an output is neither supported nor contradicted by the source. The breadth of the review is also its limitation: it inventories heterogeneous tasks and methods, but does not provide a single reasoning representation that explains how claims, premises, warrants, and conclusions relate across those tasks. It is directly relevant as a seed taxonomy for span-level labeling and as evidence that detection must be conditioned on the source and task; this project extends that foundation with an explicit argument structure and auditable provenance.

3. Rawte et al. (2023)

Vipula Rawte, Swagata Chakraborty, Agnibh Pathak, Anubhav Sarkar, S. M. Towhidul Islam Tonmoy, Aman Chadha, Amit Sheth, and Amitava Das. “The Troubling Emergence of Hallucination in Large Language Models — An Extensive Definition, Quantification, and Prescriptive Remediations.” *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2541–2573, 2023. DOI: 10.18653/v1/2023.emnlp-main.155 (https://doi.org/10.18653/v1/2023.emnlp-main.155); ACL Anthology (https://aclanthology.org/2023.emnlp-main.155/). Rawte et al. propose a fine-grained framework that profiles hallucination by orientation, degree, and category, distinguishing “factual mirage” from “silver lining,” subdividing each by intrinsic and extrinsic status and by severity, and introducing six named error types; they also release the 75,000-sample HILT dataset and propose a Hallucination Vulnerability Index for comparing models. The paper’s strength is its insistence that hallucination is a family of mismatches requiring differentiated measurement and remediation rather than one binary defect. Its limitation for this project is that broad orientation, severity, and category labels do not by themselves identify which premise, warrant, or conclusion failed, and the proposed index compresses heterogeneous behaviors into aggregate vulnerability. The project can therefore use Rawte et al. as a taxonomy-first precedent while making each label mechanically inspectable through span-level evidence, inferential relation, and verification status.

4. Huang et al. (2023 arXiv; 2025 ACM publication)

Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. “A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions.” arXiv:2311.05232, 2023; subsequently published in *ACM Transactions on Information Systems* with DOI 10.1145/3703155 (https://doi.org/10.1145/3703155); arXiv (https://arxiv.org/abs/2311.05232). Huang et al. develop an LLM-specific taxonomy and review causes, detection methods, benchmarks, mitigation strategies, and open problems, including the limits of retrieval-augmented generation and the role of knowledge boundaries, long-tail knowledge, and changing world knowledge. The survey’s method is a structured synthesis of the rapidly expanding LLM hallucination literature, and its main finding is that plausible nonfactual generation persists even when models are augmented with retrieval, because retrieval quality, interpretation, and generation can each introduce unsupported content. Its limitation is that the organization is primarily by causes, methods, and benchmarks rather than by an explicit representation of the reasoning structure of individual claims. For this project, it is the closest general survey: its treatment of support, knowledge boundaries, and RAG failure motivates representing premise availability and claim status as entailed, contradicted, supported only by external evidence, or unresolved before generation.

5. Tonmoy et al. (2024)

S. M. Towhidul Islam Tonmoy, S. M. Mehedi Zaman, Vinija Jain, Anku Rani, Vipula Rawte, Aman Chadha, and Amitava Das. “A Comprehensive Survey of Hallucination Mitigation Techniques in Large Language Models.” arXiv:2401.01313, 2024. arXiv (https://arxiv.org/abs/2401.01313). Tonmoy et al. survey more than 32 techniques for reducing hallucination in LLMs and classify them across dimensions including dataset use, task, feedback mechanism, and retriever type; the review covers interventions such as retrieval augmentation, knowledge retrieval, contrastive or feedback-based methods, and verification-oriented approaches. Its main contribution is a vocabulary for locating mitigation in the pipeline—before generation through grounding and retrieval, during generation through decoding or control, or after generation through checking and correction—and a discussion of the limitations that remain across these strategies. Because it is a mitigation survey, it does not supply a detailed account of the argument content that each intervention is intended to protect, and its technique-centered categories can obscure whether a method addresses a faulty premise, an invalid inference, or an unsupported conclusion. The project uses this work to evaluate compact context construction as a pre-generation grounding intervention while adding a deductive skeleton and span-level provenance so mitigation effects can be tied to specific reasoning units.

6. Venkit et al. (2024)

Pranav Narayanan Venkit, Tatiana Chakravorti, Vipul Gupta, Heidi Biggs, Mukund Srinath, Koustava Goswami, Sarah Rajtmajer, and Shomir Wilson. “An Audit on the Perspectives and Challenges of Hallucinations in NLP.” *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 6528–6548, 2024. DOI: 10.18653/v1/2024.emnlp-main.375 (https://doi.org/10.18653/v1/2024.emnlp-main.375); ACL Anthology (https://aclanthology.org/2024.emnlp-main.375/). Venkit et al. audit how hallucination is characterized in 103 peer-reviewed NLP publications and complement that literature analysis with a survey of 171 NLP and AI practitioners. They find substantial disagreement over the meaning and scope of “hallucination,” and argue that inconsistent definitions create downstream problems for evaluation, communication, and responsible deployment; they therefore call for explicit definitions and frameworks. The audit’s limitation is that it diagnoses field-level conceptual instability rather than delivering a compact, operational grammar for classifying the relation between a claim and its evidence. Its relevance to this project is foundational: it justifies establishing an explicit ontology and schema before classification, recording the textual evidence and argument relation behind every decision so labels remain auditable across traditions and evaluators.

7. Sahoo et al. (2024)

Pranab Sahoo, Prabhash Meharia, Akash Ghosh, Sriparna Saha, Vinija Jain, and Aman Chadha. “A Comprehensive Survey of Hallucination in Large Language, Image, Video and Audio Foundation Models.” *Findings of the Association for Computational Linguistics: EMNLP 2024*, 11709–11724, 2024. DOI: 10.18653/v1/2024.findings-emnlp.685 (https://doi.org/10.18653/v1/2024.findings-emnlp.685); ACL Anthology (https://aclanthology.org/2024.findings-emnlp.685/). Sahoo et al. broaden hallucination research beyond text by surveying definitions, taxonomies, detection methods, and mitigation across language, image, video, and audio foundation models. The paper’s central finding is that hallucination is a cross-modal reliability problem with modality-specific symptoms but recurring grounding and evidence-alignment concerns, and it proposes a framework for organizing detection and mitigation across those modalities. Its breadth necessarily leaves less room for fine-grained analysis of textual inference, and modality-specific benchmarks and metrics are not interchangeable without additional normalization. Although this project is text-centered, the survey supports a modality-neutral abstraction in which a generated claim is checked against an evidence channel and an intended task relation; the project specializes that abstraction by recovering language-level spans, premises, and conclusions for compact, high-signal context.

8. van Deemter (2024)

Kees van Deemter. “The Pitfalls of Defining Hallucination.” *Computational Linguistics* 50(2), 807–816, 2024. DOI: 10.1162/coli_a_00509 (https://doi.org/10.1162/coli_a_00509); MIT Press (https://direct.mit.edu/coli/article/50/2/807/119144/The-Pitfalls-of-Defining-Hallucination); arXiv:2401.07897 (https://arxiv.org/abs/2401.07897). van Deemter compares existing analyses of hallucination, omission, and other factual errors in data-to-text generation and proposes a logic-based synthesis using the two directions of logical consequence between input and output: whether the input entails the output and whether the output entails the input, with additional distinctions for contradiction and cases where neither side follows from the other. He shows that common taxonomies can conflate materially different errors—for example, unsupported additions, omissions, and contradictions—and argues that ambiguity, vagueness, pragmatics, and open-ended LLM tasks limit any purely logical account. The paper is limited as an operational system because it does not provide a scalable cross-tradition span extractor or context compressor, and natural-language entailment is itself difficult for ambiguous or underspecified text. It is nevertheless the most direct theoretical bridge to this project: its entailment structure supplies a principled backbone for classifying additions, omissions, contradictions, and logical independence while the project contributes the extraction, evidence, and compact-representation layers.

9. Xu, Jain, and Kankanhalli (2024)

Ziwei Xu, Sanjay Jain, and Mohan Kankanhalli. “Hallucination Is Inevitable: An Innate Limitation of Large Language Models.” arXiv:2401.11817, 2024. arXiv (https://arxiv.org/abs/2401.11817). Xu, Jain, and Kankanhalli formalize hallucination as inconsistency between a computable LLM and a computable ground-truth function, use learning-theoretic results to argue that no LLM can learn all computable functions, and conclude that hallucination cannot be eliminated for general problem solving; they further characterize hallucination-prone tasks under time-complexity constraints, validate aspects of the analysis empirically, and discuss what existing mitigations can and cannot achieve. The formal result is intentionally high-level and depends on an idealized world and assumptions about computability, so it does not constitute a complete empirical taxonomy of the diverse errors observed in deployed systems. Its relevance is the design implication that reliability systems must represent uncertainty and abstention rather than force every generated span into a binary true/false or supported/contradicted class. This project operationalizes that implication as an explicit unknown or insufficient-support status attached to a compact evidence record and reasoning skeleton.

02 Logic Hallucination

This set contains paragraph summaries for the 11 retained primary studies identified in `agents/02_logic_hallucination.md`. Each paragraph states the study's contribution, principal finding, limitation, and relevance to the project.

1. Xu et al. (2023), "Are Large Language Models Really Good Logical Reasoners? A Comprehensive Evaluation and Beyond"

Xu et al. evaluate seven LLMs across 15 datasets covering deductive, inductive, abductive, and mixed-form reasoning. Rather than treating answer accuracy as sufficient, they assess answers and explanations, classify logical flaws in evidence selection and reasoning, and propose a six-dimensional evaluation scheme that includes correctness, rigor, self-awareness, activity, orientation, and "no hallucination." The study's central contribution is therefore methodological: it makes unsupported or flawed reasoning an explicit evaluation dimension alongside final-answer correctness. Its limitation for this project is that the abstract-level benchmark description does not establish a reusable proof-span representation, a dedicated hallucination mechanism model, or coverage of the project's cross-tradition argument forms. It nevertheless provides a strong precedent for scoring intermediate explanations and unsupported claims separately from final labels, supporting a deductive skeleton in which each generated edge can be checked rather than relying on answer accuracy alone. Xu et al. 2023 (https://arxiv.org/abs/2306.09841)

2. Yuan et al. (2024), "Whispers that Shake Foundations: Analyzing and Mitigating False Premise Hallucinations in Large Language Models"

Yuan et al. directly study false-premise hallucination, defined as generating hallucinated text when confronted with a false-premise question. Their analysis identifies a small subset of attention heads—"false premise heads"—that disrupt knowledge extraction, and they introduce FAITH, an inference-time method that constrains those heads. The paper reports that constraining approximately 1% of attention heads yields nearly a 20% performance improvement in extensive experiments. The limitation is that the mechanism is an attention-head intervention rather than a general proof-structure or argument-scheme representation, and the task setting centers on false-premise questions rather than the full range of deductive forms. For this project, the paper is the closest direct precedent for placing premise validity before downstream generation: premise spans should be explicitly represented, checked for contradiction or falsity, and prevented from silently becoming inputs to later inference. Yuan et al. 2024 (https://aclanthology.org/2024.emnlp-main.155/)

3. Varshney et al. (2025), "Investigating and Addressing Hallucinations of LLMs in Tasks Involving Negation"

Varshney et al. examine hallucination when models must process negation, an operator central to logical reasoning and inference. They evaluate four tasks—false-premise completion, constrained fact generation, multiple-choice question answering, and fact generation with negation—and report considerable hallucination by open-source models including LLaMA-2-chat, Vicuna, and Orca-2 across these settings; they also test mitigation strategies. The study's limitation is that it is a workshop paper focused mainly on factual-generation and QA tasks, so it does not establish behavior across complete formal deduction systems or diverse argument schemes. Its relevance is direct: polarity and negation cannot be treated as incidental wording during context compression or proof extraction. The project should preserve explicit positive/negative markers, attach them to premise and conclusion spans, and validate that a downstream inference has not dropped or reversed the negation that licensed the original claim. Varshney et al. 2025 (https://aclanthology.org/2025.trustnlp-main.37/)

4. McKenna et al. (2023), “Sources of Hallucination by Large Language Models on Inference Tasks”

McKenna et al. use controlled natural-language-inference experiments with LLaMA, GPT-3.5, and PaLM families to identify pretraining biases that produce hallucination-like false entailments. They show that models can label a sample as entailed regardless of its premise when the hypothesis is attested in training data, with entities functioning as indices into memorized material; they also identify a corpus-level predicate-frequency bias in which relative frequency predicts false entailment behavior. Models perform worse on test samples that do not conform to these biases, providing useful controls for evaluation. NLI is only a proxy for proof and does not capture long multi-step deduction or every form of invalid derivation, so the paper’s mechanism account should not be generalized into a complete reasoning theory. For the project, the result demonstrates that local premise–hypothesis relations can be overridden by memorization and distributional cues, making premise-conditioned entailment and adversarially controlled distractors necessary evaluation targets. McKenna et al. 2023 (https://aclanthology.org/2023.findings-emnlp.182/)

5. Hu et al. (2023), “Won’t Get Fooled Again: Answering Questions with False Premises”

Hu et al. introduce FalseQA, a dataset of 2,365 human-written false-premise questions with explanations of the false premise and revised true-premise questions. They find that pretrained language models are easily deceived by tricky questions even though they often possess the knowledge needed to rebut them; fine-tuning on moderate amounts of data, such as 256 examples, improves false-premise discrimination, and models can generate reasonable rebuttal explanations. The study is limited by its question-answering focus, its emphasis on premise correction rather than general derivation, and its dependence on PLM-era supervised fine-tuning conditions. Its project relevance lies in separating knowledge availability from premise-validation behavior: a model may know the relevant fact but still answer inside an invalid premise. A proof-relevant context layer should therefore preserve the correction or rebuttal signal and give the model an explicit route to reject or repair an invalid premise before applying rules. Hu et al. 2023 (https://aclanthology.org/2023.acl-long.309/)

6. Zhu et al. (2025), “KG-FPQ: Evaluating Factuality Hallucination in LLMs with Knowledge Graph-based False Premise Questions”

Zhu et al. propose an automated pipeline for generating false-premise questions from knowledge graphs. The method modifies true knowledge-graph triples to create false premises, uses GPT-based generation to produce semantically rich questions, and yields KG-FPQ, a benchmark of approximately 178,000 questions spanning three domains, six confusability levels, and two task formats. The paper’s main contribution is scalable, controlled evaluation of factuality hallucination under structured premise corruption, rather than a new formal proof calculus. Its limitation is that knowledge-graph triples are narrower and more atomic than natural-language arguments; the benchmark therefore does not directly measure multi-step proof validity, defeasible reasoning, or cross-tradition argument forms. For this project, the triple-corruption design is a useful template for generating controlled counterfactual and contradiction fixtures: perturb one premise or relation while holding the rest of a reasoning chain fixed, then test whether the model preserves provenance and refuses the unsupported deduction. Zhu et al. 2025 (https://aclanthology.org/2025.coling-main.698/)

7. Li et al. (2025), “Don’t Take the Premise for Granted: Evaluating the Premise Critique Ability of Large Language Models”

Li et al. define premise critique ability as the capacity to proactively identify and articulate errors in input premises and introduce PCBench, which combines four error types, three difficulty levels, and multifaceted evaluation. Across 15 representative LLMs, they report that most models rely heavily on explicit prompts and show limited autonomous critique; direct contradictions are easier than complex or procedural errors; reasoning ability does not consistently predict premise-critique ability; and flawed premises cause reasoning models to overthink and produce substantially longer responses. The abstract does not specify which formal logics dominate the benchmark, and “overthinking” is a reliability and efficiency symptom rather than a complete hallucination taxonomy. The project should nevertheless treat premise critique as a separate stage and label, not as an implicit consequence of general reasoning strength. This supports a pipeline that validates input premises before expansion and records whether a failure arose from accepting, repairing, or rejecting the premise. Li et al. 2025 (https://aclanthology.org/2025.findings-emnlp.44/)

8. Hoppe, Ilievski, and Kalo (2025), “Investigating the Robustness of Deductive Reasoning with Large Language Models”

Hoppe, Ilievski, and Kalo study whether informal and autoformalization-based LLM deductive methods remain reliable under perturbation. They construct seven perturbed datasets from two families of interventions—adversarial noise and counterfactual statements—and organize reasoners by reasoning format, formalization syntax, and error-recovery feedback. The abstract reports that adversarial noise particularly affects autoformalization, counterfactual statements influence all approaches, and detailed feedback reduces syntax errors without improving overall accuracy, indicating limited self-correction. The study measures robustness and deduction accuracy rather than labeling every invalid inference as hallucination, and it does not provide a cross-tradition span classifier or a complete taxonomy of derivation failures. Its project relevance is strong: counterfactual and irrelevant spans should be treated as distinct from proof-relevant context, and evaluations should test distractor insertion, polarity changes, and order/dependency preservation rather than only clean instances. Hoppe et al. 2025 (https://arxiv.org/abs/2502.04352)

9. Mi et al. (2026), “Logical Consistency as a Bridge: Improving LLM Hallucination Detection via Label Constraint Modeling between Responses and Self-Judgments”

Mi et al. propose LaaB, a hallucination detector that connects implicit neural uncertainty features with explicit symbolic or verbalized self-judgments through logical label constraints. The framework uses a meta-judgment process to map symbolic labels into feature space and aligns the two views through mutual learning; experiments on four public datasets, four LLMs, and eight baselines report that LaaB outperforms the comparison systems. The limitation is conceptual as well as empirical: logical consistency functions here as a detection signal linking response and self-judgment labels, not necessarily as a formal proof checker or argument-form classifier, and exact benchmark-level error taxonomies require the full paper for deeper analysis. The study is nevertheless a direct precedent for using symbolic consistency after generation or retrieval. It supports adding a post-generation validation layer that checks whether the response, its stated judgment, and the retained context agree under explicit constraints. Mi et al. 2026 (https://aclanthology.org/2026.acl-long.286/)

10. Tyen et al. (2024), “LLMs Cannot Find Reasoning Errors, But Can Correct Them Given the Error Location”

Tyen et al. separate logical-error localization from error correction across five reasoning tasks using BIG-Bench Mistake, a dataset of LLM-generated logical mistakes. They find that state-of-the-art LLMs generally struggle to locate even simple and objective reasoning errors, while correction improves when the error location is supplied; they further show that a small classifier trained on out-of-domain data can locate mistakes more effectively than prompting a large model. The benchmark’s mistakes are intentionally unambiguous and therefore may not capture the ambiguity of commonsense or philosophical reasoning, and successful correction after localization does not establish faithful internal reasoning. The project implication is architectural: the generation model should not be assumed to reliably critique its own deductive skeleton. Separate span-level localization, verifier, or classifier components should identify the failed edge or premise before asking a model to repair the chain. Tyen et al. 2024 (https://arxiv.org/abs/2311.08516)

11. Li and Ng (2025), “Reasoning Models Hallucinate More: Factuality-Aware Reinforcement Learning for Large Reasoning Models”

Li and Ng study hallucination in long reasoning traces and report that outcome-based reinforcement learning can increase hallucination, with many errors originating in incorrect intermediate reasoning even when final-answer evaluation alone can obscure the problem. Their analysis attributes the effect to sparse outcome rewards and develops Factuality-aware Step-wise Policy Optimization (FSPO), which adds step-wise factuality verification to token-level advantages; the paper also reports that incorrect reasoning steps are the dominant cause in a manual analysis of cases where a reasoning model failed while a baseline succeeded. The limitation is that factuality is broader than formal validity: evidence-based verification may miss a formally invalid but factually plausible derivation, and the experiments center on mathematical and QA-style benchmarks rather than the project’s cross-tradition logic forms. The project should nonetheless score and provenance-track every intermediate proof step, not just the final answer, because a correct endpoint can coexist with an unsupported chain. Li and Ng 2025
(https://proceedings.neurips.cc/paper_files/paper/2025/file/ddd50f29fa472095515fa0df31749e6c-Paper-Conference.pdf)

Scope note

The retained set is a targeted subtopic review rather than an exhaustive bibliography. The inspected papers do not provide a single cross-tradition span-classification pipeline covering categorical syllogism, modus ponens/tollens, abduction, analogy, qiyas, and Ny■ya inference while also constructing compact proof-relevant context for downstream generation. That gap is a project hypothesis, not a proof that no such paper exists.

03 Context Degradation

This set contains paragraph summaries for the 10 retained primary studies identified in `agents/03_context_degradation.md`. Each paragraph states what the study examines, its main result, a limitation, and its relevance to the subtopic.

1. Liu et al. (2024), “Lost in the Middle: How Language Models Use Long Contexts”

Liu et al. study how language models use long inputs on multi-document question answering and key-value retrieval, varying the position of the information needed to answer. They find a pronounced positional effect: performance is often strongest when relevant information appears near the beginning or end of the context and drops substantially when it is placed in the middle, even for models explicitly designed to accept long contexts. The study’s limitation is that it uses two controlled information-identification tasks and the model generation available at the time, so it does not establish the same curve for every form of long-context reasoning, agent trace, modality, or current frontier model. Its relevance is foundational: it demonstrates that a nominally large context window does not imply uniform access to its contents and provides a direct evaluation protocol for detecting “lost in the middle.” Liu et al. 2024 (https://aclanthology.org/2024.tacl-1.9/)

2. Sun et al. (2021), “Do Long-Range Language Models Actually Use Long-Range Context?”

Sun et al. perform a fine-grained analysis of two long-range Transformer language models that accept sequences up to 8K tokens, asking whether context beyond the recent local window improves prediction. They report that context beyond roughly the previous 2K tokens helps only a small subset of tokens, such as tokens that can be copied from distant context, and does not improve sentence-level prediction; the benefit also varies by document type, helping literary novels more than textbooks or magazines. The limitation is that the experiments concern earlier long-range language-model architectures and perplexity-style prediction rather than modern instruction-tuned LLMs performing retrieval, reasoning, or tool use. The result is nevertheless an early direct warning that architectural support for long sequences and effective use of long-range information are different properties, making it a precursor to later context-degradation work. Sun et al. 2021 (https://aclanthology.org/2021.emnlp-main.62/)

3. Tay et al. (2021), “Long Range Arena: A Benchmark for Efficient Transformers”

Tay et al. introduce Long Range Arena (LRA), a unified benchmark for evaluating efficient Transformer architectures on long-context sequence-processing tasks rather than judging models only by their computational complexity or advertised sequence length. The benchmark spans several task families, including document-level classification, retrieval-like matching, and structured sequence problems, with sequence lengths extending from approximately 1K to 16K tokens, thereby exposing differences between architectures that may look similar on short inputs. Its limitation is that LRA is primarily an efficiency benchmark built around synthetic or structured tasks, not a direct measure of modern instruction-following LLMs, natural document workflows, positional “lost in the middle” curves, or context rot. Its relevance is methodological: it established the need for task-based, length-sensitive evaluation and helped separate nominal long-context capability from demonstrated ability to process long dependencies. Tay et al. 2021 (https://www.iclr.cc/virtual/2021/poster/3225)

4. Xu et al. (2023), “Retrieval meets Long Context Large Language Models”

Xu et al. compare retrieval augmentation with extended context windows using two large pretrained LLMs and long-context tasks covering question answering, query-based summarization, and in-context few-shot learning. They find that a 4K-context model with simple retrieval at generation can achieve performance comparable to a finetuned 16K-context model using positional interpolation while requiring much less computation; retrieval also improves performance when the model’s context window has already been extended, and their best retrieval-augmented 32K Llama2 model outperforms its non-retrieval baseline and several comparison systems. The limitation is that the findings depend on particular models, retrieval settings, task distributions, and evaluation choices, so they do not show that retrieval is universally better than long context or eliminate retrieval misses and ranking errors. The paper is relevant as a mitigation-oriented counterpoint to context rot: selectively retrieving relevant material can reduce distractors, computation, and the amount of context the model must use at once. Xu et al. 2023 (https://arxiv.org/abs/2310.03025)

5. Bai et al. (2024), “LongBench: A Bilingual, Multitask Benchmark for Long Context Understanding”

Bai et al. introduce LongBench, a bilingual benchmark containing 21 English and Chinese datasets across six categories: single-document question answering, multi-document question answering, summarization, few-shot learning, synthetic tasks, and code completion. The benchmark standardizes the tasks and evaluates eight LLMs at substantially longer inputs than typical short-context tests; the authors find that GPT-3.5-Turbo-16k outperforms the evaluated open models but still struggles as contexts lengthen, while scaled position embeddings and long-sequence fine-tuning improve performance and retrieval helps weaker long-context models without closing the gap to models with stronger native long-context ability. Its limitation is that results are sensitive to dataset construction, prompting, scoring, language, and model versions, while aggregate task scores can hide position-specific failures and do not identify the mechanism behind degradation. Its relevance is that it broadens the evidence beyond single-needle retrieval to heterogeneous language tasks and makes cross-task comparisons of long-context capability possible. Bai et al. 2024 (https://aclanthology.org/2024.acl-long.172/)

6. Zhang et al. (2024), “∞Bench: Extending Long Context Evaluation Beyond 100K Tokens”

Zhang et al. propose ∞Bench, a benchmark whose average example length exceeds 100K tokens and whose English and Chinese tasks include both synthetic and realistic settings requiring long dependencies rather than merely locating a small number of passages. Their evaluation shows that existing long-context LLMs still need substantial improvement to process 100K-plus contexts effectively, despite recent systems claiming support for such lengths; the benchmark therefore tests the gap between advertised capacity and usable ultra-long-context understanding. The limitation is that evaluations at this scale are expensive and sensitive to inference infrastructure, tokenization, truncation, and task design, and even strong benchmark results would not prove robust use of every token in arbitrary real-world inputs. Its relevance is boundary-pushing: it extends degradation research from conventional 8K–32K contexts to the 100K-plus regime where context-window claims are most likely to overstate practical reliability. Zhang et al. 2024 (https://aclanthology.org/2024.acl-long.814/)

7. Hsieh et al. (2024), “Found in the middle: Calibrating Positional Attention Bias Improves Long Context Utilization”

Hsieh et al. investigate why relevant information in the middle of long inputs is underused and connect the lost-in-the-middle pattern to an intrinsic U-shaped positional attention bias: tokens near the beginning and end receive more attention than middle tokens regardless of their relevance. They introduce a calibration method, found-in-the-middle, that encourages attention to follow relevance rather than position and report improved retrieval of middle-position information as well as retrieval-augmented-generation gains of up to 10 percentage points over existing methods across tested tasks. The limitation is that the calibration is evaluated under specific model and task conditions, may trade off other behaviors, and cannot by itself explain all degradation caused by distractors, multi-hop reasoning, or excessive length. The paper is relevant because it turns a descriptive positional failure into a concrete mechanistic hypothesis and mitigation target, linking context degradation to attention allocation rather than treating it only as a benchmark artifact. Hsieh et al. 2024 (https://aclanthology.org/2024.findings-acl.890/)

8. Hsieh et al. (2024), “RULER: What’s the Real Context Size of Your Long-Context Language Models?”

Hsieh et al. argue that the standard needle-in-a-haystack test measures only a superficial form of long-context understanding and introduce RULER to vary sequence length and task complexity through multiple needles, multi-hop tracing, and aggregation tasks. Evaluating 17 long-context LMs on 13 tasks, they find that models can achieve nearly perfect accuracy on vanilla needle retrieval while suffering large performance drops as context length increases; although all evaluated models claim at least 32K-token contexts, only about half maintain satisfactory performance at 32K, and even a model advertised for 200K shows worsening results with longer and more complex inputs. The limitation is that RULER remains a synthetic, task-family-specific benchmark, its notion of “satisfactory” depends on a chosen threshold, and its model snapshot is from 2024. Its relevance is direct to the “context rot” framing: it operationalizes length-dependent degradation and shows why simple NIAH success can substantially overestimate general long-context competence. Hsieh et al. 2024 (https://arxiv.org/abs/2404.06654)

9. Modarressi et al. (2025), “NoLiMa: Long-Context Evaluation Beyond Literal Matching”

Modarressi et al. examine a weakness in needle-in-a-haystack evaluations: lexical overlap between the question and inserted answer can make retrieval much easier than genuine semantic search. NoLiMa minimizes literal overlap and requires latent associations to locate the relevant information; across 13 LLMs claiming at least 128K-token contexts, models perform well below 1K tokens but degrade sharply with length, with 11 models falling below half of their short-context baselines at 32K and GPT-4o dropping from 99.3% to 69.7%. The authors attribute the decline to the increasing difficulty of attention-based retrieval when direct lexical cues are absent, and report that reasoning capability or chain-of-thought prompting does not remove the problem. The limitation is that controlled needle construction is still not open-ended document reasoning, and the results depend on the selected model set and benchmark design. Its relevance is measurement-critical: it shows that apparent long-context robustness may be an artifact of literal matching and that semantic retrieval exposes context degradation more sharply. Modarressi et al. 2025 (https://proceedings.mlr.press/v267/modarressi25a.html)

10. Tian et al. (2025), “Distance between Relevant Information Pieces Causes Bias in Long-Context LLMs”

Tian et al. extend positional-bias analysis from one relevant span to multiple relevant information pieces, introducing LongPiBench with tasks and input lengths designed to vary the spacing among pieces that must be combined. Experiments with three commercial and six open-source models show that, although many current systems are more robust to the original single-piece lost-in-the-middle pattern, they still exhibit noticeable biases associated with the distance between relevant pieces. The limitation is that LongPiBench remains a controlled benchmark focused on spacing and multi-piece positional effects; it does not fully represent noisy conversation histories, tool traces, multimodal inputs, or evolving agent state. Its relevance is strong for realistic multi-document reasoning: degradation may depend not only on where each relevant item occurs, but also on how far apart the items are and whether the model can integrate dispersed evidence into one answer. Tian et al. 2025 (https://aclanthology.org/2025.findings-acl.28/)

Scope note

Across the retained studies, long-input acceptance is consistently separable from reliable information use. The evidence progresses from early findings about weak long-range dependency use, through lost-in-the-middle and positional-bias analyses, to broader benchmarks that vary task type, semantic difficulty, and context length. The exact phrase “context rot” is not used consistently in the scholarly papers, but the retained literature operationalizes the same concern through length-dependent performance loss, positional bias, distractor sensitivity, incomplete long-range dependency use, and failures to combine dispersed relevant information.

04 Context Compression

1. LLMLingua: Compressing Prompts for Accelerated Inference of Large Language Models

Jiang et al. (EMNLP 2023) propose LLMLingua, a coarse-to-fine prompt compressor that uses a budget controller to allocate compression across prompt components, demonstration-level reduction to preserve semantic integrity, iterative token-level compression to model conditional dependencies among retained tokens, and instruction tuning to align the small compressor with the target LLM. Across GSM8K, BBH, ShareGPT, and Arxiv-March23, the paper reports up to 20× compression with little performance loss, including strong preservation of reasoning and in-context-learning behavior. Its main limitation is that importance is estimated from token-level likelihood and compression budgets rather than from explicit claims, premises, warrants, proof steps, or logical dependencies; compression can therefore preserve task behavior without exposing why a span is inferentially necessary. It is foundational prior art for efficient prompt shortening and overlaps with a proposed reasoning-aware system at the selection/compression stage, but leaves a clear gap at the level of an inspectable argument representation and provenance-preserving dependency selection. Verified source: https://aclanthology.org/2023.emnlp-main.825/

2. LongLLMLingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt Compression

Jiang et al. (ACL 2024) extend LLMLingua to long-context settings with query-aware coarse- and fine-grained compression, document reordering, dynamic compression ratios, and subsequence recovery, targeting the interaction between key-information density, position bias, cost, and latency. Experiments across long-context question answering, multi-document reasoning, and related benchmarks show substantial gains: for example, the paper reports up to a 21.4% improvement on NaturalQuestions with roughly four times fewer GPT-3.5-Turbo input tokens, 94% cost reduction on LooGLE, and 1.4–2.6× latency acceleration for approximately 10k-token prompts compressed 2–6×. The limitation is that its query-conditioned importance signal is based on model likelihood/contrastive perplexity and positional or document-level heuristics, not on an explicit logical form or argument graph. It therefore overlaps with reasoning-aware context selection through query conditioning and protected information density, while leaving open the extraction of claims, premises, warrants, support/attack relations, and other inferential dependencies before compression. Verified source: https://aclanthology.org/2024.acl-long.91/

3. LLMLingua-2: Data Distillation for Efficient and Faithful Task-Agnostic Prompt Compression

Pan et al. (Findings ACL 2024) recast prompt compression as token classification: an LLM generates compression supervision, and a smaller bidirectional encoder is distilled to decide which tokens to retain or remove using full contextual information. The resulting compressor is faster and more faithful than repeatedly invoking a large autoregressive model, and the paper reports strong task-agnostic compression performance across multiple benchmarks. The central limitation is representational: the classifier predicts token retention labels rather than identifying argument components, logical relations, proof dependencies, or source-grounded propositions, so it may retain a useful lexical surface while discarding a low-frequency but globally essential premise. LLMLingua-2 is the closest overlap for a system that classifies before selecting context, and it establishes token-level classification as prior art; the gap is an explicit reasoning-structure intermediate representation whose selection objective preserves inferential dependencies rather than only token utility. Verified source: https://aclanthology.org/2024.findings-acl.57/

4. RECOMP: Improving Retrieval-Augmented LMs with Compression and Selective Augmentation

Xu, Shi, and Choi (ICLR 2024) introduce RECOMP, a retrieval-context compressor trained with downstream language-model performance as the objective, with both extractive and abstractive variants and the ability to return an empty string when retrieved evidence is not useful. The method improves retrieval-augmented generation by reducing distractors and selectively augmenting the generator with evidence that helps answer the task, demonstrating that compression can be learned as an end-task utility decision rather than as generic summarization. Its limitation is that extractive selection operates at the sentence level and abstractive compression generates a summary; neither explicitly identifies claims, premises, warrants, evidence roles, support/attack relations, or a logical skeleton. RECOMP overlaps with a proposed approach in downstream, task-oriented selection and in the possibility of dropping irrelevant evidence entirely, but it leaves the argument-aware representation and provenance-preserving reasoning selection problem unresolved. Verified source: https://arxiv.org/abs/2310.04408/

5. Compressing Context to Enhance Inference Efficiency of Large Language Models (Selective Context)

Li et al. (EMNLP 2023) present Selective Context, a model-agnostic method that removes redundant lexical units—tokens, phrases, or sentences—using surprisal and information-based criteria, then evaluates the shortened context on long documents, news, and conversations. The paper provides a simple and broadly applicable baseline for reducing input length while retaining generation quality and faithfulness, making explicit that local information content can guide inference-time pruning without retraining the target model. Its limitation is the mismatch between local redundancy and global reasoning value: a premise may be locally predictable or low-surprisal yet indispensable to a conclusion, and the method does not recover argument roles or dependencies. Selective Context is therefore important baseline prior art for lexical pruning, but a reasoning-aware compressor would need to add structure extraction, dependency protection, and source-span provenance rather than relying only on token- or sentence-level information scores. Verified source: https://aclanthology.org/2023.emnlp-main.391/

6. Adapting Language Models to Compress Contexts (AutoCompressors)

Chevalier et al. (EMNLP 2023) train language models to convert long text segments into compact continuous summary vectors or soft prompts, recursively passing compressed memory forward so later computation can condition on earlier content without rereading the full text. The paper demonstrates that latent compressed context can support long-context language modeling, compressed demonstrations, retrieval-augmented use, and passage reranking, showing that raw text is not always necessary at inference time. Its key limitation is opacity: the latent vectors have no directly inspectable argument units, logical relations, or span-level provenance, and compression is representational rather than an explicit extract-then-select pipeline. AutoCompressors thus overlap with a proposed system in the goal of replacing bulky context with compact memory, but they leave a gap for auditable reasoning scaffolds in which each retained proposition and dependency can be traced back to source text. Verified source: https://arxiv.org/abs/2305.14788

7. In-context Autoencoder for Context Compression in a Large Language Model (ICAE)

Ge et al. (ICLR 2024) introduce ICAE, a lightweight compressor that maps long contexts into compact memory slots consumed directly by a base LLM, using autoencoding/language-modeling pretraining followed by instruction tuning. The method reports substantial reductions in context length, latency, and GPU-memory use while retaining useful task behavior, establishing a practical latent-memory route that does not require changing the full target model. Its limitation is that the memory slots are opaque and there is no explicit extraction of claims, premises, warrants, proof steps, or logical relations, nor a selective retention stage that identifies which source spans support the downstream answer. ICAE is strong prior art for latent context compression and efficient inference, but it does not subsume an argument-skeleton approach whose contribution is an inspectable, provenance-aware intermediate representation and reasoning-specific selection objective. Verified source: https://proceedings.iclr.cc/paper_files/paper/2024/hash/0b276510ec2d3f6613a8b60c41ff0438-Abstract-Conference.html

8. Learning to Compress Prompts with Gist Tokens

Mu et al. (NeurIPS 2023) propose gist tokens, a small set of learned tokens into which a model encodes prompts, instructions, or demonstrations so that the compressed representation can be reused in subsequent generation. Experiments with decoder-only and encoder-decoder models show large prompt-length reductions while preserving task behavior, making gist tokens a reusable primitive for few-shot examples and repeated instructions. The limitation is that the representation is learned end to end and is not required to expose or preserve explicit argument components, logical relations, negation, numerical qualifiers, or provenance links to the original prompt. Gist tokens overlap with any proposed method that serializes selected reasoning information into a compact context, but they do not establish argument-aware selection; the open gap is whether an inspectable logical or argumentative scaffold can provide better faithfulness and controllability than opaque learned tokens. Verified source:
https://proceedings.neurips.cc/paper_files/paper/2023/hash/3d77c6dcc7f143aa2154e7f4d5e22d68-Abstract.html

9. SelfCP: Compressing Over-Limit Prompt via the Frozen Large Language Model Itself

Gao, Cao, and Li (2024) propose SelfCP, which uses the target frozen LLM to encode over-limit prompt regions into dense vectors and a small learnable connector and embedding layer to convert those vectors into dense tokens, while leaving the allowed portion of the prompt unmodified. Training updates only the lightweight connector and uses language-model supervision on long texts plus instruction data; experiments with two backbones report substituting roughly 12× over-limit prompts with dense tokens, reducing memory costs and improving throughput while maintaining or improving response quality. Its limitation is that the compression boundary is driven by hard context overflow and positional partitioning, while the dense tokens remain opaque and lack argument-level provenance or logical dependency information. SelfCP overlaps with hybrid designs that preserve some raw context while compressing overflow, but it leaves open a reasoning-aware policy for deciding which claims or dependencies deserve verbatim protection and which can be compressed. Verified source:
https://arxiv.org/abs/2405.17052

10. Concise and Precise Context Compression for Tool-Using Language Models

Xu et al. (Findings ACL 2024) address tool-documentation compression with two complementary strategies: selective compression retains execution-critical tool and parameter names as raw tokens, while block compression partitions documentation and maps blocks to fixed-length soft summaries, allowing the effective compression ratio to vary with documentation length. On API-Bank and APIBench, the approach reaches performance comparable to an upper-bound baseline under compression ratios of up to 16×, demonstrating that mixed raw-plus-soft context is useful when exact identifiers must survive compression. The limitation is domain-specificity: protected spans are schema-critical names rather than claims, premises, warrants, or proof dependencies, and the method does not extract an argumentative or logical representation. It offers a valuable design analogy and partial precedent for protecting reasoning-critical spans exactly, but the gap is a principled mechanism for identifying those spans from inferential structure rather than tool schemas. Verified source:
https://aclanthology.org/2024.findings-acl.974/

11. Compressed Context Memory for Online Language Model Interaction

Kim et al. (ICLR 2024) propose a recurrent key/value-memory compressor for online interaction, continually folding accumulated Transformer attention keys and values into a compact memory using a lightweight conditional LoRA inserted into the forward pass, without fine-tuning all model weights. The method is trained through a parallelized formulation of recursive compression and performs at the level of a full-context model with approximately 5× smaller context memory; it also supports streaming interaction with unbounded context and outperforms a sliding-window baseline. Its limitation is that compression occurs inside model state rather than through an inspectable textual or symbolic representation, so users cannot audit which propositions were retained or recover source-span provenance. This is important prior art for persistent and streaming context engineering, but it does not overlap with an explicit extract/classify/select argument pipeline beyond sharing the general objective of preserving useful history under a memory budget. Verified source:
https://proceedings.iclr.cc/paper_files/paper/2024/hash/97dc07f1253ab33ee514f395a82fa7cc-Abstract-Conference.html

12. Effective In-Context Example Selection through Data Compression

Sun et al. (Findings ACL 2024) frame few-shot example selection as data compression and introduce a two-stage method: BM25 first recalls examples relevant to the query, then an influence-aware reranking stage uses a meta-gradient/Fisher approximation to select examples that preserve useful information from the broader training set. Across five real-world datasets and four GPT-2 model sizes, the method reports an average improvement of 5.90% and consistently outperforms its BM25 comparison under three- and four-example settings. Its limitation is that it compresses the example set rather than the internal content of each example, and the selection signal is relevance plus parameter influence rather than argument roles, logical form, or proof dependency. The paper is relevant precedent for compact reasoning-scaffold selection at the demonstration level, but it leaves a gap for extracting and ranking the reasoning-critical components within examples or source documents before serialization into the final LLM context. Verified source: https://aclanthology.org/2024.findings-acl.50/

Cross-paper conclusion

The twelve papers establish six main compression families: lexical extractive pruning, query- or task-aware token/sentence selection, abstractive textual compression, learned latent or soft-token memory, recurrent key/value-memory compression, and example-set selection. Their common limitation for the present research question is that they optimize likelihood, redundancy, task loss, retrieval utility, memory efficiency, or end-task performance without first recovering an explicit, inspectable argument or logical representation. LLMLingua-2 provides the closest classification precedent, LongLLMLingua and RECOMP provide strong query/task-conditioned selection precedents, and the tool-documentation work provides a useful protected-span design pattern; however, no verified paper in this set combines claim/premise/warrant or logical-form extraction with downstream compression whose objective is to preserve inferential dependencies for LLM reasoning. A differentiated system should therefore make the structure representation, dependency-aware selection objective, and source-span provenance central, and should compare against these methods with ablations for structure extraction, edge preservation, protected qualifiers, and provenance.

05 Rag Grounding

Guu et al. (2020), “REALM: Retrieval-Augmented Language Model Pre-Training”

Guu et al., 2020, arXiv:2002.08909 (https://arxiv.org/abs/2002.08909) introduce REALM, which trains a latent document retriever jointly with a language model so that external text—such as Wikipedia—can be retrieved during pretraining, fine-tuning, and inference; on three open-domain QA benchmarks, the method improves absolute accuracy by roughly 4–16 points over prior systems while making knowledge access more modular and inspectable. For scholastic context engineering, the paper establishes the architectural case for separating an evidence store from parametric memory and retrieving evidence before structured claim or argument extraction. Its central limitation is that retrieval is trained through a language-modeling/task-performance signal rather than claim-level entailment: a passage may be relevant but incomplete, ambiguous, biased, outdated, or contradictory, and inspectable retrieval is therefore not equivalent to a supported conclusion.

Lewis et al. (2020), “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”

Lewis et al., 2020, arXiv:2005.11401 (https://arxiv.org/abs/2005.11401) define the canonical RAG architecture by combining a parametric sequence-to-sequence generator with a dense-vector index of Wikipedia, comparing a sequence-level formulation that keeps passages fixed with a token-level formulation that can vary retrieval across generation; the models achieve state-of-the-art results on three open-domain QA tasks and produce more specific, diverse, and factual language than a parametric-only baseline. The work directly supports a retrieve-then-generate context-engineering layer in which candidate documents remain explicit inputs to later extraction. However, the authors identify provenance and knowledge updating as open problems, and benchmark gains do not show that each generated sentence is supported: the generator may omit, distort, or combine retrieved facts without traceable claim-to-source links.

Izacard et al. (2022), “Atlas: Few-shot Learning with Retrieval Augmented Language Models”

Izacard et al., 2022, arXiv:2208.03299 (https://arxiv.org/abs/2208.03299) present Atlas, a retrieval-augmented language model jointly pretrained with a retriever and generator for knowledge-intensive few-shot learning; with only 64 examples, Atlas-11B exceeds 42% accuracy on Natural Questions and outperforms a 540B-parameter PaLM baseline by about three points, while the document index can be updated. This supports treating the evidence corpus as a replaceable, versionable context-engineering component rather than embedding all domain knowledge in model weights, a useful design for domain-specific argument extraction. The limitation is that few-shot QA and index updateability measure answer performance and adaptability, not source-faithful reasoning, citation completeness, argumentative validity, or robustness to incorrect and adversarial documents.

Ma et al. (2023), “Query Rewriting for Retrieval-Augmented Large Language Models”

Ma et al., 2023, arXiv:2305.14283 (https://arxiv.org/abs/2305.14283) propose Rewrite–Retrieve–Read, inserting a prompted or trainable query-rewriting stage between the user input and retrieval; a small rewriter can be optimized from reader feedback, and experiments on open-domain QA and multiple-choice QA report consistent improvements. The result is directly relevant to structured argument extraction because an initial request often fails to specify the entities, claims, time bounds, or relationships needed to retrieve the right evidence, so query planning should be explicit and auditable while preserving the original request. The authors note trade-offs between task specialization and generalization, the limits of one-turn retrieval compared with multi-step agents, and the reduced controllability of web search versus filtered dense indexes; moreover, a rewritten query can silently change scope and is not evidence that the retrieved material answers the original question.

Gao et al. (2023), “Enabling Large Language Models to Generate Text with Citations” (ALCE)

Gao et al., 2023, arXiv:2305.14627 (https://arxiv.org/abs/2305.14627) introduce ALCE, an automatic benchmark requiring systems to retrieve supporting passages and generate answers with citations, with metrics for fluency, correctness, and citation quality that correlate with human judgments; current systems still have substantial room to improve, and on ELI5 even the best models lack complete citation support about half the time. ALCE provides the clearest bridge from RAG to evidence-grounded argument extraction: correctness, citation relevance, and citation completeness must be evaluated separately, and an argument artifact should preserve claim-level support edges rather than merely append a bibliography. Its metrics depend on automatically generated claims and NLI models, which miss partial support and can be sensitive to output length; the benchmark also omits harder multi-hop, mathematical, and code settings, so citation presence is not sufficient proof of entailment.

Liu et al. (2023), “Lost in the Middle: How Language Models Use Long Contexts”

Liu et al., 2023, arXiv:2307.03172 (https://arxiv.org/abs/2307.03172) use controlled multi-document QA and key-value retrieval experiments to show that language-model performance depends strongly on where relevant information appears in a long context: accuracy is generally highest when evidence is near the beginning or end and falls when the answer is placed in the middle, including for models advertised as long-context systems. For context engineering, this means that adding more retrieved passages does not automatically improve grounding; evidence should be ranked, segmented, ordered, and localized, and argument-extraction evaluations should test position sensitivity. The study uses controlled QA and retrieval tasks rather than full argument extraction and does not directly measure citation faithfulness, so it establishes a recurring context-use failure mode rather than a universal quantitative law for every model, corpus, or prompt layout.

Asai et al. (2023), “SELF-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection”

Asai et al., 2023, arXiv:2310.11511 (https://arxiv.org/abs/2310.11511) propose SELF-RAG, which trains a single language model to decide when retrieval is needed, assess retrieved passages, generate answers, and critique its own outputs through special reflection tokens; 7B and 13B models outperform ChatGPT and retrieval-augmented Llama2-chat on reported open-domain QA, reasoning, fact-verification, long-form factuality, and citation-accuracy tasks. The design motivates adaptive evidence acquisition and an explicit critique stage in structured argument workflows: retrieve conditionally, evaluate relevance, and check extracted claims against the selected passages before acceptance. The limitation is that retrieval decisions, relevance judgments, and critiques are all generated by the same learned system, so the model can fail to retrieve, misclassify evidence, or approve an unsupported claim; self-reflection is a control aid, not an independent verifier.

Niu et al. (2024), “RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models”

Niu et al., 2024, arXiv:2401.00396 (https://arxiv.org/abs/2401.00396) introduce RAGTruth, a corpus of nearly 18,000 naturally generated RAG responses annotated at case and word/span levels for unsupported or contradictory content, and benchmark hallucination detectors across tasks, models, and contexts; fine-tuning a smaller Llama model on the corpus can approach prompt-based GPT-4 detection performance, while span-level detection remains difficult. The dataset offers a practical audit resource for checking extracted premises, claims, warrants, and conclusions individually rather than treating a response as wholly grounded or ungrounded. Its limitation is coverage: the corpus samples selected tasks, models, and application settings, and the authors note that important real-world scenarios remain unaddressed; detector performance on RAGTruth therefore cannot by itself establish that a production argument is grounded.

Tang and Yang (2024), “MultiHop-RAG: Benchmarking Retrieval-Augmented Generation for Multi-Hop Queries”

Tang and Yang, 2024, arXiv:2401.15391 (https://arxiv.org/abs/2401.15391) introduce MultiHop-RAG, a news-based knowledge base with multi-hop queries, ground-truth answers, and supporting evidence, and show that existing systems perform unsatisfactorily both when retrieving the multiple pieces of evidence and when answering after those pieces are supplied. The result is important for structured arguments because many conclusions depend on chains of premises distributed across documents; evaluation should therefore measure recall of the full support set and preservation of the reasoning chain, not only final-answer similarity. The benchmark restricts answers to simple forms such as yes/no, entity names, or temporal relations, caps support at four pieces, and evaluates a basic LlamaIndex RAG setup, so it does not fully represent open-ended argumentative synthesis, longer evidence chains, or stronger agentic retrieval systems.

Yan et al. (2024), “Corrective Retrieval Augmented Generation” (CRAG)

Yan et al., 2024, arXiv:2401.15884 (https://arxiv.org/abs/2401.15884) propose CRAG, which adds a lightweight retrieval evaluator that estimates document quality and triggers different actions—using the retrieved material, augmenting it with web search, or discarding/refining it through decomposition and recomposition; across four short- and long-form datasets, CRAG improves RAG-based generation. For context engineering, this supports a retrieval-quality gate before argument extraction, with evidence routed into states such as accepted, weak, conflicting, or unresolved instead of being passed through indiscriminately. The method depends on an externally trained evaluator whose mistakes can send the pipeline down the wrong path, while web search adds source-quality, temporal, and adversarial-content risks; corrective retrieval is thus not independent truth verification.

Jiang et al. (2024), “LongRAG: Enhancing Retrieval-Augmented Generation with Long-context LLMs”

Jiang et al., 2024, arXiv:2406.15319 (https://arxiv.org/abs/2406.15319) propose LongRAG, grouping related documents into approximately 4K-token retrieval units and pairing a long-context retriever with a long-context reader; on Wikipedia-based Natural Questions and HotpotQA, grouping reduces the corpus from 22 million to 600,000 units and improves retrieval recall, while zero-shot end-to-end results are competitive with trained systems, with additional gains on Qasper and MultiFieldQA-en. The paper is relevant where short chunks lose definitions, exceptions, or cross-sentence relations needed to reconstruct arguments, but the final artifact should still localize exact supporting spans within long units. Larger units also increase prompt load and may include irrelevant material, and the evaluation is primarily QA-oriented; it does not establish improved claim-level grounding, citation completeness, or resistance to long-context position effects.

06 Cot Faithfulness

1. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

Wei et al. (2022) evaluate few-shot chain-of-thought prompting on arithmetic, commonsense, and symbolic reasoning tasks, including GSM8K, across three large language models, comparing ordinary prompting with demonstrations that expose intermediate steps. They find that CoT produces large accuracy gains at sufficient scale, including state-of-the-art GSM8K performance for a 540B-parameter model with eight exemplars. The study's central limitation for faithfulness research is that improved performance and readable intermediate text are not tests of causal fidelity: it does not intervene on the trace, insert misleading factors, or determine whether the stated steps drove the answer. Its relevance is foundational: it established the capability benefit and the now-common assumption that visible reasoning traces reflect reasoning, thereby motivating later work that separates usefulness from faithful explanation. Source: https://arxiv.org/abs/2201.11903

2. Faithful Reasoning Using Large Language Models

Creswell and Shanahan (2022) design a structured reasoning process for multi-step logical deduction and scientific question answering in which two fine-tuned language models perform selection and inference, while beam search constructs candidate traces. The method makes the causal structure of the computation mirror the logical structure of the problem and yields checkable intermediate steps; it improves final-answer accuracy over baselines on ProofWriter-style deduction and EntailmentBank-style scientific QA. Its limitation is scope: the faithfulness guarantee is relative to the explicitly engineered selection/inference procedure, not a proof that an unconstrained language model's free-form CoT reveals its hidden computation, and it requires task-specific fine-tuning and structured domains. The paper is relevant as an early constructive alternative to trusting natural-language rationales: faithful reasoning is obtained by constraining the computation and exposing a verifiable procedure. Source: https://arxiv.org/abs/2208.14271

3. Talking About Large Language Models

Shanahan (2022) provides a conceptual analysis of language models as probabilistic sequence models and examines prompt patterns that produce argument-like or apparently reasoned text, including chain-of-thought behavior. The paper argues that a model can generate coherent, correct-seeming arguments through learned continuation patterns without that text necessarily constituting genuine reasoning or a faithful window into the computation that produced the answer; it contrasts unconstrained prompting with systems whose causal structure is aligned with logic. Because it is theoretical rather than an experimental study, it does not supply a quantitative faithfulness test or causal intervention result, and its conclusions depend on conceptual interpretation. Its relevance is diagnostic: it supplies the conceptual distinction between plausible argumentative form and a mechanism that actually tracks the factors responsible for an answer, a distinction later empirical papers operationalize. Source: https://arxiv.org/abs/2212.03551

4. Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting

Turpin et al. (2023) test CoT faithfulness by injecting controlled biasing features into inputs across 13 BIG-Bench Hard tasks and social-bias settings, using Claude 1.0 and GPT-3.5, then checking whether generated explanations acknowledge the factors that influence the answer. The models often follow injected biases, sometimes producing wrong answers, while giving plausible rationales that omit the decisive bias; under some interventions accuracy falls by as much as 36%. The main limitation is that the study uses a small set of older/proprietary models and deliberately constructed interventions, while textual acknowledgment remains an operational proxy rather than direct access to internal computation. The paper is directly relevant because it demonstrates that CoT can be systematically post-hoc and misleading: a rationale may sound coherent while concealing a causal influence that actually changed the prediction. Source: https://arxiv.org/abs/2305.04388

5. Faithful Chain-of-Thought Reasoning

Lyu et al. (2023) propose a two-stage Faithful CoT framework in which a language model translates natural-language inputs into a symbolic reasoning chain and a deterministic solver maps that chain to the final answer. They evaluate the approach on ten benchmarks covering math word problems, planning, multi-hop question answering, and relational inference, reporting a faithfulness guarantee for the chain-to-answer computation and performance gains over standard CoT on nine of ten benchmarks. The guarantee does not extend to the translation stage: the language model can still omit, distort, or misrepresent information before the deterministic solver executes it, and performance depends on task structure and solver quality. Its relevance is practical and architectural: moving final inference into an externally checkable procedure narrows the gap between the exposed reasoning and the computation that determines the answer. Source: https://arxiv.org/abs/2301.13379

6. Reasoning or Reciting? Exploring the Capabilities and Limitations of Language Models Through Counterfactual Tasks

Wu et al. (2024 version) evaluate whether apparent reasoning abilities transfer when default assumptions are changed, using eleven counterfactual variants spanning arithmetic, logic, syntax, code, spatial reasoning, music, chess, and related tasks, with CoT among the evaluation conditions. Models retain some ability but show substantial and consistent degradation on assumption-violating versions, indicating that default-task success can combine abstract competence with narrow, non-transferable procedures or memorized conventions. The limitation for direct faithfulness claims is that counterfactual generalization measures robustness of the underlying capability, not whether a particular textual CoT causally reports the computation for one answer; task construction and familiarity with the counterfactuals also matter. The paper is relevant as an adjacent diagnostic: fluent traces and high default accuracy can coexist with dependence on default procedures that the trace does not reveal or test. Source: https://arxiv.org/abs/2307.02477

7. Measuring Faithfulness in Chain-of-Thought Reasoning

Lanham et al. (2023) develop an intervention-based framework for testing whether models actually use their CoT by truncating traces early, inserting mistakes, paraphrasing them, replacing them with filler, and comparing answer sensitivity across tasks and model scales. They find wide task- and model-level variation: some answers depend little on the generated reasoning, while others depend substantially on it; filler does not reproduce the benefit of genuine CoT, and larger or more capable models are less faithful on most studied tasks. The limitation is interpretive: answer sensitivity is a necessary diagnostic but not a complete causal account, because textual perturbations can alter fluency, distributional likelihood, or other hidden cues without exposing all internal computation. Its relevance is methodological: it establishes reusable behavioral tests and shows that generating a rationale is not itself evidence that the rationale faithfully represents the model's reasoning. Source: https://arxiv.org/abs/2307.13702

8. Towards Faithful Chain-of-Thought: Large Language Models are Bridging Reasoners

Li et al. (2024) analyze how context, generated CoT, and final answers interact at the step level, distinguishing centralized reasoning from distributed pathways in which information reaches the answer without being verbalized in the chain. They use causal-relevance and attribution analyses and propose inferential bridging that supplies attribution-derived hints followed by semantic-consistency filtering; on the paper's evaluation tasks, the approach improves faithfulness. The method is limited by the quality of attribution scores, the selected benchmarks, and the fact that bridging does not establish faithfulness to every internal mechanism. The paper is relevant because it broadens the failure model beyond simple post-hoc rationalization: a model may retrieve correct information from context through a pathway that bypasses or supplements the visible CoT, yielding a correct answer with an incomplete or misleading explanation. Source: https://arxiv.org/abs/2405.18915

9. On the Hardness of Faithful Chain-of-Thought Reasoning in Large Language Models

Tanneru, Ley, Agarwal, and Lakkaraju (2024) test whether common steering approaches—novel in-context-learning prompts, fine-tuning, and activation editing—can improve CoT faithfulness across multiple reasoning and truthful-question-answering benchmarks and language models. They find that gains are generally marginal, controlled, and poorly transferable: fine-tuning and ICL sometimes improve faithfulness, but not reliably alongside accuracy, while activation editing has minimal success and methods do not generalize consistently across datasets. The conclusions are constrained by the chosen faithfulness metric, benchmark set, model coverage, and the limited range of activation-editing interventions. The paper is relevant as negative evidence against assuming that ordinary prompting or light steering solves the problem; faithful CoT behaves like a difficult alignment and interpretability target rather than a property that can be reliably induced with simple controls. Source: https://arxiv.org/abs/2406.10625

10. Chain-of-Thought Reasoning in the Wild Is Not Always Faithful

Arcuschin et al. (2026) investigate whether CoT unfaithfulness appears on naturally worded, non-adversarial prompts by pairing logically related questions, such as a question and its converse, and by studying implicit post-hoc rationalization, restoration errors, and shortcuts on difficult mathematics across frontier and reasoning models. They report substantial unfaithfulness in realistic settings, including implicit bias rationalization and illogical shortcuts, and find no tested model that is perfectly faithful. The evidence remains behavioral and proxy-based: headline rates depend on operational definitions, prompt design, and model versions, and the experiments do not directly reveal the full internal causal story. Its relevance is ecological validity: it extends concerns beyond deliberately injected laboratory biases to ordinary-looking prompts that resemble deployment and safety-monitoring conditions. Source: https://arxiv.org/abs/2503.08679

11. FAITHCOT-BENCH: Benchmarking Instance-Level Faithfulness of Chain-of-Thought Reasoning

Shen et al. (2026) formulate unfaithfulness detection as an instance-level discrimination problem and introduce FINE-CoT, an expert-annotated collection of more than 1,000 trajectories from four language models across four domains, including more than 300 unfaithful instances with fine-grained causes and step-level evidence. They compare eleven detectors spanning counterfactual, logit-based, and LLM-as-judge approaches and find that detection is especially difficult in knowledge-intensive domains and for stronger models, with step skipping and selective-explanation bias among recurring failure signals. The benchmark is limited by the validity and agreement of expert labels, the domain/model coverage, and the fact that detecting an unfaithful trajectory is not equivalent to proving the model's internal causal process. Its relevance is operational: it turns aggregate claims about CoT faithfulness into a concrete trajectory-level evaluation problem and shows why average faithfulness rates need instance-level validation. Source: https://arxiv.org/abs/2510.04040

12. Lie To Me: How Faithful Is Chain-of-Thought Reasoning in Open-Weight Reasoning Models?

Young (2026) evaluates twelve open-weight reasoning models from nine architecture families, ranging from 7B to 685B parameters, on 498 MMLU and GPQA Diamond questions with six classes of injected hints, measuring whether the model acknowledges a hint when it changes the answer. Across 41,832 inference runs, reported faithfulness ranges from 39.7% to 89.9% by model family; consistency and sycophancy hints receive the lowest acknowledgment rates, and keyword analysis shows a large gap between acknowledgment in thinking tokens and in answer text. The study is limited by its single-author preprint status, the possibility that keyword/acknowledgment criteria conflate lexical disclosure with genuine causal explanation, and sensitivity to prompts, implementations, and model versions. Its relevance is comparative and safety-oriented: it extends intervention-based acknowledgment tests across open-weight reasoning models and indicates that faithfulness is not a fixed property of "reasoning models," while visible answer text may systematically suppress influences recognized internally. Source: https://arxiv.org/abs/2603.22582

07 Neurosymbolic

Paper summaries

1. **ProofWriter: Generating Implications, Proofs, and Abductive Statements over Natural Language** (Tafjord, Dalvi, and Clark, Findings of ACL-IJCNLP 2021). ProofWriter trains and evaluates generative systems that answer questions over rule theories while producing natural-language implications and proof chains, and it also treats abductive statement generation as a task. Its evidence is controlled but substantive: the paper reports a new state of the art for proof generation, with significantly more correct proofs across the tested proof depths and evaluations of generalization to deeper and out-of-domain cases. The main limitation is that the theories are synthetic and RuleTaker-like, so the natural-language semantics, ontology, and inference regime are tightly constrained; a generated proof string is also not equivalent to validity in a rich typed or defeasible logic. For this project, ProofWriter is a direct precedent for separating answer generation from proof-producing context construction: passages can be normalized into explicit premises and rule chains before a natural-language rendering is produced, while its controlled setting highlights the need to preserve ambiguity and alternative interpretations when moving to cross-tradition material. Primary source (https://aclanthology.org/2021.findings-acl.317/)

2. **ReAct: Synergizing Reasoning and Acting in Language Models** (Yao et al., ICLR 2023). ReAct interleaves language-model reasoning traces with actions against external sources or environments, allowing a model to update plans, handle exceptions, and gather information instead of solving in one pass. In the reported experiments, interaction with a Wikipedia API reduced hallucination and error propagation on HotpotQA and FEVER, while the method also improved success on ALFWorld and WebShop relative to the cited baselines. Its limitation for the present agenda is that the actions are chiefly retrieval or environment operations rather than explicit validation of argument form; the visible reasoning trajectory is not guaranteed to be a faithful proof, and results depend on tool quality and benchmark affordances. ReAct nevertheless supplies an important architecture for context construction: an argument engine could hypothesize a claim or warrant, retrieve an authority, test a relation, record the result, and revise its context graph iteratively rather than treating retrieval as a single preliminary step. Primary source (https://arxiv.org/abs/2210.03629)

3. **PAL: Program-aided Language Models** (Gao et al., ICML 2023). PAL asks an LLM to read a natural-language problem and emit a runnable program as the intermediate reasoning artifact, then delegates computation to an interpreter such as Python. Across 13 mathematical, symbolic, and algorithmic tasks, the authors report higher accuracy than substantially larger models using ordinary language reasoning; the abstract specifically reports a 15-point absolute GSM8K advantage for Codex-based PAL over PaLM with chain-of-thought. The limitation is that execution checks the program's behavior, not whether the LLM's semantic decomposition correctly captured the problem, so a syntactically valid program may implement the wrong interpretation; the evaluated domains also do not address disagreement between traditions or implicit warrants. The project can adapt PAL's clean division of labor by using an LLM for semantic parsing and compilation into a project-specific argument/context representation, while reserving deterministic components for normalization, constraint checks, relation tests, and other operations that should not be left to token prediction. Primary source (https://proceedings.mlr.press/v202/gao23f.html)

4. **Program of Thoughts Prompting: Disentangling Computation from Reasoning for Numerical Reasoning Tasks** (Chen, Ma, Wang, and Cohen, arXiv 2022). Program of Thoughts (PoT) represents the reasoning process as executable code and delegates computation to an external computer, explicitly separating linguistic problem decomposition from arithmetic execution. The method was evaluated on five mathematical word-problem datasets and three financial-QA datasets in few-shot and zero-shot settings; the authors report an average gain of about 12% over chain-of-thought and state-of-the-art or near-state-of-the-art results when combined with self-consistency. Its limitation is domain scope: external execution can catch arithmetic and runtime failures but not framing, source-selection, interpretation, or warrant errors, and the program representation is poorly suited to defeasible, rhetorical, or tradition-specific argumentation without additional structure. PoT is relevant as a methodological precedent for compiling a natural-language context plan into executable or checkable intermediate artifacts, with the project needing to extend that separation from numeric computation to claim normalization, argument-scheme checking, provenance operations, and counterexample search. Primary source (https://arxiv.org/abs/2211.12588)

5. **Logic-LM: Empowering Large Language Models with Symbolic Solvers for Faithful Logical Reasoning** (Pan, Albalak, Wang, and Wang, Findings of EMNLP 2023). Logic-LM translates a natural-language problem into a symbolic formulation, runs a deterministic solver, and feeds solver error messages into a self-refinement loop that revises the formalization. It evaluates this pipeline on ProofWriter, PrOntoQA, FOLIO, LogicalDeduction, and AR-LSAT, reporting average gains of 39.2% over standard prompting and 18.4% over chain-of-thought prompting. The central limitation is that translation

remains the principal error source: a solver can return a valid result for a silently distorted formalization, while the benchmarks generally assume one intended logic and do not represent competing interpretive traditions, ambiguous warrants, or historically situated meanings. Logic-LM is the closest general template for this project because it motivates a typed intermediate representation for claims, premises, and argument-form labels, plus verifier feedback that repairs parses without erasing the source passage or alternative candidate formalizations. Primary source (https://aclanthology.org/2023.findings-emnlp.248/)

6. **LINC: A Neurosymbolic Approach for Logical Reasoning by Combining Language Models with First-Order Logic Provers** (Olausson et al., EMNLP 2023). LINC makes the LLM a semantic parser from natural language into first-order logic and delegates deductive inference to an external theorem prover. On FOLIO and a balanced ProofWriter subset, it reports substantial gains across models; notably, StarCoder+ augmented with LINC outperformed GPT-3.5 and GPT-4 with chain-of-thought on ProofWriter by the reported margins, while the analysis found complementary failure modes between LINC and chain-of-thought. The approach is limited by the fit of first-order logic: quantifier scope, negation, implicit assumptions, defeasible inference, abduction, pragmatics, and rhetorical force can all be lost or misrepresented, and theorem-prover validity does not establish historical or interpretive adequacy. For this project, LINC directly supports modularizing “understand the argument” from “check the inference,” and its complementary-failure result argues for retaining multiple candidate formalizations and uncertainty annotations rather than collapsing early to a single parse. Primary source (https://aclanthology.org/2023.emnlp-main.313/)

7. **Coupling Large Language Models with Logic Programming for Robust and General Reasoning from Text** (Yang, Ishay, and Lee, Findings of ACL 2023). This paper uses an LLM as a few-shot semantic parser into Answer Set Programming (ASP), then applies reusable declarative logic modules to answer questions and plan actions. The same general coupling is evaluated across bAbI, StepGame, CLUTRR, gSCAN, and robot-planning tasks; the authors report state-of-the-art performance on several benchmarks and successful plans in cases where an LLM alone fails, while the ASP layer makes errors more inspectable. Its limitation is that reusable modules still require domain engineering, and parser errors or missing coverage remain decisive; benchmark performance does not establish robustness to open-ended texts, conflicting authorities, rhetorical implicature, or historically situated concepts. ASP is especially relevant to the project because it offers a substrate for reusable argument/context operators, constraints, alternatives, and nontrivial rule interactions, while the project would need a tradition-aware layer that maps source-specific concepts into such modules without treating that mapping as lossless. Primary source (https://aclanthology.org/2023.findings-acl.321/)

8. **LeanDojo: Theorem Proving with Retrieval-Augmented Language Models** (Yang et al., arXiv 2023). LeanDojo provides an open toolkit, dataset, annotations, and interactive environment for Lean theorem proving, and introduces ReProver, which retrieves accessible formal premises before proposing proofs. The resource includes fine-grained premise annotations, program analysis for accessible premises and hard negatives, and a benchmark of 98,734 theorems and proofs with splits requiring generalization to novel premises; the reported experiments show ReProver improving over non-retrieval baselines and GPT-4. Its limitation is that Lean supplies a crisp correctness oracle that most interpretive arguments lack, while theorem-library conventions can make the setting unlike open-world reasoning; retrieval can still miss the needed premise or select a misleading one. The project can borrow LeanDojo's strongest design principle: context construction should be constrained by the authorities and formal resources actually accessible in the current environment, and the final structured artifact should preserve premise provenance and verifier state, even when the analogue of a theorem prover is only a partial consistency or entailment checker. Primary source (https://arxiv.org/abs/2306.15626)

9. **ToRA: A Tool-Integrated Reasoning Agent for Mathematical Problem Solving** (Gou et al., arXiv 2023). ToRA trains tool-integrated reasoning agents that alternate natural-language reasoning with computation libraries and symbolic solvers, using curated interactive trajectories, imitation learning, and output-space shaping. Across ten mathematical reasoning datasets, the authors report average absolute gains of 13–19% over open-source models; ToRA-7B reached 44.6% on MATH, and ToRA-Code-34B exceeded 50% according to the abstract. The limitation is that tool-use trajectories are costly and domain-specific, and the principal evaluation target is final answer accuracy rather than validity of intermediate context, provenance of selected authorities, or faithfulness of the reasoning trace; tool availability can also inflate apparent capability. ToRA extends PAL and PoT toward iterative tool selection, suggesting that the project should let a model choose among argument parsers, ontology lookups, citation retrieval, counterexample generation, and consistency checkers, with every invocation and result attached to the evolving context graph. Primary source (https://arxiv.org/abs/2309.17452)

10. **Verification and Refinement of Natural Language Explanations through LLM-Symbolic Theorem Proving** (Quan, Valentino, Dennis, and Freitas, EMNLP 2024). Explanation-Refiner integrates LLMs with theorem provers to generate and formalize explanatory sentences, propose inference strategies for natural-language inference, verify their logical validity, and

feed formal feedback into iterative refinement. The paper evaluates explanatory reasoning, autoformalization, error correction, and explanation enhancement across datasets and domains, using theorem-prover output as a formal validity signal rather than relying only on language-generation metrics or crowdsourced explanations. Its limitation is the same boundary that makes the guarantee useful: theorem-prover validity applies only to the encoded representation, so LLM formalization errors, incomplete explanations, normativity, ambiguity, and non-deductive force remain outside the guarantee; NLI datasets also do not represent the full complexity of historical or cross-tradition argumentation. This is the closest retained precedent for verifying explanations rather than only answers and supports a project pipeline with three linked artifacts: a source-grounded natural-language rendering, a formal or semi-formal claim/warrant graph where expressible, and verifier feedback attached to uncertain or failed edges. Primary source (https://aclanthology.org/2024.emnlp-main.172/)

Cross-paper implication

Across the ten papers, the convergent architecture is neural interpretation followed by an explicit intermediate representation, constrained or deterministic execution, feedback/refinement, and provenance-preserving natural-language rendering. The literature provides strong methodological precedent for separating semantic parsing from checking and for making retrieval or tool use iterative, but it does not yet solve cross-tradition adequacy: a symbolic system can validate an encoded deduction without establishing that the encoding captured a source's genre, historical use, local concept, implicit warrant, or normative force. The project's distinctive opportunity is therefore the missing middle layer—provenance-aware, uncertainty-preserving representations that retain source text and alternative parses while hosting multiple argument forms and marking exactly what symbolic verification can and cannot establish.

08 Logic Theorem Provers

Research date: 2026-10-11. Each retained paper below was opened at its ACL Anthology, ICLR Proceedings, or arXiv source page; the Explanation-Refiner paper was additionally checked from the published PDF. The novelty role is stated specifically against the question of whether prior systems classify or extract argument forms for context selection or context construction.

1. Logic-LM: Empowering Large Language Models with Symbolic Solvers for Faithful Logical Reasoning — Pan, Albalak, Wang, and Wang (2023)

ACL Anthology source (https://aclanthology.org/2023.findings-emnlp.248/) · arXiv source (https://arxiv.org/abs/2305.12295). Logic-LM uses an LLM to translate a natural-language logical problem into a symbolic formulation, sends that formulation to a deterministic symbolic solver for inference, and feeds solver error messages back into an iterative self-refinement loop. The framework is evaluated on ProofWriter, PrOntoQA, FOLIO, LogicalDeduction, and AR-LSAT, where solver-backed reasoning improves over ordinary prompting baselines. For novelty analysis, Logic-LM is a strong adjacent threat because it constructs a formal reasoning context from premises and a conclusion and validates the resulting inference externally; however, the verified paper does not make argument-form classification, proof-relevant sentence ranking, or minimal natural-language context selection a distinct component. A contribution focused on selecting or constructing context from candidate prose must therefore distinguish its relevance objective and selection representation from Logic-LM's whole-problem formalization-and-repair pipeline.

2. LINC: A Neurosymbolic Approach for Logical Reasoning by Combining Language Models with First-Order Logic Provers — Olausson et al. (2023)

ACL Anthology source (https://aclanthology.org/2023.emnlp-main.313/) · arXiv source (https://arxiv.org/abs/2310.15164). LINC, or Logical Inference via Neurosymbolic Computation, treats the LLM as a semantic parser that converts natural-language premises and conclusions into first-order-logic expressions, after which an external theorem prover performs deductive inference. The authors report gains on FOLIO and a balanced subset of ProofWriter across multiple language models, demonstrating that separating semantic parsing from symbolic deduction can improve logical reliability. Its role in novelty threat analysis is a strong adjacent precedent for extracting the content of an argument into formal premises and a formal conclusion, so claims about LLM-to-FOL argument extraction cannot be presented as new without a sharper distinction. LINC does not, in the verified description, introduce a dedicated taxonomy of argument forms or a proof-relevance stage that chooses a minimal or sufficient subset of natural-language candidate sentences for context construction.

3. Verification and Refinement of Natural Language Explanations through LLM-Symbolic Theorem Proving (Explanation-Refiner) — Quan, Valentino, Dennis, and Freitas (2024)

ACL Anthology source (https://aclanthology.org/2024.emnlp-main.172/) · published PDF (https://aclanthology.org/2024.emnlp-main.172.pdf) · arXiv source (https://arxiv.org/abs/2405.01379). Explanation-Refiner is the closest verified prior art for natural-language context construction: an LLM generates a rough inference, elicits sufficient and necessary facts, distinguishes relevant, redundant, and unrelated explanatory sentences, formalizes selected facts as axioms, and uses Isabelle/HOL theorem-proving feedback to iteratively refine the explanation. The framework combines event semantics and first-order logic and is evaluated on e-SNLI, QASC, and WorldTree, with the prover supplying formal validity guarantees and error feedback. In novelty analysis, this paper is the strongest direct threat because it operationalizes proof-relevant sentence selection and turns the selected explanatory context into a verified formal theory rather than merely translating an entire input problem. Any new system claiming argument-form classification or extraction for context selection must specify what is different in its argument-form taxonomy, representation, selection objective, proof/context interface, domain, or evaluation, and must compare directly against Explanation-Refiner.

4. LeanDojo: Theorem Proving with Retrieval-Augmented Language Models — Yang et al. (2023)

arXiv source (https://arxiv.org/abs/2306.15626). LeanDojo is an open-source Lean environment, dataset, toolkit, and benchmark that extracts proof data programmatically and annotates which premises are used in formal proofs. Building on those annotations, its ReProver system retrieves accessible premises from a large Lean mathematical library and conditions an LLM-based prover on the retrieved material; the benchmark contains 98,734 theorems and proofs and includes splits requiring generalization to novel premises. Its novelty role is a direct context-selection threat when “context construction” includes retrieving formal lemmas or premises for a theorem prover: it provides a mature mechanism for selecting compact proof context from a formal library. The overlap is weaker for a contribution that classifies or extracts argument forms from natural-language prose, because LeanDojo’s inputs are formal proof states and library declarations, not candidate explanatory sentences or unstructured arguments.

5. Magnushammer: A Transformer-Based Approach to Premise Selection — Miku■a et al. (2024)

ICLR Proceedings source (https://proceedings.iclr.cc/paper_files/paper/2024/hash/ab5f5f22e3e09f4424592ffb06840ab0-Abstract-Conference.html) · arXiv source (https://arxiv.org/abs/2303.04488). Magnushammer applies contrastive transformer training to premise selection for interactive automated theorem proving, replacing hand-engineered symbolic retrieval with learned retrieval of premises relevant to a proof state. The paper reports improvements over Sledgehammer on PISA and miniF2f, further improves proof success when combined with a language-model prover, and releases a large Isabelle dataset of proof-state/relevant-premise pairs. For novelty analysis, it is a direct formal-premise-selection precedent: a claim that a new system constructs prover context by learning which formal premises matter must explain its difference from Magnushammer’s retrieval objective, training data, proof assistant, and evaluation. It is only an indirect threat to natural-language argument-form extraction because it selects from formal library premises and does not classify argumentative relations or relevance among ordinary-language sentences.

6. Towards Large Language Models as Copilots for Theorem Proving in Lean (Lean Copilot) — Song, Yang, and Anandkumar (2024)

arXiv source (https://arxiv.org/abs/2404.12534). Lean Copilot integrates LLM inference natively into Lean and provides a framework for proof-step suggestion, proof-goal completion, and relevant-premise selection, supporting both pretrained models and user-supplied models running locally or in the cloud. The reported Mathematics in Lean experiments show improvements over rule-based automation such as aesop for both human-assisted and automated proving. Its role in novelty threat analysis is a direct baseline for interactive formal-context selection and for claims that LLMs can identify premises relevant to a Lean goal. The overlap is limited when the proposed contribution begins with natural-language argument forms, because Lean Copilot operates inside an already formalized Lean goal and library environment rather than extracting, classifying, or pruning candidate prose before formalization.

7. Harnessing the Power of Large Language Models for Natural Language to First-Order Logic Translation (MALLS/LogicLLaMA) — Yang et al. (2024)

ACL Anthology source (https://aclanthology.org/2024.acl-long.375/). This paper introduces MALLS, a 28K sentence-level dataset of verified natural-language/first-order-logic pairs, created through FOL rule parsing, human annotation, and automatic filtering, and trains LogicLLaMA models based on LLaMA2 for NL-to-FOL translation and correction. The dataset and models target the fine-grained formalization bottleneck that affects solver-augmented reasoning, including correction of rules generated by other LLMs. In novelty analysis, MALLS/LogicLLaMA is an important component-level precedent for sentence-level argument or logical-form extraction: it weakens claims that translating individual natural-language statements into formal rules is itself novel. The verified source does not establish an end-to-end mechanism that classifies argument forms, selects proof-relevant sentences, constructs a minimal context for a target conclusion, or validates the selected context with a theorem prover, so the threat is moderate and mainly concerns the extraction layer.

8. LOGIC-LM++: Multi-Step Refinement for Symbolic Formulations — Kirtania, Gupta, and Radhakrishnan (2024)

arXiv source (https://arxiv.org/abs/2407.02514). Logic-LM++ extends Logic-LM by using multi-step refinement and pairwise LLM comparisons to evaluate candidate symbolic formulations and select better refinements for complex reasoning tasks. The abstract reports improvements over Logic-LM and other methods on FOLIO, ProofWriter, and AR-LSAT. Its novelty role is an adjacent refinement-and-selection threat: it shows that comparing alternative formalizations and retaining a better candidate can be part of a logic–LM pipeline, so a proposed selection mechanism should not describe candidate-formulation comparison as unprecedented. The overlap remains narrower than explicit argument-form/context construction because Logic-LM++ selects among formal specifications generated for a problem; the verified source does not identify a natural-language argument-form classifier, a proof-relevance ranking over candidate sentences, or a minimal-context objective.

9. Automated Theorem Provers Help Improve Large Language Model Reasoning — McGinness and Baumgartner (2024)

arXiv source (https://arxiv.org/abs/2408.03492) · LPAR 2024 reference (https://doi.org/10.29007/2n9m). This work uses an LLM as a front end that translates PRONTOQA steamroller problems into a formal logic language, invokes logic-programming systems and first-order automated theorem provers, and analyzes translation failures using syntactic and semantic error categories. Its main extension uses theorem provers to correct semantic translation errors, reducing those errors and improving reasoning accuracy. For novelty analysis, it is a correction and validation precedent: it threatens claims that prover feedback can diagnose or repair faulty formalized arguments, and it establishes the importance of separating translation correctness from downstream proving. The verified paper does not report a dedicated argument-form classifier or a context selector that chooses relevant natural-language facts; its system formalizes the given problem and repairs errors, so the overlap with selective context construction is moderate and indirect.

10. LeanReasoner: Boosting Complex Logical Reasoning with Lean — Jiang, Fonseca, and Cohen (2024)

ACL Anthology source (https://aclanthology.org/2024.naacl-long.416/). LeanReasoner formalizes logical reasoning problems as Lean theorems and solves them by proving or disproving the corresponding formal statements, leveraging Lean’s symbolic checking and theorem library. The authors report state-of-the-art performance on FOLIO and near-state-of-the-art performance on ProofWriter while using fewer than 100 in-domain samples per dataset. Its novelty role is a formal-context and proof-assistant baseline: it shows that converting natural-language logical tasks into a theorem-prover representation and relying on library-backed symbolic verification is established practice. The overlap is weaker for argument-form classification or selective context construction because the abstract describes theorem formalization and proof/disproof rather than a separate stage that extracts argumentative relations, ranks candidate facts, or constructs a minimal proof-relevant context.

11. Language Models can be Deductive Solvers (LoGiPT) — Feng et al. (2024)

ACL Anthology source (https://aclanthology.org/2024.findings-naacl.254/) · arXiv source (https://arxiv.org/abs/2311.06158). LoGiPT addresses a weakness of solver-augmented language models in which an LLM’s parsing error prevents the external solver from executing: instead of relying on an explicit parser-to-solver interface, it trains a language model to internalize and emulate deductive-solver reasoning while adhering to strict solver syntax and grammar. The model is trained on a dataset derived from revealed and refined solver reasoning and is evaluated on two deductive reasoning benchmarks, outperforming solver-augmented and few-shot alternatives in the reported experiments. For novelty analysis, LoGiPT is a weak direct threat to context construction because it intentionally removes the explicit parsing, selection, and external-prover stages; it is nonetheless relevant as an alternative architecture for formal-syntax adherence and logical execution. A system whose novelty depends on explicit argument-form extraction, proof-relevant context selection, or prover-verified construction remains distinguishable from LoGiPT’s end-to-end solver imitation.

Cross-paper novelty conclusion

The retained literature separates into three context regimes. Explanation-Refiner is the closest precedent for selecting relevant, sufficient, necessary, redundant, and unrelated natural-language facts and converting the selected material into a prover-checked theory. Logic-LM and LINC are the main precedents for converting whole natural-language logical problems into formal premises and conclusions, while MALLS/LogicLLaMA strengthens the sentence-level translation component. LeanDojo/ReProver, Magnushammer, and Lean Copilot are the principal formal-library premise-retrieval precedents. Logic-LM++, Automated Theorem Provers Help Improve LLM Reasoning, LeanReasoner, and LoGiPT cover refinement, correction, Lean formalization, and solver imitation, respectively. A differentiated contribution should therefore state whether its novelty lies in a new argument-form taxonomy, extraction of argumentative relations rather than sentence-level FOL translation, proof-relevance selection under an explicit sufficiency/minimality/coverage/faithfulness objective, a cross-domain or long-context setting, or a new interface from selected natural-language context to formal verification; it should compare directly against Explanation-Refiner, Logic-LM/LINC, and at least one formal-premise retriever.

Sources consulted

1. Pan et al. (2023), ACL Anthology: https://aclanthology.org/2023.findings-emnlp.248/

2. Olausson et al. (2023), ACL Anthology: https://aclanthology.org/2023.emnlp-main.313/

3. Quan et al. (2024), ACL Anthology and PDF: https://aclanthology.org/2024.emnlp-main.172/

4. Yang et al. (2023), arXiv: https://arxiv.org/abs/2306.15626

5. Miku■a et al. (2024), ICLR Proceedings: https://proceedings.iclr.cc/paper_files/paper/2024/hash/ab5f5f22e3e09f4424592ffb06840ab0-Abstract-Conference.html

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9. McGinness and Baumgartner (2024), arXiv/LPAR DOI: https://arxiv.org/abs/2408.03492

10. Jiang et al. (2024), ACL Anthology: https://aclanthology.org/2024.naacl-long.416/

11. Feng et al. (2024), ACL Anthology: https://aclanthology.org/2024.findings-naacl.254/

09 Argument Mining

The ten retained papers below were checked against their primary PDFs or proceedings pages. Each paragraph records the paper's method and principal finding, its main limitation, and its relation to the target novelty question: whether mined argument structure or schemes are used to select/compress proof-relevant context for LLM reasoning.

Feng and Hirst (2011), "Classifying Arguments by Scheme." Feng and Hirst formulate argumentation-scheme classification as a supervised task over arguments whose premises and conclusions are already identified, using scheme-specific lexical and structural features to distinguish five common schemes; they report 63–91% one-against-others accuracy and 80–94% pairwise accuracy. The motivating downstream use is enthymeme reconstruction: classify the scheme, fit explicit propositions to its template, and infer unstated premises. The study is limited by its small Araucaria-derived corpus, restricted five-scheme inventory, upstream assumption that components are already detected, and low or incompletely reported annotation agreement. Relative to the target, it is a near-direct conceptual precedent for converting argument form into proof-relevant latent structure, but it neither retrieves or compresses context nor evaluates LLM reasoning. Feng & Hirst 2011 (https://aclanthology.org/P11-1099/)

Lawrence and Reed (2016), "Argument Mining Using Argumentation Scheme Structures." Lawrence and Reed exploit the internal structure of argumentation schemes to classify proposition types and assign component roles, showing that scheme components can be recognized from segmented natural language without requiring a pre-existing argumentative graph; the reported component F-scores are approximately 0.78–0.91. This is more informative than a binary scheme label because it identifies how individual propositions instantiate a reasoning pattern and therefore supplies material for a deductive skeleton. Its limitations are the restricted scheme set and corpus, noisy or only partially analyzed input, and evaluation focused on component-role recognition rather than proof validity or downstream retrieval utility. It is one of the closest pre-LLM representation precedents for component detection plus scheme structure, but it does not use that representation to select or compress context for reasoning. Lawrence & Reed 2016 (https://discovery.dundee.ac.uk/ws/files/12346360/FAIA287_0379.pdf)

Stab and Gurevych (2017), "Parsing Argumentation Structures in Persuasive Essays." Stab and Gurevych present a global argumentation-structure parser that jointly identifies token-level components and directed relations, using integer linear programming to impose coherence across component types and support/refutation links; the joint model improves over local or heuristic baselines on persuasive-essay data. The resulting graph is a strong substrate for tracing support and attack paths and for identifying candidate proof-relevant nodes. However, the work is domain-specific to persuasive essays, mainly represents claims/premises and support/attack rather than fine-grained inference schemes, and global coherence does not guarantee logical or proof validity; error analysis also shows substantial false positives caused by class imbalance and non-argumentative prose. Relative to the target, it is a core relation-mining precedent but not a scheme classifier, context compressor, or LLM reasoning system. Stab & Gurevych 2017 (https://aclanthology.org/J17-3005/)

Lawrence and Reed (2020), "Argument Mining: A Survey." Lawrence and Reed survey the field's foundations, datasets, and methods for argument/non-argument detection, component identification, relation mining, scheme analysis, and applications including retrieval and decision support. The survey establishes that claims, premises, directed relations, and scheme-level representations are mature and complementary strands rather than a single unified pipeline, and it catalogs recurring data and annotation challenges. As a survey, it introduces no new evaluated context-selection or reasoning method; moreover, its publication predates the current LLM and RAG wave, so it cannot by itself settle novelty against later prompt-compression systems. Its relation to the target is therefore architectural and historical: it confirms the availability of the representation layers that an integrated proof-oriented selector could combine, but contains no such closed loop. Lawrence & Reed 2020 (https://aclanthology.org/J19-4006/)

Lauscher et al. (2021), "Argument Mining for Scholarly Document Processing: Taking Stock and Looking Ahead." Lauscher and colleagues survey argument mining for scientific and scholarly documents, covering argumentative discourse units, component types, and relations such as support, contradiction, same-claim, detail, and sequence, while emphasizing that scientific writing interprets evidence and persuades readers about conclusions. The paper's principal contribution is to frame scholarly argument structure as a potentially useful representation for evidence interpretation, scientific retrieval, and knowledge construction, while identifying domain-specific data scarcity and annotation inconsistency. It is limited by sparse coverage of scientific discourse, heterogeneous label schemes across corpora, and the absence of an experiment showing that mined structure improves proof selection, context compression, or prompting. It is thus a near miss: it motivates structure-aware scientific context selection, but does not operationalize it and does not evaluate LLM reasoning. Lauscher et al. 2021 (https://aclanthology.org/2021.sdp-1.7.pdf)

Fan et al. (2024), "FRVA: Fact-Retrieval and Verification Augmented Entailment Tree Generation for Explainable Question Answering." Fan et al. introduce FRVA, which first retrieves and filters irrelevant facts to shrink the proof-search space, then combines deductive and abductive bidirectional reasoning with cross-verification and contrastive learning to generate entailment trees. The method reports strong fact-selection and structural-correctness results on EntailmentBank and is the clearest inspected precedent for selecting only facts that participate in an explicit proof skeleton. Its limitations are the controlled EntailmentBank fact setting, concatenation of filtered facts into long model inputs, a termination condition that can miss proofs with multiple intermediate conclusions, and evaluation difficulties when several valid tree structures exist. Relative to the target, FRVA directly overlaps on proof-relevant context selection and verified deductive structure, but it does not mine natural-language argument components or classify Walton-style schemes; its inputs are already fact statements and entailment trees. Fan et al. 2024 (https://aclanthology.org/2024.findings-acl.540/)

Yue et al. (2024), "Retrieval Augmented Fact Verification by Synthesizing Contrastive Arguments (RAFTS)." Yue et al. combine demonstration retrieval, document retrieval and reranking from verifiable sources, and in-context generation of supporting and refuting argument branches before an LLM makes a fact-verification prediction and explanation; the paper reports improvements over GPT-based baselines, including strong results with a smaller 7B model. RAFTS therefore demonstrates that retrieved evidence can be transformed into argument-shaped context rather than passed as undifferentiated passages. Its limitations are that retrieval robustness and reliability are not fully studied, domain mismatch between retrieval and fact-checking is not addressed, generated contrastive branches are not formally verified argument graphs, and no prompt-compression or proof-utility objective is measured. It is a direct overlap on argument-shaped RAG for LLM reasoning, but not on mined components, relations, schemes, or verified proof-skeleton selection. Yue et al. 2024 (https://aclanthology.org/2024.acl-long.556/)

Ruiz-Dolz, Kikteva, and Lawrence (2025), "Mining Complex Patterns of Argumentative Reasoning in Natural Language Dialogue." Ruiz-Dolz et al. create QT-SCHEMES, a corpus of 891 extracted dialogue arguments, 441 of which receive one of 24 argumentation-scheme labels, and evaluate Transformer models and prompted/fine-tuned LLMs for scheme and scheme-family classification, including transfer from textbook-like scheme data to political debate. The work supplies a current fine-grained representation of premise-to-conclusion reasoning in natural dialogue and reports state-of-the-art scheme-mining results, making it the closest current precedent for the scheme layer of a proof-oriented system. Its main limitations are the small effective annotated set, severe class imbalance, expensive annotation, unstable scores across many classes, and weaker transfer when textbook-like patterns meet natural debate. It does not use predicted schemes to retrieve or compress context and does not test downstream proof construction or LLM reasoning over selected context, so it remains a representation precedent rather than an integrated selector. Ruiz-Dolz et al. 2025 (https://aclanthology.org/2025.acl-long.368/)

Bezou-Vrakatseli, Cocarascu, and Modgil (2025), "Can Large Language Models Understand Argument Schemes?" Bezou-Vrakatseli et al. systematically evaluate seven LLMs on Walton-taxonomy scheme classification under zero-shot, few-shot, and chain-of-thought prompting, comparing prompts augmented with formal scheme definitions against prompts containing LLM-generated descriptions and testing manually annotated arguments, automatically generated arguments, and enthymemes. They find that larger models can classify schemes satisfactorily in suitable few-shot settings, while smaller models struggle and real, variable arguments are harder than structured generated examples. The study is limited to English, one taxonomy, a selected subset of schemes, and classification-only evaluation; it does not establish that scheme predictions improve reasoning, retrieval efficiency, or proof construction. Its novelty relation is direct for scheme classification in an LLM prompting regime, but the missing step is precisely the target integration: use scheme predictions to select or compress externally supplied proof context. Bezou-Vrakatseli et al. 2025 (https://aclanthology.org/2025.findings-acl.702/)

Mushtaq and Cabessa (2025), "Argument Mining with Fine-Tuned Large Language Models." Mushtaq and Cabessa cast argument component classification, relation identification, relation classification, and joint end-to-end argument mining as text-generation tasks and fine-tune eight open-weight LLM configurations on the PE, AbstRCT, and CDCP benchmarks; they report state-of-the-art results across the principal AM subtasks and datasets, with structural/context tags helping some settings. The work demonstrates that a single fine-tuned LLM can recover component and support/attack structure without bespoke architectures or feature engineering, making it a practical baseline for constructing an argument graph. Its limitations include dependence on benchmark domains, model and hyperparameter choices, post-processing and formatting guardrails, incomplete statistical robustness analysis, and no evaluation of proof validity, scheme classification, retrieval, or context compression. Relative to the target, it directly covers LLM-based component/relation extraction but leaves the scheme layer and proof-relevant context-selection loop unimplemented. Mushtaq & Cabessa 2025 (https://aclanthology.org/2025.coling-main.442/)

10 Walton Schemes Nlp

Verification date: October 12, 2026 (US/Eastern). I checked the linked publisher, ACL Anthology, repository, arXiv, PMC, or full-text PDF pages available for each retained paper. The 12 papers below are the retained core set from the subtopic-10 review. Each paragraph states the paper's contribution and its direct novelty threat or near-miss relation to scheme/form classification for context selection, retrieval, compression, or reasoning.

1. **Feng & Hirst (2011), "Classifying Arguments by Scheme."** This paper establishes the core NLP task: after argument detection and proposition classification, classify a natural-language argument into one of five common argumentation schemes using scheme-specific and general linguistic features. Its motivation is especially relevant to reasoning systems because scheme templates can support reconstruction of implicit premises (enthymemes), and it reports substantially better-than-chance one-against-others and pairwise classification accuracy. It is a **direct predecessor** for any claim to introduce Walton-scheme classification in NLP, and its enthymeme-reconstruction framing overlaps with scheme-guided reasoning; however, it has no LLM, retrieval, context-selection, compression, or token-budget component. Source: https://aclanthology.org/P11-1099/ and https://aclanthology.org/P11-1099.pdf

2. **Walton & Macagno (2015), "A Classification System for Argumentation Schemes."** This foundational paper is a taxonomy and methodology contribution rather than an NLP experiment: it surveys scheme classification, articulates desiderata for useful classifications, compares five classification approaches, and presents a refined graph-like organization of widely used defeasible schemes. Its most important relevance is conceptual: it distinguishes classifying schemes as a taxonomy from classifying textual arguments into schemes, while also separating both from deductive validity or formal logical-form classification. It is therefore a **foundational boundary-setting source and near miss**, not prior art for a learned context router; any system using scheme labels should state which taxonomy and ambiguity policy it adopts. Source: https://journals.sagepub.com/doi/10.1080/19462166.2015.1123772

3. **Lawrence & Reed (2016), "Argument Mining Using Argumentation Scheme Structures."** This paper uses argumentation-scheme structure as an intermediate representation for mining complex argumentative structures from segmented natural language, including proposition types and role relations rather than only assigning a final scheme label. That structural decomposition is the closest older precedent for routing later reasoning from inferred argument roles or scheme components: a proposed system that identifies premises, conclusions, and scheme positions before selecting evidence should compare against this lineage. It remains a **direct structural predecessor but not a direct context-selection system**, because it does not use LLM prompting, retrieval ranking, context compression, or an explicit context-budget objective. Source: https://doi.org/10.3233/978-1-61499-686-6-379

4. **Green (2018), "Towards Mining Scientific Discourse Using Argumentation Schemes."** Green proposes seven domain-specific argumentation schemes implemented as Prolog logic programs over semantic predicates that could be extracted by biomedical NLP tools, with the goal of recovering scheme names, explicit or implicit premises, conclusions, and relations within scientific-discovery narratives. The work is a **near miss with downstream reasoning and search/summarization implications**: it shows how scheme-guided semantic extraction can organize nonlocal scientific evidence and handle enthymemes, but its schemes are manually designed for biomedical discourse and are not a generic Walton classifier or learned LLM context policy. A context-routing proposal should distinguish domain-specific symbolic scheme programs from generic scheme-conditioned retrieval and test whether its gains transfer beyond one scientific genre. Source: https://doi.org/10.3233/AAC-180038

5. **Lawrence, Visser & Reed (2019), "An Online Annotation Assistant for Argument Schemes."** This paper integrates the Argument Scheme Key into the OVA argument-analysis environment to assist annotators and improve the construction of scheme-labelled corpora. Its contribution is **annotation infrastructure rather than inference-time reasoning**: it is a near miss for context selection, but an important prerequisite for it because reliable labels, explicit annotation procedures, and dataset construction determine whether a learned scheme-aware retriever can be evaluated. It does not rank retrieved evidence, compress context, route dialogue history, or control an LLM's token budget, so novelty claims should not treat annotation assistance as an existing context-selection baseline. Source: https://aclanthology.org/W19-4012/

6. **Jo, Bang, Reed & Hovy (2021), "Classifying Argumentative Relations Using Logical Mechanisms and Argumentation Schemes."** This paper classifies support, attack, and neutral relations between two statements using four theory-informed mechanisms—factual consistency, sentiment coherence, causal relation, and normative relation—and shows that these mechanisms improve unsupervised and supervised relation classifiers. It is a **near miss at the form-versus-scheme boundary**: the mechanisms provide interpretable logical signals inspired by argumentation schemes,

but the task is relation classification, not assigning a complete Walton scheme to an argument. It should be used as a negative control or comparison when claiming that scheme features outperform generic logical-form or relation features; it does not perform retrieval, context selection, or compression. Source: https://doi.org/10.1162/tacl_a_00394

7. **Visser, Lawrence, Reed, Wagemans & Walton (2021), "Annotating Argument Schemes."** This paper operationalizes Walton's taxonomy for annotation, describes the Argument Scheme Key and related procedures, and reports a corpus in which most inference relations receive a scheme assignment, while documenting the difficulty and ambiguity of the task. It is a **direct dataset and annotation basis** for scheme-aware models: it establishes how labels can be applied, how uncertain or overlapping classifications arise, and why a system should report multilabel or ambiguity behavior rather than assume every argument has one obvious class. The paper does not itself select or compress context, but any new routing method needs to evaluate on such human-annotated material and preserve the paper's distinction between argument form and argument substance. Source: https://doi.org/10.1007/s10503-020-09519-x

8. **Ruiz-Dolz, Taverner, Lawrence & Reed (2024), "NLAS-multi: A Multilingual Corpus of Automatically Generated Natural Language Argumentation Schemes."** NLAS-multi introduces a large multilingual, multi-topic corpus of automatically generated natural-language arguments instantiated from 20 Walton schemes, together with baselines and fine-tuned scheme-identification models. It is a **direct corpus/model resource and evaluation near threat**: it makes multilingual transfer and broad scheme coverage testable, but synthetic generation and validation can underrepresent the ambiguity, ellipsis, and contextual irregularity of human arguments. A context-selection contribution should not rely only on NLAS-multi or report gains from a single generated distribution; it should test human-authored data as well and measure downstream reasoning and token cost rather than classification alone. Source: https://doi.org/10.1016/j.dib.2024.111087

9. **Bezou-Vrakatseli, Cocarascu & Modgil (2024), "EthiX: A Dataset for Argument Scheme Classification in Ethical Debates."** EthiX provides 686 manually curated Kialo arguments across 22 ethical topics, annotated into eight Walton-based classes, and evaluates fine-tuned pretrained models including generalization to unseen or new topics. It is a **direct dataset/model resource** and a stronger contextual test than purely synthetic data because it captures human-authored ethical debate, dialogical context, enthymemes, and topic shift. The paper still stops at scheme classification: it does not use predicted schemes to retrieve evidence, select dialogue history, compress prompts, or optimize a reasoning budget. Any novelty claim about scheme-conditioned context should therefore compare against EthiX-style human data and show downstream gains beyond classification accuracy. Source: https://doi.org/10.3233/FAIA240919

10. **Trajano, Engelmann, Bordini, Sarkadi, Mumford & Panisson (2024), "Translating Natural Language Arguments to Computational Arguments Using LLMs."** This is the **strongest direct novelty threat** in the retained set. The paper classifies a natural-language argument by its associated argumentation scheme, supplies scheme descriptions/examples as context to an LLM, and uses retrieval-augmented generation to support translation into a computational argument representation; it also discusses short versus larger context and the risk that semantic retrieval returns examples from the wrong scheme. A new proposal cannot claim novelty merely for "classify the scheme, then provide scheme-relevant context to an LLM." The plausible gap is a demonstrably new adaptive mechanism—such as uncertainty-aware scheme-conditioned retrieval, learned evidence ranking, critical-question-aware compression, or explicit token-budget optimization—evaluated against this scheme-conditioned RAG/translation baseline. Source: https://doi.org/10.3233/FAIA240329

11. **Heinrich, Al Khatib & Stein (2025), "Multi-Class versus Means-End: Assessing Classification Approaches for Argument Patterns."** This paper compares ordinary multi-class scheme classification with a means-end approach that guides language models through an expert decision tree, using both prompting and supervised learning on two datasets. The staged approach improves explainability by exposing which decision-tree steps cause errors, while supervised means-end classification performs only slightly below the standard multi-class approach. It is a **direct classifier and hierarchical-guidance overlap, but not a retrieval/compression system**: it threatens claims of novel decision-tree or staged reasoning guidance, yet the paper does not on the inspected evidence rank external context, compress retrieved evidence, or allocate a token budget. A context-routing method should compare its controller against this decision-tree scaffold and isolate any benefit due to retrieval or compression rather than decomposition alone. Source: https://aclanthology.org/2025.argmining-1.19/ and https://aclanthology.org/2025.argmining-1.19.pdf

12. **Bezou-Vrakatseli, Cocarascu & Modgil (2025), "Can Large Language Models Understand Argument Schemes?"** This paper provides the strongest current benchmark-style reference for LLM Walton-scheme classification, evaluating seven models under zero-shot, few-shot, and chain-of-thought prompting with formal and LLM-generated scheme descriptions, on both manually annotated and automatically generated arguments including enthymemes. It is a **direct benchmark overlap** for any method that uses an LLM to infer a scheme or argument form, and its findings imply that model size, prompting, description quality, and data origin must be controlled. It does not retrieve, select, or compress context, so a new contribution

can remain distinct only by demonstrating downstream reasoning or evidence-selection value beyond these prompting baselines and by reporting the cost of the added context mechanism. Source: https://aclanthology.org/2025.findings-acl.702/ and https://aclanthology.org/2025.findings-acl.702.pdf

Cross-paper conclusion

The literature establishes argument-scheme identification, annotation, and LLM classification as mature enough that “classify Walton schemes” is not a novel contribution. The most consequential prior art is Trajano et al. (2024), which already conditions an LLM translation pipeline on scheme information and uses RAG, while Heinrich et al. (2025) covers hierarchical decision-tree guidance and Bezou-Vrakatseli et al. (2025) covers modern LLM prompting baselines. The remaining defensible gap is a separately evaluated mechanism that uses uncertain scheme/form predictions to select, rank, compress, or budget context for a downstream reasoning task, with explicit comparisons against generic embedding retrieval, direct prompting, decision-tree guidance, and scheme-conditioned RAG, and with both human-authored and generated/multilingual evaluation data.

11 Fallacy Detection

Verified paper summaries

1. Jin et al. (2022), "Logical Fallacy Detection." Jin et al. introduce the LOGIC dataset and the LOGICCLIMATE challenge set for detecting logical fallacies in general text and climate-change claims. Their central methodological claim is that fallacy detection requires modeling an argument's underlying logical structure rather than treating the task as ordinary sentence classification; a structure-aware classifier outperformed the strongest evaluated pretrained language model by 5.46 F1 on LOGIC and 4.51 F1 on LOGICCLIMATE. The main limitation is that the examples are relatively short and educational, while the challenge set is narrowly focused on climate claims, so the benchmark does not test long-range discourse, evidence retrieval, or faithful proof generation. For this project, the paper establishes the foundational design principle that context should be organized around premises, conclusions, and intended relations before an LLM is asked to identify a fallacy. **Verified source:** ACL Anthology (https://aclanthology.org/2022.findings-emnlp.532/) and the linked PDF.

2. Alhindi, Chakrabarty, Musi, and Muresan (2022), "Multitask Instruction-based Prompting for Fallacy Recognition." Alhindi et al. treat heterogeneous fallacy-recognition corpora as related tasks and train a multitask T5 prompting system across differences in input format, genre, and label inventory. The approach recognizes 28 fallacy types across domains and examines the effects of model size, prompt formulation, annotation quality, and external knowledge. Its main limitation is that it is primarily supervised T5-based recognition of supplied text: it does not reconstruct missing premises, recover a discourse graph, or retrieve evidence that establishes why an inference fails. The project relevance is direct because the work shows that context representation and task instructions must normalize heterogeneous argument formats; its external-knowledge analysis also motivates typed context augmentation rather than a single undifferentiated prompt. **Verified source:** ACL Anthology (https://aclanthology.org/2022.emnlp-main.560/) and the linked PDF.

3. Sourati et al. (2023), "Robust and Explainable Identification of Logical Fallacies in Natural Language Arguments." Sourati et al. formalize fallacy identification as a three-stage pipeline—binary detection, coarse-grained classification, and fine-grained classification—and evaluate prototype-based reasoning, instance-based reasoning, knowledge injection, data augmentation, and curriculum learning. The study finds that performance varies substantially by task component and fallacy class, and that data augmentation and curriculum learning can improve robustness while case- and knowledge-based methods provide explainability mechanisms. The limitation is that the framework consolidates heterogeneous prior datasets and its explanations are not necessarily faithful reconstructions of the exact premise-to-conclusion failure; injected knowledge is not selected by an explicit argument-graph or evidence-retrieval policy. For this project, it supplies a useful evaluation decomposition and supports treating fallacy classification as specialized reasoning over examples, prototypes, and relevant background knowledge rather than as a single label prediction. **Verified source:** Knowledge-Based Systems (https://www.sciencedirect.com/science/article/pii/S0950705123001685), with the matching arXiv record (https://arxiv.org/abs/2212.07425).

4. Goffredo et al. (2023), "Argument-based Detection and Classification of Fallacies in Political Debates." Goffredo et al. extend the ElecDeb60To16 political-debate corpus with token-level fallacy spans, claims, premises, support and attack relations, and six fallacy categories, then compare Transformer architectures that combine text representations with argumentative and engineered features. The reported result is that adding non-text argumentative features improves fallacy detection and classification over text-only alternatives. The limitation is domain concentration in U.S. presidential debates and a small six-class inventory; the relations are used as largely static model features rather than dynamically selected context or retrieved proof evidence, and the method does not generate an auditable object explaining why an edge is invalid. The paper is highly relevant here because it makes premise/claim boundaries and support/attack edges explicit, providing a concrete schema for selecting the surrounding context that a proof-oriented fallacy detector should preserve. **Verified source:** ACL Anthology (https://aclanthology.org/2023.emnlp-main.684/) and the linked PDF.

5. Payandeh et al. (2024), "How Susceptible Are LLMs to Logical Fallacies?" Payandeh et al. introduce LOGICOM, a multi-round persuader–debater benchmark in which an LLM must update or defend a position while being exposed either to logically valid reasoning or to fallacious arguments. GPT-3.5 and GPT-4 changed their opinions through reasoning, but were erroneously convinced 41% and 69% more often, respectively, under fallacious than logical conditions; the authors also release more than 5,000 paired logical/fallacious arguments. The limitation is that the benchmark measures susceptibility and decision change rather than accurate localization, classification, or explanation of a fallacy, and its paired debate scenarios may not represent naturally occurring discourse or domain-specific evidence. Its project relevance is that it demonstrates why local sentence context is insufficient: fallacy effects depend on prior turns, debate roles, claims, and the persuader's reasoning strategy, making multi-round discourse context an important evaluation condition. **Verified source:** ACL Anthology

(https://aclanthology.org/2024.lrec-main.726/) and the linked PDF.

6. Glockner et al. (2024), "MISSCI: Reconstructing Fallacies in Misrepresented Science." Glockner et al. introduce MISSCI, a dataset and argumentation-theoretic task in which a misinformation claim misrepresents a cited biomedical publication. The model must classify the implicit fallacy and verbalize the invalid reasoning that connects the publication's relevant content to the inaccurate claim; GPT-4 performs promisingly in zero-shot experiments, but human evaluation confirms that the task remains difficult. The limitation is the biomedical setting and curated, simplified study context, which do not fully test general discourse-graph recovery, and generated explanations can be inconsistent with the predicted label. For this project, MISSCI is a particularly strong template: proof-relevant context consists of the accurate study premise, publication-specific qualifiers such as species or population, and the reconstructed bridge that turns that premise into an unsupported claim. **Verified source:** ACL Anthology (https://aclanthology.org/2024.acl-long.240/) and the linked PDF.

7. Lei and Huang (2024), "Boosting Logical Fallacy Reasoning in LLMs via Logical Structure Tree." Lei and Huang construct an unsupervised logical-structure tree from constituency structure and a taxonomy of connective words, representing relation connectives as internal nodes and textual arguments as leaves. They inject this structure into LLMs either as a textualized hard prompt or as a relation-aware soft embedding, and report improved precision and recall for both fallacy detection and classification on benchmark datasets. The limitation is that connective-driven parsing can miss implicit relations, pragmatic context, and nonlocal discourse dependencies, while the recovered tree is not guaranteed to be the correct argument graph. The project relevance is exceptionally direct: the tree provides an explicit intermediate representation for selecting the premises and connective relation that should enter a fallacy-classification prompt instead of passing the entire flat string to the model. **Verified source:** ACL Anthology (https://aclanthology.org/2024.emnlp-main.730/) and the linked PDF.

8. Glockner et al. (2025), "Grounding Fallacies Misrepresenting Scientific Publications in Evidence." Glockner et al. extend MISSCI into MISSCI PLUS by grounding fallacious reasoning in real passages from the misrepresented biomedical studies and adding passage retrieval, fallacy verbalization, and fact-checking evaluations. The work shows that retrieval models must identify passages that only appear to support a claim through fallacious reasoning, that LLMs must verbalize the bridge from evidence to misinformation, and that current fact-checking models struggle to use such passages; the passages can even mislead LLMs into accepting false claims. The limitation is the curated biomedical setting and the partial separation between retrieval, classification, and fact-checking, leaving an end-to-end policy for selecting a minimal proof set unresolved. For this project, it is the strongest existing analogue for evidence-aware context construction: retrieve the passage supporting the accurate premise, preserve the domain qualifiers, identify the unsupported bridge, and test the resulting explanation against the source evidence. **Verified source:** ACL Anthology (https://aclanthology.org/2025.naacl-long.491/) and the linked PDF.

9. Li et al. (2024), "Reason from Fallacy: Enhancing Large Language Models' Logical Reasoning through Logical Fallacy Understanding." Li et al. define logical-fallacy understanding through five tasks spanning WHAT, WHY, and HOW: identifying the fallacy, explaining the violated reasoning pattern, and testing whether that understanding improves broader logical reasoning. They construct the LFUD dataset with GPT-4 assistance and limited human effort, then report that the dataset can evaluate and fine-tune LLMs, with fine-tuned models showing improved logical-reasoning performance. The limitation is that the taxonomy and generated explanations are broader than an explicit context-retrieval problem, and better task performance does not guarantee that a generated rationale is a valid or faithful proof. The project relevance is the decomposition into typed explanation slots: a proof-oriented system can separately represent the fallacy label (WHAT), the violated relation or missing assumption (WHY), and the downstream reasoning consequence (HOW). **Verified source:** ACL Anthology PDF (https://aclanthology.org/2024.findings-naacl.192.pdf) and the corresponding Findings of NAACL 2024 volume record.

10. Jeong, Jang, and Park (2025), "Large Language Models Are Better Logical Fallacy Reasoners with Counterargument, Explanation, and Goal-Aware Prompt Formulation." Jeong et al. enrich fallacy-detection inputs with model-generated contextual queries about counterarguments, explanations, and goals, assess those queries for validity within the argument context, and rank them by confidence before classification. Across datasets from five domains covering 29 fallacy types, the method reports gains of up to 0.57 F1 in zero-shot settings and 0.45 F1 in fine-tuned settings in the official abstract. The limitation is that query confidence and ranking are model-internal heuristics, so an improvement in benchmark F1 does not establish that the elicited context is causally sufficient or that the explanation is faithful. The project relevance is direct: counterargument, goal, and explanation are useful typed context slots for distinguishing an invalid inference from a merely controversial claim and for exposing what evidence is missing. **Verified source:** ACL Anthology (https://aclanthology.org/2025.findings-naacl.384/) and the linked PDF.

11. Papadopulos, Alam, and Da San Martino (2026), "Beyond Logical Forms: LLM-Extracted Patterns for Fallacy Classification." Papadopulos et al. investigate whether abstract logical forms become more useful when combined with context-level linguistic cues. Their framework uses LLMs to inductively extract reusable patterns from fallacious examples and their explanations, then evaluates those patterns in zero-shot and one-shot settings across multiple models and datasets; the reported experiments show statistically significant gains over zero-shot baselines and cross-dataset generalization. The limitation is that this is very recent preprint/workshop-stage evidence, so independent replication, adversarial paraphrase testing, and robustness to demonstration artifacts remain open; extracted patterns may inherit errors or label-specific surface cues. For this project, the paper supports storing a fallacy signature in two layers—an abstract argument form plus contextual and rhetorical cues—rather than discarding surface context as irrelevant noise. **Verified source:** arXiv (https://arxiv.org/abs/2606.26698); the arXiv record was available, while the workshop listing was used only as a publication lead.

12. Xu et al. (2026), "Socrates or Smartypants: Testing Logic Reasoning Capabilities of Large Language Models with Logic Programming-Based Test Oracles." Xu et al. introduce SMARTYPAT-BENCH, a benchmark of 502 naturally expressed Reddit posts with fine-grained fallacy labels, and SMARTYPAT, a generation framework that uses Prolog rules as logic-programming test oracles before LLMs realize controlled fallacies in fluent natural language. The authors report that the generated fallacies are comparable in subtlety and quality to human-authored examples, while structured reasoning improves categorization but excessive reasoning steps can reduce detection accuracy. The limitation is that synthetic coverage depends on the rule inventory and the quality of LLM realization, and Reddit/English selection limits generalization; a Prolog oracle verifies the intended formal pattern but not all pragmatic or domain-specific aspects of a real fallacy. The project relevance is the clean separation of formal validity checking from linguistic classification: symbolic rules can validate the intended argument form, while naturally varied text supplies the contextual stress test. **Verified source:** AAAI (https://ojs.aaai.org/index.php/AAAI/article/view/39021) and the official PDF link on that page.

13. Dore, Damo, Cabrio, and Villata (2026), "Retrieving Relations, Detecting Fallacies: A RAG Approach to Political Debate Analysis." Dore et al. propose a retrieval-augmented method that uses support and attack relations in political debates to dynamically steer retrieval from a 15 GB political knowledge base, replacing purely static argumentative features with relation-guided evidence selection. On the ElecDeb60to20 benchmark, evaluated across 42 retrieval configurations and 14 models, the paper reports improvements up to 0.864 macro-F1 for fallacy detection and 0.725 for classification over non-retrieval baselines. The limitation is that it is a very recent arXiv preprint focused on English U.S. presidential debates and a domain-tailored knowledge base; the results therefore require replication outside that setting, and retrieval/classification success does not itself prove that the selected passages form a faithful minimal proof. This is the clearest recent lead for the project's dynamic context-selection goal because it makes argument relations operational query controls for retrieving external evidence rather than merely adding them as classifier features. **Verified source:** arXiv (https://arxiv.org/abs/2608.27471), published August 18, 2026; no DOI was verified.

Verification notes

All 13 retained entries were checked against an official publisher or repository record available by October 12, 2026. ACL Anthology pages/PDFs were used for ACL/EMNLP/NAACL/LREC-COLING papers, ScienceDirect and arXiv were used for Sourati et al., AAAI's official article page/PDF were used for Xu et al., and arXiv was used for the 2026 Papadopulos and Dore preprints. The two least mature records are the 2026 arXiv papers: their metadata and abstracts were verified, but they should be treated as preprint evidence rather than settled consensus. The title/authorship for "Reason from Fallacy" is corrected here to Li et al.; the official PDF identifies Yanda Li, Dixuan Wang, Jiaqing Liang, Guochao Jiang, Qianyu He, Yanghua Xiao, and Deqing Yang.

12 Deductive Benchmarks

1. Johnson-Laird & Bara (1984), “Syllogistic Inference”

Johnson-Laird and Bara’s benchmark is a foundational human experimental regime for categorical syllogisms and three-term series problems, varying premise figure—the arrangement of terms—and the number of mental models required by an inference. Their experiments found systematic effects of figure and model count on performance and supported a mental-model account in which reasoners construct a model, derive a candidate conclusion, and search for counterexamples. Its limitation is that it is a small, classical human-cognition study rather than an LLM benchmark, and its theory and stimuli predate contemporary natural-language evaluation protocols. It remains directly relevant because any LLM syllogism benchmark that manipulates term order, figure, validity, or model complexity is extending a test structure already established in this work. Source: https://www.sciencedirect.com/science/article/abs/pii/0010027784900350

2. Oakhill, Johnson-Laird & Garnham (1989), “Believability and Syllogistic Reasoning”

This paper tests belief bias in syllogistic reasoning by examining whether believability affects premise interpretation, the examination of alternative representations, or the final filtering of candidate conclusions. Across three experiments involving both conclusion generation and conclusion evaluation, participants were influenced by beliefs at the alternative-model and conclusion-filtering stages, not only during initial premise interpretation. The limitation is that the evidence concerns human subjects and controlled syllogistic materials, so it does not directly quantify LLM reasoning or distinguish all modern failure modes such as parsing and format errors. Its relevance is methodological: benchmark items should cross logical validity with believable, unbelievable, and neutral content, because an accuracy score alone can hide whether a system is following logic or accepting plausible conclusions. Source: https://pubmed.ncbi.nlm.nih.gov/2721132/

3. Clark, Tafjord & Richardson (2020), “Transformers as Soft Reasoners over Language” (RuleTaker)

RuleTaker evaluates whether transformers can infer the truth of assertions from natural-language facts and rules generated from a controlled theory, with theorem-prover-derived labels and explicit reasoning-depth splits. The reported models reached about 99% accuracy on standard settings, above 95% on substantially deeper chains than seen during training, and transferred to hand-authored and paraphrased rulebases. Its limitations are the synthetic grammar, restricted vocabulary, and narrow rule language; it tests rule chaining rather than categorical syllogistic moods, open-ended argument interpretation, or rich proof quality. The benchmark is nevertheless a canonical baseline for deductive LLM evaluation because it makes depth, closed-world assumptions, and generalization measurable and establishes the soft-theorem-prover framing. Source: https://www.ijcai.org/Proceedings/2020/537

4. Tafjord, Dalvi Mishra & Clark (2021), “ProofWriter: Generating Implications, Proofs, and Abductive Statements over Natural Language”

ProofWriter extends RuleTaker-style natural-language theories from classifying preselected claims to generating logical implications, natural-language proofs, and constrained abductive explanations. It evaluates depth-controlled datasets, including unseen-depth and out-of-domain settings, and reports proof accuracy exceeding earlier reconstruction methods by 9 percentage points on RuleTaker while also providing strong baselines for implication enumeration and abduction. The limitation is that the theories remain largely templated and Datalog-like, so exact proof correctness does not test nuanced interpretation of real arguments, quantifier ambiguity, or human content effects. The work is highly relevant to benchmark design because it demonstrates that answer labels, proof certificates, consequence enumeration, and abduction should be treated as distinct evaluation targets rather than collapsed into one accuracy number. Source: https://aclanthology.org/2021.findings-acl.317.pdf

5. Tian et al. (2021), “Diagnosing the First-Order Logical Reasoning Ability Through LogicNLI”

LogicNLI is an NLI-style first-order-logic benchmark built from facts, rules, and hypotheses, covering combinations of core logical operators while minimizing commonsense interference; it adds a “Paradox” relation for cases where both a proposition and its negation can be derived through different paths. BERT, RoBERTa, and XLNet are evaluated across accuracy, robustness, generalization, and interpretability/traceability, and the reported models perform above random but below human performance, exposing weaknesses in FOL reasoning. Its limitation is the controlled logical language and pre-generative-LLM model regime, with no direct coverage of classical syllogistic figure or belief-bias effects. Its relevance is the diagnostic separation of entailment, contradiction, neutral, and inconsistency-like cases, which supports richer labels and robustness tests in a new deductive benchmark. Source: https://aclanthology.org/2021.emnlp-main.303/

6. Han et al. (2022/2024), “FOLIO: Natural Language Reasoning with First-Order Logic”

FOLIO evaluates deductive validity over human-authored natural-language premise sets and conclusions paired with first-order-logic annotations; its 1,430 unique conclusions are drawn from 487 premise sets, and an inference engine automatically verifies the formal annotations. The dataset supports both natural-language reasoning and NL-to-FOL translation, and the paper reports systematic experiments across multiple language models, including evidence that a difficult subset challenges GPT-4. Its limitations are that conclusion-level validity and translation accuracy still do not fully measure proof structure, minimal premise use, or alternative valid derivations, while formal annotation imposes an intended interpretation on potentially ambiguous language. FOLIO is relevant as a strong naturalness and logical-diversity baseline and as precedent for evaluating parsing/formalization separately from downstream deduction. Source: https://arxiv.org/abs/2209.00840

7. Saparov & He (2022), “Language Models Are Greedy Reasoners: A Systematic Formal Analysis of Chain-of-Thought” (PrOntoQA)

PrOntoQA generates question-answering problems from synthetic first-order world models so that a model’s chain of thought can be parsed into symbolic proof steps and analyzed by proof depth. Experiments with GPT-3 and InstructGPT found that models can often execute individual deduction steps correctly, including in fictional settings, but struggle with proof planning when multiple valid next steps must be explored. Its limitation is the restricted ontology and largely repeated modus-ponens-style structure, which makes it a controlled proof-depth test rather than a broad benchmark of categorical syllogisms, natural argumentation, or belief effects. It is relevant because it provides a clean precedent for separating final-answer accuracy from proof validity, search/planning behavior, and chain-of-thought faithfulness. Source: https://arxiv.org/abs/2210.01240

8. Srivastava et al. (2022), “Beyond the Imitation Game: Quantifying and Extrapolating the Capabilities of Language Models” (BIG-bench)

BIG-bench is a collaborative suite of 204 tasks evaluated across model families and scales, with human-rater baselines and task families covering logical deduction, categorical/formal syllogisms, and logical fallacies. Its results show broad scale-related improvement but generally poor absolute performance and calibration relative to human raters, while some multistep or brittle tasks display non-smooth capability changes. The limitation is that task-level binary or multiple-choice scoring can conflate reasoning with parsing and format compliance, and some tasks do not expose unknown, contradiction, proof, or error-type distinctions. BIG-bench is relevant as an ecosystem-level precedent for standardized zero-/few-shot evaluation and for placing syllogistic validity, formal fallacies, and general logical deduction in a common test suite. Source: https://arxiv.org/abs/2206.04615

9. Sanyal, Singh & Ren (2022), “FaiRR: Faithful and Robust Deductive Reasoning over Natural Language”

FaiRR evaluates deductive reasoning over natural-language rulebases by decomposing the task into independently modeled rule selection, fact selection, and knowledge composition components, thereby creating an explicit causal route from selected premises to generated inferences. On existing reasoning datasets, the authors report robustness to novel language perturbations, faster inference than prior approaches, and more interpretable errors than black-box generative proof systems. Its limitation is that the underlying rule-based regime remains controlled and does not directly address categorical quantifier moods, belief bias, premise-order effects, or open-ended argument structure. The work is relevant to benchmark construction because it motivates scoring evidence selection and derivation faithfulness separately from the final label and provides a model-side template for diagnosing whether a failure arose from premise selection or composition. Source: https://aclanthology.org/2022.acl-long.77/

10. Pan et al. (2023), “Logic-LM: Empowering Large Language Models with Symbolic Solvers for Faithful Logical Reasoning”

Logic-LM evaluates a pipeline in which an LLM translates a natural-language problem into a symbolic representation, a deterministic solver performs inference, and solver feedback drives self-refinement. It is tested on ProofWriter, PrOntoQA, FOLIO, LogicalDeduction, and AR-LSAT, achieving an average improvement of 39.2% over standard prompting and 18.4% over chain-of-thought prompting. Its limitations are dependence on the expressiveness of the selected symbolic solver and on in-context translation/self-refinement, which can fail for intricate or poorly expressible logical grammars; solver success also does not guarantee that the initial formalization captured the intended meaning. Its relevance is as a direct precedent for reporting parser/formalization errors separately from solver errors and for evaluating deductive benchmarks with hybrid symbolic verification. Source: https://aclanthology.org/2023.findings-emnlp.248/

11. Wu et al. (2023), “Hence, Socrates is mortal: A Benchmark for Natural Language Syllogistic Reasoning” (SylloBase)

SylloBase is a direct natural-language syllogism benchmark designed to cover a complete taxonomy of syllogistic reasoning patterns and both generation and understanding tasks. It contains 50,000 template-based samples mined from Wikidata and ConceptNet, GPT-3 paraphrases, and 1,000 manually rewritten test examples; T5 obtains a best generation ROUGE-L of 38.72 and RoBERTa a best multiple-choice accuracy of 72.77%. Its limitations are artifact risk from template construction and GPT-3 paraphrasing, plus an emphasis on answer prediction rather than proof generation, minimal-premise use, or invariance under irrelevant premise/order changes. It is highly relevant as an existing baseline against which any new categorical-syllogism benchmark must demonstrate added value beyond taxonomy coverage and surface naturalness. Source: https://aclanthology.org/2023.findings-acl.148/

12. Ando et al. (2023), “Evaluating Large Language Models with NeuBAROCO: Syllogistic Reasoning Ability and Human-like Biases”

NeuBAROCO repurposes a dataset originally created for psychological experiments into English- and Japanese-language LLM evaluations of categorical syllogisms, including belief bias, conversion errors, and atmosphere effects. The experiments find that current LLMs struggle more on problems involving these human-associated bias conditions, linking model behavior to established cognitive-science phenomena rather than only measuring aggregate validity. Its limitations are the need to disentangle dataset provenance, translation/language effects, and genuine bias resemblance; reproducing a human error pattern is not evidence of logically valid reasoning. The paper is relevant because it establishes multilingual, psychologically grounded bias testing as a benchmark dimension and supports evaluating content effects and indeterminate relations alongside correctness. Source: https://aclanthology.org/2023.naloma-1.1.pdf

13. Ozeki et al. (2024), “Exploring Reasoning Biases in Large Language Models Through Syllogism: Insights from the NeuBAROCO Dataset”

This follow-up NeuBAROCO study evaluates English and Japanese syllogistic problems using entailment, contradiction, and neutral cases, and adds a Translate-and-Explain chain-of-thought procedure that asks models to formalize the syllogism before explaining their reasoning. The results report human-like reasoning biases, additional model-specific error tendencies, and especially poor performance when the relation is neither entailment nor contradiction; analysis suggests that reasoning-process failures are more important than basic syllogism interpretation errors. Its limitation is that it is closely related to the 2023 NeuBAROCO report rather than an independent benchmark family, and the translation-plus-explanation protocol may itself introduce prompt and language artifacts. It is relevant as evidence that neutral/indeterminate cases and separate parsing-versus-reasoning diagnostics are essential for a modern syllogistic benchmark. Source: https://arxiv.org/abs/2408.04403

14. Eisape et al. (2023/2024), “A Systematic Comparison of Syllogistic Reasoning in Humans and Language Models”

This study compares human behavior with PaLM 2 and related language-model variants on two-premise categorical syllogisms, probing model scale, premise/term ordering, and classic syllogistic fallacies. Larger PaLM 2 models are reported to be more logical than smaller models and, on the studied items, more logical than humans, but even the largest models remain sensitive to irrelevant variable ordering and make confident invalid inferences that resemble human biases. Its limitations are the time-specific model family and prompt protocol, and human-like behavior is not necessarily the target for an LLM benchmark because humans can be systematically wrong. Its relevance is particularly strong for paired invariance and human/model evaluation: it shows that aggregate accuracy must be supplemented with order perturbations, fallacy profiles, calibration, and confidence/error analysis. Source: https://arxiv.org/abs/2311.00445

15. Abe et al. (2024/2025), “Abductive Reasoning with Syllogistic Forms in Large Language Models”

This paper converts a syllogistic dataset into an abductive task in which models choose which candidate premise best explains an observed conclusion, while also comparing abductive and deductive judgments. The released data encode English premises and hypotheses, gold abductive and deductive labels, syllogistic figure/form and term-order metadata, contradiction labels, and belief-consistency categories; the work frames abduction as a complementary form of reasoning rather than a replacement for deduction. Its limitation is conceptual as well as empirical: explanatory adequacy is not deductive validity, and the publication metadata differ across the Springer/HAR record and the later arXiv version, so bibliography details require care. It is relevant as an adjacent contrast benchmark that can test whether an LLM distinguishes necessary consequence from plausible explanation instead of silently pooling both under “reasoning accuracy.” Source: https://arxiv.org/abs/2603.06428

13 Knowledge Graphs

Verified paper summaries

1. ERNIE: Enhanced Representation through Knowledge Integration — Sun et al. (2019)

ERNIE adapts BERT-style masked-language pretraining so that the masking units reflect linguistic and knowledge-bearing structure rather than isolated tokens. Its entity-level masking hides multi-token entities as a unit, while phrase-level masking hides contiguous conceptual phrases, encouraging the model to recover semantically meaningful knowledge from context. The paper reports improvements over baseline models on five Chinese NLP tasks, including natural-language inference, semantic similarity, named-entity recognition, sentiment analysis, and question answering, as well as stronger cloze-style knowledge inference. As a scaffold, ERNIE is internal rather than inspectable at inference time: structured knowledge is absorbed into model representations during pretraining, which improves downstream competence but does not expose a query-specific evidence graph or guarantee that a generated claim can be traced to a particular fact. [10]

2. K-BERT: Enabling Language Representation with Knowledge Graph — Liu et al. (AAAI 2020)

K-BERT injects knowledge-graph triples directly into an input sentence, expanding the original text into a knowledge-rich tree that a BERT encoder can process without full domain-specific pretraining. The method addresses two practical problems created by injection: heterogeneous representations between textual tokens and graph entities, and knowledge noise caused by irrelevant expansions. It uses soft positions to preserve the original sentence structure and a visible matrix to restrict which tokens may attend to injected knowledge, thereby limiting interference with the sentence's meaning. The paper reports gains across twelve NLP tasks and emphasizes that the injected knowledge remains manually editable. K-BERT therefore establishes a clear side-context scaffold, but its sentence expansion and encoder-oriented design leave open the harder problem of selecting and verifying evidence for unconstrained, open-ended generation. [9]

3. KEPLER: A Unified Model for Knowledge Embedding and Pre-trained Language Representation — Wang et al. (2020)

KEPLER unifies language-model pretraining with knowledge-graph embedding by encoding textual descriptions of entities with a pretrained language model and jointly optimizing language-modeling and relational knowledge-embedding objectives. This design lets the model use textual information to improve entity representations while also imposing graph-based constraints that capture relations among entities. The authors introduce Wikidata5M, a large-scale knowledge graph aligned with entity descriptions, and report strong results on language tasks and knowledge-graph link prediction, including inductive settings involving entities not seen during training. KEPLER is important as a shared-representation precursor to graph-conditioned language systems: it places language and structured knowledge in a common latent space rather than constructing a runtime prompt. Its limitation for context engineering is that joint embeddings do not themselves expose which graph facts support a particular answer or enforce faithful multi-hop generation. [11]

4. QA-GNN: Reasoning with Language Models and Knowledge Graphs for Question Answering — Yasunaga et al. (NAACL 2021)

QA-GNN addresses knowledge-intensive multiple-choice question answering by combining language-model relevance estimation with graph reasoning over a retrieved knowledge-graph subgraph. Given a question and answer choice, it first identifies topic entities and extracts a bounded subgraph, then scores graph nodes according to their relevance to the QA context. The model represents the textual QA context as a node in a joint graph connected to the retrieved KG entities, and an attention-based graph network updates the language and graph representations together. Evaluations on CommonsenseQA and OpenBookQA show improvements over language-model-only and prior LM-plus-KG systems, including better handling of negation and entity substitution; the paper also presents reasoning subgraphs as a form of explanation. QA-GNN is a strong query-specific scaffold, although its evidence selection and reasoning are specialized to bounded QA rather than free-form generation. [12]

5. GreaseLM: Graph REASoning Enhanced Language Models for Question Answering — Zhang et al. (ICLR 2022)

GreaseLM builds a deeper fusion between language models and graph neural networks for questions that require both textual interpretation and structured world knowledge. Instead of adding a graph representation only before or after language encoding, it stacks cross-modal interaction layers in which information propagates in both directions: graph representations ground the language context, while linguistic features such as negation and hedging modify the graph-side reasoning. The authors evaluate the approach on CommonsenseQA, OpenBookQA, and the medical MedQA-USMLE benchmark, reporting more reliable reasoning over situational constraints and structured knowledge and performance that can exceed models substantially larger than GreaseLM. The method demonstrates that a KG can function as an active reasoning partner rather than a static prompt supplement. However, it depends on curated graph construction and task-specific architecture, and its learned fusion does not by itself provide claim-level provenance for generated answers. [13]

6. Knowledge Graph Prompting for Multi-Document Question Answering — Wang et al. (AAAI 2024)

Knowledge Graph Prompting (KGP) targets multi-document question answering, where the relevant evidence is distributed across passages and document structures rather than contained in one independently retrievable chunk. The method constructs a graph whose nodes represent passages or structural units such as pages and tables, with edges encoding semantic or lexical similarity and document relations. An LLM-based graph-traversal agent then navigates this global graph to collect supporting passages, using the graph as a global regulator of transitions and the agent as a local navigator that progressively approaches the question. The paper reports improved retrieval and QA performance and argues that graph organization can reduce retrieval latency while preserving cross-document associations. KGP is especially close to context engineering because it separates graph construction, traversal, and prompt formation, but its final evidence selection still depends on learned extraction and LLM navigation rather than formal entailment checks. [3]

7. MindMap: Knowledge Graph Prompting Sparks Graph of Thoughts in Large Language Models — Wen, Wang, and Sun (ACL 2024)

MindMap presents a plug-and-play prompting pipeline that uses knowledge graphs not only as external evidence but also as a scaffold for organizing the model’s intermediate reasoning. The system extracts keywords from the question, identifies corresponding KG entities, retrieves path and neighborhood evidence, merges those subgraphs into reasoning structures, and prompts an LLM to combine external KG facts with its implicit knowledge. It additionally elicits a “mind map” of the inference process, intended to make the reasoning pathway more transparent and graph-structured. Experiments across question-answering tasks, with particular attention to medical settings, report improvements over prompting baselines and introduce a hallucination-oriented evaluation benchmark. MindMap’s distinctive contribution is topological organization of evidence and thoughts, but its reasoning transformations remain model-mediated: graph completeness, consistency, and correspondence between the displayed mind map and the actual causal basis of the answer are not guaranteed. [2]

8. KnowGPT: Knowledge Graph based Prompting for Large Language Models — Feng et al. (NeurIPS 2024)

KnowGPT develops a knowledge-graph prompting framework for augmenting closed or black-box language models with domain knowledge using hard prompts. It is motivated by the large search spaces, API costs, and manual prompt-engineering burden that arise when an LLM must decide how to use KG information. The framework includes a knowledge-extraction module that selects informative facts from a graph and a context-aware prompt-construction module that converts those facts into prompts suitable for the target LLM. The authors evaluate KnowGPT on three benchmarks and report substantial gains, including 92.6% accuracy on OpenBookQA. The paper’s main relevance to scaffolding is its explicit separation of graph knowledge selection from prompt realization, allowing structured domain facts to reach an otherwise unchanged LLM. The remaining risks are entity-linking and path-selection errors, sensitivity to KG quality, and the absence of a guarantee that the model’s response is entailed by the prompted facts. [1]

9. KAPALM: Knowledge grAPh enhAnced Language Models for Fake News Detection — Ma et al. (Findings EMNLP 2023)

KAPALM applies graph-enhanced language modeling to fake-news detection, showing that structured entity relations can support credibility-sensitive classification beyond simple entity matching. The system links entities in a news item to a knowledge graph, extracts a relevant subgraph, and computes a coarse-grained representation with a graph neural network. It then prunes the subgraph and processes the resulting fine-grained knowledge with attentive graph and graph-pooling layers before combining both graph representations with the textual representation of the news. On two benchmark datasets, the authors report results superior to state-of-the-art baselines and competitive performance in a few-shot setting. KAPALM broadens the evidence for graph scaffolds beyond QA and generation: graph structure can supply relational signals for a decision task. Its direct transfer to general-purpose LLM context construction is limited, however, because the architecture, supervision, and evaluation are tightly tied to fake-news classification and do not test generation faithfulness. [8]

10. Knowledge Graph-Enhanced Large Language Models via Path Selection — Liu et al. (Findings ACL 2024)

This paper proposes KELP, a framework for selecting knowledge-graph paths before incorporating them into LLM processing. It responds to two weaknesses in earlier KG-enhanced methods: reliance on the LLM to make coarse binary decisions about whether a path should be used, and a tendency to select only paths with direct lexical or semantic overlap with the input. KELP assigns finer-grained scores to candidate paths through latent semantic matching and uses trained encodings to preserve useful paths whose relations to the input are indirect. Experiments on real-world datasets support the effectiveness of this more principled selection process. For structured context engineering, KELP isolates a key control point between graph retrieval and model consumption: the scaffold should be compact, relevance-ranked, and capable of preserving inferentially useful indirect relations. The remaining concern is that learned path scoring can encode selection bias and may omit conflicting or complementary evidence outside the chosen paths. [7]

11. GRAG: Graph Retrieval-Augmented Generation — Hu et al. (Findings NAACL 2025)

GRAG extends retrieval-augmented generation from isolated documents to networked textual graphs such as citation networks, social graphs, and knowledge graphs. It treats retrieval as a textual-subgraph problem and proposes a divide-and-conquer strategy intended to identify a useful subgraph efficiently, with linear-time retrieval in the paper's formulation. For generation, GRAG exposes the retrieved graph through complementary text and graph views so that the LLM can use both node content and topology. Experiments on graph-reasoning benchmarks show especially strong gains over conventional RAG in multi-hop settings where relationships among documents are essential. GRAG's central contribution is to make topology part of the retrieved context rather than treating edges as incidental metadata. This broadens KG scaffolding to arbitrary networked corpora, but neighborhood expansion can still introduce noise and context cost, and the reported benefits depend on graphs whose topology is meaningfully aligned with answer relevance. [4]

12. Knowledge Graph-Guided Retrieval Augmented Generation — Zhu et al. (NAACL 2025)

KG²RAG improves retrieval-augmented generation by using a knowledge graph to connect otherwise isolated text chunks. It begins with semantic retrieval to obtain seed chunks, then performs KG-guided chunk expansion to find factually related evidence and KG-based organization to arrange the selected material into coherent paragraphs for the generator. The framework is designed to improve both retrieval quality and response quality by preserving fact-level relationships, increasing evidence diversity, and supplying the LLM with a more structured context than a ranked list of semantically similar chunks. Experiments on HotpotQA and variants report advantages over existing RAG methods on both retrieval and answer measures. KG²RAG is a direct model of graph-mediated context construction, but it remains primarily a retrieval optimization layer: it assumes reliable graph construction and entity/chunk linking, can propagate graph errors, and does not eliminate unsupported claims produced after the organized context reaches the LLM. [5]

13. Graph-constrained Reasoning: Faithful Reasoning on Knowledge Graphs with Large Language Models — Luo et al. (ICML 2025)

Graph-constrained reasoning (GCR) moves from presenting graph evidence to constraining the language model's reasoning process so that generated paths remain grounded in a knowledge graph. Its KG-Trie is a trie-based index of valid KG reasoning paths that is integrated into decoding, restricting the model toward graph-valid continuations rather than allowing unconstrained free-form reasoning. The framework combines a lightweight KG-specialized LLM for graph-constrained reasoning with a stronger general LLM for inductive reasoning across multiple candidate paths. On several KGQA benchmarks, the paper reports state-of-the-art results, zero reasoning hallucination under its graph-grounded formulation, and zero-shot generalization to unseen KGs without additional training. GCR represents a stronger scaffold than retrieval or prompting because graph structure becomes a decoding constraint. The tradeoff is brittleness: incomplete, ambiguous, or poorly matched graphs can reduce recall and flexibility, so faithfulness is bounded by graph coverage and entity/linking quality. [6]

Verification sources

[1] https://arxiv.org/abs/2312.06185

[2] https://aclanthology.org/2024.acl-long.558/

[3] https://ojs.aaai.org/index.php/AAAI/article/view/29889

[4] https://aclanthology.org/2025.findings-naacl.232/

[5] https://aclanthology.org/2025.naacl-long.449/

[6] https://proceedings.mlr.press/v267/luo25t.html

[7] https://aclanthology.org/2024.findings-acl.376/

[8] https://aclanthology.org/2023.findings-emnlp.263/

[9] https://aaai.org/papers/02901-k-bert-enabling-language-representation-with-knowledge-graph/

[10] https://arxiv.org/abs/1904.09223

[11] https://arxiv.org/abs/1911.06136

[12] https://aclanthology.org/2021.naacl-main.45/

[13] https://arxiv.org/abs/2201.08860

14 Premise Context Selection

Paper summaries

1. **DeepMath — Deep Sequence Models for Premise Selection** (Irving et al., NeurIPS 2016). DeepMath treats premise selection as a central bottleneck in automated theorem proving: a conjecture must be paired with a small subset of useful axioms from a much larger formal library. Its two-stage neural sequence-model approach is evaluated on the Mizar corpus and is designed to avoid much of the hand-engineered feature machinery used by earlier systems, establishing an early large-scale precedent for learned proof-relevant retrieval. The task regime is formal theorem proving rather than open-ended language-model reasoning, and its correctness target is whether retrieved premises help an automated prover. For this project, the direct overlap is foundational: context is not selected for topical similarity alone but for its expected utility in constructing a proof. The gap is that DeepMath does not classify argument forms, preserve natural-language provenance, or model a downstream LLM's need for interpretable premises and explicit proof obligations. Primary source (https://papers.neurips.cc/paper_files/paper/2016/hash/f197002b9a0853eca5e046d9ca4663d5-Abstract.html)

2. **Premise Selection for Theorem Proving by Deep Graph Embedding** (Wang, Tang, Wang, and Deng, NeurIPS 2017). This paper represents higher-order-logic formulas as graphs and learns embeddings for premise selection, explicitly preserving syntactic and semantic structure while remaining invariant to variable renaming and retaining function-argument order. On the HolStep benchmark, the graph method improves reported classification accuracy from 83% to 90.3%, showing that formula structure can provide a stronger proof-relevance signal than sequential token representations. The task regime is candidate-premise classification for HOL theorem proving, where the selected statements are subsequently useful to a formal prover; it is therefore a direct rather than merely semantic-retrieval precedent. Its limitation for the project is that the graph is a formal expression representation, not an argument-form representation over heterogeneous natural-language sources, and the benchmark does not require preserving rhetorical role, provenance, or alternative interpretations. The project can extend the structural insight by making premise selection depend on explicit argument/proof forms and their obligations, not only formula topology. Primary source (https://papers.nips.cc/paper_files/paper/2017/hash/18d10dc6e666eab6de9215ae5b3d54df-Abstract.html)

3. **Graph Representations for Higher-Order Logic and Theorem Proving** (Paliwal et al., ICLR/AAAI publication record, 2019–2020). Paliwal et al. study several graph encodings for higher-order logic and feed them to graph neural networks for theorem-proving search in the HOList environment. The paper's important result is architectural: context-aware message passing over formula graphs substantially improves proof search, with the best reported representation approaching 50% proof closure on the validation setting and outperforming sequence-oriented baselines. The task regime combines premise relevance and tactic selection inside an interactive higher-order theorem prover, so the selected context is tightly coupled to proof-state action rather than being a static retrieval result. For this project, the direct overlap lies in treating subexpression context and structural relations as evidence for what belongs in a proof-relevant working set. The gap is that HOList supplies a formal grammar and a crisp proof oracle; it does not address classification of natural-language argument forms, source-grounded provenance, defeasible or abductive warrants, or context compression that must remain faithful to an interpreted argument. Primary source (https://openreview.net/pdf?id=iUGUc3HJ_3f)

4. **Learning to Prove Theorems via Interacting with Proof Assistants** (Bansal et al., ICML 2019). This work frames theorem proving as learned interaction with a proof assistant and introduces CoqGym, a large dataset and environment containing roughly 71,000 human-written proofs from 123 Coq projects. ASTactic generates tactics as abstract syntax trees and interacts with Coq's formal state; the reported system proves new theorems that were not previously solved by the cited automated methods. Although its headline contribution is tactic generation rather than a standalone retriever, the context regime is intrinsically proof-state-specific: each action is conditioned on the current formal goal, local hypotheses, and the assistant's evolving state. That makes it direct overlap for project work in which context selection should depend on the current proof obligation and the next reasoning action, not on document-level similarity. The missing piece is an explicit, inspectable policy for selecting and compressing heterogeneous premises, especially when the source material is natural language and the project must preserve provenance, uncertainty, and competing interpretations rather than only a formally valid state transition. Primary source (https://arxiv.org/abs/1905.09381)

5. **Formal Premise Selection With Language Models** (Tworkowski et al., AITP 2022). This paper gives the clearest early formulation of a retrieve-then-reason pipeline: a premise-selection model first chooses a small set, such as eight premises, from a database of more than 100,000 Isabelle lemmas, and a premise-guided language model then receives those premises with the current proof state to generate a proof step. The experiments report improved proof rates over a state-only language model and Sledgehammer, with further gains when the retrieval-guided and state-only systems are combined. The task

regime is formal interactive theorem proving, but the architectural decomposition maps directly onto the project's concern: selection determines the compact context in which downstream language-model reasoning occurs. Its limitation is that relevance is learned from formal proof-state/premise relations and is evaluated by proof success; it does not expose argument roles, explain why a premise satisfies an obligation, or preserve natural-language source evidence through compression. The project gap is therefore not retrieve-then-reason itself, but an explicit argument-form-to-proof-obligation policy that selects minimally sufficient, provenance-preserving context. Primary source (https://aitp-conference.org/2022/abstract/AITP_2022_paper_32.pdf)

6. **TextGraphs-16 Natural Language Premise Selection Task: Zero-Shot Premise Selection with Prompting Generative Language Models** (Kovriguina, Teucher, and Wardenga, 2022). This paper moves premise selection toward natural-language methods by prompting GPT-3 to rerank candidate premises without additional training, then comparing the result with SBERT semantic similarity ranking. Its central finding is cautionary: vanilla prompt-based reranking does not outperform SBERT, but combining the prompt ranking with semantic ranking yields better results. The task regime remains premise selection for automated theorem proving, yet the candidates and queries are closer to natural-language retrieval than the formal graph systems, making it a useful bridge between theorem proving and LLM context selection. The direct overlap is the attempt to use a generative language model as a selector rather than assuming that selection and reasoning are the same capability. The gap is that the setup ranks candidate premises but does not identify argument form, formal proof obligations, dependency preservation, or the adequacy of a compressed context for a later reasoning process. For this project, the result supports hybrid selectors and an evaluation that separates retrieval quality from downstream proof or argument validity. Primary source (https://aclanthology.org/2022.textgraphs-1.15/)

7. **Machine-Learned Premise Selection for Lean** (Piotrowski, Fernández Mir, and Ayers, 2023). This paper implements a lightweight online random-forest model directly inside Lean 4 to recommend relevant premises through the `suggest_premises` tactic. The model is trained from Mathlib-derived data and is designed around tight proof-assistant integration, ease of installation, and low latency, allowing recommendations during interactive proof construction rather than as an offline preprocessing step. The task regime is proof-assistant-native premise recommendation: the query is the current Lean theorem state, and the utility of a retrieved premise is judged by whether it helps the user or prover advance the proof. Its direct relevance to the project is operational as well as conceptual: context selection can be an interactive service constrained by the exact local state and available library. The limitation is that the model's features and labels are tied to Lean's formal environment and do not explain premise roles in natural-language arguments or preserve source-level evidence. The project gap is an interpretable, cross-domain selector that can expose why a context item was chosen and what proof obligation it is intended to discharge. Primary source (https://arxiv.org/abs/2304.00994)

8. **LeanDojo: Theorem Proving with Retrieval-Augmented Language Models** (Yang et al., NeurIPS 2023). LeanDojo supplies an open toolkit, dataset, interactive environment, and fine-grained annotations for Lean theorem proving, then uses those resources to build ReProver, an LLM-based prover augmented with retrieval from a large Mathlib library. The paper explicitly identifies premise selection as a bottleneck, uses program analysis to determine accessible premises and hard negatives, and evaluates on a benchmark of 98,734 theorems and proofs with splits requiring generalization to novel premises. This is especially close to the project's target regime because retrieval is not merely document augmentation: it is conditioned on a formal proof state and feeds a prover whose success can be checked by Lean. The project can borrow the discipline of accessibility constraints, hard-negative construction, proof-state conditioning, and premise provenance. The gap is that Lean provides a closed formal language and exact correctness oracle, whereas the project must handle natural-language sources, argument roles, uncertain interpretations, and partial or non-deductive support. A proof-relevant context layer for the project therefore needs LeanDojo-like dependency accounting without pretending that formal verification covers semantic or historical adequacy. Primary source (https://arxiv.org/abs/2306.15626)

9. **Magnushammer: A Transformer-Based Approach to Premise Selection** (Miku■a et al., ICLR 2024). Magnushammer uses contrastive transformer training and reranking to retrieve premises for interactive theorem proving, reducing reliance on symbolic feature engineering and domain-specific hand rules. On PISA and miniF2F, it reports substantially higher success rates than Sledgehammer, and pairing the retriever with a language-model prover raises proof success further, demonstrating that retrieval quality propagates into downstream proving. The task regime is direct proof-state-to-premise retrieval in Isabelle, with a released dataset of textual `(proof state, relevant premise)` pairs that makes the selection problem more explicitly language-model compatible. For the project, this is strong evidence that context selection should be treated as an independently measurable component rather than hidden inside generation, and that retriever improvements can matter even when the final reasoner is unchanged. Its limitation is that "relevance" is learned from formal proof use and benchmark success; the method does not preserve argumentative provenance, distinguish premise roles, or certify that a compressed natural-language context retains all required dependencies. The project's novelty boundary is a selector organized around

argument form and proof obligation rather than only proof-state similarity. Primary source (https://proceedings.iclr.cc/paper_files/paper/2024/file/ab5f5f22e3e09f4424592ffb06840ab0-Paper-Conference.pdf)

10. **Towards Large Language Models as Copilots for Theorem Proving in Lean** (Song, Yang, and Anandkumar, 2024/2025). Lean Copilot embeds LLM inference natively in Lean and packages several interactive capabilities, including proof-step suggestion, goal completion, and relevant-premise selection. The evaluation on the Mathematics in Lean textbook compares the system with rule-based automation and reports fewer manually entered proof steps and higher automated proof-step coverage for Lean Copilot. The task regime is collaborative theorem proving rather than fully autonomous retrieval: a human works inside the proof assistant, and the assistant supplies context- and action-sensitive suggestions that remain subject to Lean's correctness checker. This is directly relevant to the project's envisioned reasoning loop because it shows how premise selection can be exposed as an interactive assistant function alongside generation and search. Its gap is the same boundary as other Lean systems: formal validity does not guarantee that a natural-language source was represented faithfully, that an implicit warrant is acceptable, or that a selected item is the right explanation for a human reader. The project can extend the copilot model with explicit argument-form labels, provenance, uncertainty, and proof-obligation traces. Primary source (https://arxiv.org/abs/2404.12534)

11. **Graph Sequence Learning for Premise Selection** (Holden and Korovin, Journal of Symbolic Computation 2025; conference version 2023). This work treats premise selection as an "axiom captioning" problem: a graph neural network encodes the conjecture and a language model generates the sequence of axioms needed to build its proof. The reported approach combines structural graph information with sequential modeling and achieves a substantial improvement over the baseline, with the conference abstract reporting a 17.7% gain and the retained journal record reporting a 14% improvement in solved problems. The task regime is large-theory automated theorem proving, where a sequence or set of axioms must be recovered for a proof rather than merely finding a semantically similar paragraph. The direct overlap is important for the project because it models inter-axiom relationships and joint context composition, moving beyond independent candidate scoring. The gap is that the generated sequence is still a formal axiom artifact: it does not encode argument roles, source provenance, alternative warrants, or the faithfulness of a compressed explanation for an LLM. The project can build on the sequence perspective by selecting a structured proof-relevant context whose order and dependency edges are explicit and inspectable. Primary source (https://www.sciencedirect.com/science/article/pii/S0747717124000804)

12. **Tree-Based Premise Selection for Lean4** (Wang, Dong, and Wen, NeurIPS 2025). This paper argues that semantic embeddings alone miss the precise structural alignments needed for theorem application and proposes a Lean4 premise selector built around expression trees. Its pipeline combines common-subexpression-eliminated tree representations with Weisfeiler–Lehman similarity, tree-edit distance, constant-node Jaccard similarity, collapse-match similarity, cluster-based search, and adaptive fusion under structural compatibility constraints. The task regime is large-scale premise retrieval from Mathlib4 for interactive theorem proving, where the selected theorem must be structurally applicable to a formal expression rather than merely related in topic. The direct overlap with the project is the explicit use of form and structure as retrieval criteria; it is a strong analogue for making argument form a first-class selection signal. The limitation is that Lean expression trees are formal and unambiguous, while natural-language arguments require interpretation, uncertainty handling, and preservation of rhetorical or evidential roles. The project gap is therefore a higher-level structural index: argument-form labels and proof obligations should guide retrieval while retaining the underlying source spans and avoiding a false equivalence between syntactic compatibility and substantive validity. Primary source (https://openreview.net/pdf?id=omyNP89YW6)

13. **Assisting Mathematical Formalization with A Learning-based Premise Retriever** (arXiv:2501.13959, 2025). This work develops a premise-retrieval model trained on formalized corpora and Mathlib data, using a domain-specific tokenizer, contrastive learning, fine-grained query–premise similarity, and a reranking stage. The query is a Lean proof state and the candidates are formal premises, with the aim of helping mathematicians and theorem-proving systems locate useful library statements at lower computational cost; the authors also release an open-source search engine. The task regime is direct formal-state retrieval for mathematical formalization, especially addressing the difficulty that natural-language queries require substantial mathematical expertise while formal retrieval can suffer from sparse training data. For the project, it is relevant as a recent example of domain adaptation and fine-grained matching rather than generic embedding search. Its limitation is that the retrieval unit remains a formal theorem or premise, and the evaluation centers on retrieval and proof assistance rather than argument interpretation, source-grounded explanation, or dependency-preserving compression. The project can adapt its specialized-tokenizer and fine-grained matching ideas to an argument/context index, but must add form labels, provenance, and a policy for selecting context that is sufficient for a particular proof obligation. Primary source (https://arxiv.org/abs/2501.13959)

14. **Learning an Effective Premise Retrieval Model for Efficient Mathematical Formalization** (Tao, Liu, Wang, and Xu, AI4Math@ICML 2025). This paper proposes a two-stage formal-state retriever: context-free retrieval (CFR) embeds Lean proof states and premises for efficient candidate generation, while context-aware reranking (CAR) refines the list using richer interactions. A tokenizer trained on formal corpora and contrastive learning are used to handle the syntax and data scarcity of formal mathematics, and the system is integrated with an augmented theorem-proving pipeline; the authors report improved accuracy over baselines with lower computational load. The task regime is directly aligned with proof-relevant context selection because both the current state and the candidate premise structure influence ranking, and the retrieved list is intended for downstream proof generation. The project gap is not the two-stage architecture but its domain assumptions: the method selects formal Mathlib statements, does not classify argument or proof forms, and does not preserve natural-language source evidence or alternative interpretations. It provides a useful engineering precedent for separating broad retrieval from obligation-aware reranking, which the project can generalize into a form-conditioned selector whose final context remains traceable to original passages. Primary source (https://openreview.net/pdf/ec313b0b95dba96e7bb145dc4a3b376cd9f4e8e1.pdf)

15. **LeanSearch v2: Global Premise Retrieval for Lean 4 Theorem Proving** (Gao et al., 2026). LeanSearch v2 explicitly distinguishes global premise retrieval from both semantic search for individual declarations and tactic-local premise selection: it seeks the scattered set of lemmas jointly required by an entire theorem. Its standard mode uses a hierarchy-informed Mathlib corpus with embedding and reranking, while its reasoning mode iterates through sketch–retrieve–reflect cycles; the reported benchmark results show stronger recovery of ground-truth premise groups and higher downstream proof success than comparison retrievers. This is the closest current match to project-level proof-relevant context selection because the unit of relevance is a whole proof, not a single next action. The project can borrow its joint-set objective, iterative retrieval/reflection loop, and evaluation of retrieval-to-proof propagation. The remaining gap is substantial: LeanSearch operates over formal declarations with an exact prover, whereas the project must decide which natural-language spans, premises, warrants, and counterconsiderations belong in a compact context while preserving interpretive uncertainty and provenance. The project's narrow novelty claim can therefore focus on explicit argument-form-to-proof-obligation policies for global, not merely local, context selection. Primary source (https://arxiv.org/abs/2605.13137)

16. **RECOMP: Improving Retrieval-Augmented LMs with Compression and Selective Augmentation** (Xu, Shi, and Choi, ICLR 2024). RECOMP addresses a general retrieval-augmented language-model problem: retrieved documents can be long and costly, so an extractive compressor selects useful sentences and an abstractive compressor synthesizes concise summaries before the context reaches the LM. The system can also return an empty context when retrieved material is irrelevant or adds no value, and the paper reports very high compression ratios with limited performance loss on language modeling and open-domain question answering. This is adjacent rather than direct overlap: the task regime is generic RAG context compression, not theorem proving or proof-state premise selection. Its strongest contribution for the project is the selective-context objective—compression should be judged by downstream utility, and omission can be preferable to injecting irrelevant material. Its limitation is that sentence usefulness is not constrained by a proof dependency graph, argument role, or formal validity condition; an abstractive summary can omit a necessary qualification while remaining broadly relevant. The project must therefore augment RECOMP-like compression with dependency preservation, provenance links, obligation coverage, and faithfulness checks for argument forms. Primary source (https://arxiv.org/abs/2310.04408)

17. **LLMLingua: Compressing Prompts for Accelerated Inference of Large Language Models** (Jiang et al., EMNLP 2023). LLMLingua proposes coarse-to-fine prompt compression using a budget controller, token-level iterative compression, and distribution alignment, reporting up to 20× compression with little performance loss across reasoning and other datasets. The paper's task regime is long-prompt efficiency rather than premise selection: it optimizes the amount and placement of information an LLM receives, not which formal premises are necessary for a proof. It is therefore an adjacent baseline for the project's context-budget component, especially where large source collections must be reduced before reasoning. The direct lesson is that compression can be learned and evaluated as a separate systems problem, but its limitation is decisive for proof-relevant work: token-level salience does not guarantee preservation of theorem dependencies, argument roles, polarity, qualifications, or the exact evidence supporting a conclusion. The project should use LLMLingua-family methods only as generic compression comparators and add proof-obligation recall, dependency preservation, provenance retention, and downstream validity tests to determine whether a compressed context is actually safe to reason over. Primary source (https://aclanthology.org/2023.emnlp-main.825/)

18. **LongLLMLingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt Compression** (Jiang et al., ACL 2024). LongLLMLingua extends prompt compression to long-context settings by making compression question-aware, reorganizing documents to reduce position bias, dynamically controlling compression ratios, and recovering key subsequences after compression. Across long-context benchmarks, it reports improved performance, lower token cost,

and lower latency, including substantial gains on multi-document question answering with fewer tokens. The task regime is generic long-context retrieval and question answering, so this paper is adjacent rather than direct prior art for theorem-proving premise selection. It is useful to the project because it makes key-information density and position effects explicit: a context can contain the right evidence yet fail if compression or ordering hides it from the model. The gap is that “key information” is defined by question-answering utility, not by a formal proof dependency or argument structure; reordering and subsequence recovery do not establish that all required warrants and qualifiers survived. A project system can use LongLLMLingua as a budget and ordering baseline while evaluating the stronger requirement of proof-obligation coverage and provenance-preserving, argument-form-aware compression. Primary source (https://aclanthology.org/2024.acl-long.91.pdf)

Verification note

All 18 retained records were checked against the linked scholarly or first-party page/PDF where accessible. The NeurIPS, arXiv, ACL Anthology, AITP, ICLR proceedings, and search-indexed OpenReview records supplied title/abstract or paper text. Several OpenReview PDF URLs and the ScienceDirect page were protected by browser verification or returned HTTP 403 during direct opening; for those items, the linked records were cross-checked through indexed scholarly results and the paper metadata/abstract preserved in the agent file. The retained literature supports a narrow gap claim: premise selection and generic context compression are well established separately, but the explicit combination of argument/proof-form classification, proof-obligation tracking, minimal context selection, and provenance-preserving LLM reasoning is not represented by these 18 papers.

15 Llm Philosophy

Verification date: 2026-10-12 (US/Eastern). The 12 retained records in `agents/15_llm_philosophy.md` were rechecked against accessible arXiv, ACL Anthology, publisher, institutional-repository, or journal records. Each entry below gives the paper's contribution, evidence, limitations, and relevance to philosophical LLM generation or evaluation.

1. Schwitzgebel, Schwitzgebel & Strasser — "Creating a Large Language Model of a Philosopher" (Mind & Language; 2023/2024)

This paper fine-tunes GPT-3 on Daniel Dennett's writings to test whether an LLM can reproduce a particular philosopher's intellectual and stylistic profile. Its central evidence is a comparison involving Dennett, the model, and ten philosophical questions, followed by discrimination experiments in which experts and non-experts attempt to distinguish real-Dennett from model-generated answers; the study reports that model answers can be difficult to identify, especially when multiple answers are considered rather than a single cherry-picked example. The main limitations are the narrow single-philosopher setting, dependence on textual imitation, possible contamination or familiarity with Dennett's published positions, and the fact that indistinguishable prose does not establish understanding, authorship, consciousness, or valid argumentation. For philosophical LLM research, it is the clearest precedent for philosopher-style generation and sets a high bar: new work must add stronger grounding, argument-structure evaluation, longitudinal revision, source faithfulness, or independent expert assessment rather than claiming novelty from fluent philosopher-like text alone. Source: arXiv 2302.01339 (https://arxiv.org/abs/2302.01339)

2. Yang et al. — "InterIDEAS: Philosophical Intertextuality via LLMs" (EMNLP 2025)

InterIDEAS contributes a dataset and LLM-assisted workflow for studying how ideas circulate through authentic philosophical texts. The authors combine literary theories of intertextuality, bibliometric relations, GPT-4, retrieval-augmented generation, prompt engineering, and iterative expert review; the pipeline chunks and embeds texts, retrieves relevant passages, extracts reference attributes such as content type, intertextual function, and sentiment, and produces a dataset containing 2,236 reference–attitude pairs for downstream evaluation. Evidence includes ablation and human-in-the-loop validation of the extraction workflow, plus demonstrations of quantitative and qualitative analyses of philosophical relations and a sentiment-classification benchmark. Limitations include dependence on prompt design and retrieval quality, the interpretive instability of labels such as sentiment and intertextual function, residual hallucination risk, and the danger that structured extraction simplifies subtle philosophical reasoning. The paper is directly relevant to philosophical generation and evaluation because it establishes prior art for corpus-scale philosophical RAG and intertextual analysis; a new context engine must therefore distinguish itself through argument-form awareness, proof-relevant retrieval, faithfulness tests, or better independent evaluation. Source: ACL Anthology (https://aclanthology.org/2025.emnlp-main.1180/)

3. Millière & Buckner — "A Philosophical Introduction to Language Models—Part I: Continuity With Classic Debates" (2024)

Part I is a philosophical survey and technical primer that situates contemporary LLMs within debates about compositionality, language acquisition, semantic competence, grounding, world models, and cultural transmission. Its contribution is argumentative rather than a new benchmark: it uses the capabilities of systems such as GPT-4 to challenge strong assumptions about neural networks while preserving the "Blockhead" or shallow-pattern-matching hypothesis as a skeptical alternative and calling for empirical investigation of internal mechanisms. The evidence is a systematic engagement with classic philosophy of language, cognitive science, AI, and computational-linguistics findings, including the observation that statistical learners can acquire substantial syntactic structure from exposure. Its limitations are those of an opinionated survey: it does not independently settle the competence-versus-performance dispute, and its conclusions depend on contested philosophical background commitments and the state of empirical evidence reviewed. It is essential for philosophical LLM work because it prevents category errors: fluent output, benchmark success, and apparent semantic competence must be separated from grounding, internal representation, and genuine philosophical understanding. Source: arXiv 2401.03910 (https://arxiv.org/html/2401.03910v1)

4. Millière & Buckner — “A Philosophical Introduction to Language Models—Part II: The Way Forward” (2024)

Part II extends the survey toward empirical and methodological questions, covering mechanistic and causal-intervention approaches, multimodal and modular systems, agency, consciousness, cognitive modeling, secrecy, reproducibility, and benchmark validity. Its strongest contribution to evaluation is the warning that benchmark saturation may reflect gamification, contamination, or ill-defined constructs rather than a genuine increase in competence; the paper argues that interventions and interpretability can provide evidence beyond surface behavior. The evidence consists of philosophical analysis tied to emerging empirical methods and case discussions of how internal mechanisms, architecture, and training conditions bear on claims about cognition. Limitations include the survey’s dependence on rapidly changing technical literature and the fact that causal interpretability methods themselves remain incomplete and theory-laden. For philosophical text generation and evaluation, this paper requires explicit construct definitions, contamination controls, robustness to prompt variation, and—where claims concern reasoning or understanding—tests that go beyond answer accuracy or an LLM judge’s impression. Source: arXiv 2405.03207 (https://arxiv.org/html/2405.03207v1)

5. Oka — “Evaluating Large Language Models on the Frame and Symbol Grounding Problems: A Zero-shot Benchmark” (2025)

Oka operationalizes two classical AI-philosophy problems as zero-shot LLM tasks: a frame-problem scenario requiring relevant-information filtering and plan revision in a changing urban environment, and a symbol-grounding task involving a novel object called a “kluben,” whose properties must be integrated into coherent meanings and relations. The study evaluates 13 closed and open models across repeated trials with multiple criteria aimed at contextual reasoning, semantic coherence, information filtering, and consistency; the reported pattern is that leading closed models perform more stably, while many open models vary substantially and symbol-grounding scores often exceed frame-problem scores. Its limitations are substantial: the prompt reformulations may capture only selected behavioral analogues of the philosophical problems, custom criteria may not be fully independent of the author’s theoretical assumptions, and high scores do not demonstrate intrinsic grounding or commonsense cognition. The paper is a direct precedent for philosophical-problem benchmarking, so new work should improve construct validity through preregistered human rubrics, adversarial paraphrases, independent adjudication, contamination checks, and separate measurement of relevance filtering, semantic integration, and philosophical interpretation. Source: arXiv 2506.07896 (https://arxiv.org/html/2506.07896v1)

6. Li et al. — “I Think, Therefore I am: Benchmarking Awareness of Large Language Models Using AwareBench” (2024)

AwareBench introduces a taxonomy and dataset for evaluating purported LLM awareness across five dimensions: capability, mission, emotion, culture, and perspective. AwareEval combines binary, multiple-choice, and open-ended questions generated through a human–AI collaborative process, and the authors evaluate 13 LLMs; the reported evidence shows that most models struggle with capability and mission awareness while performing better on social-interaction-related dimensions. The benchmark’s contribution is to make self-model and social-awareness claims behaviorally testable and to connect them to alignment and safety. Its limitations are conceptual and methodological: “awareness” is defined behaviorally, the question sets partly rely on model-generated material, anthropomorphic interpretations remain possible, and answering self-referential questions cannot establish phenomenal consciousness, stable self-knowledge, or a philosophically substantive subject. It is directly relevant to AI-philosopher projects involving self-reflection or personhood: self-report-like philosophical answers should be treated as outputs to evaluate for calibration, consistency, and capability awareness—not as evidence that the model is conscious or genuinely self-aware. Source: arXiv 2401.17882 (https://arxiv.org/html/2401.17882v2)

7. Smart, Clowes & Clark — “ChatGPT, extended: large language models and the extended mind” (Synthese, 2025)

This paper develops “Digital Andy,” a system that uses retrieval-augmented generation over Andy Clark’s writings to answer questions about the extended mind, and then analyzes the system through the philosophical framework of active externalism and extended cognition. Its contribution is twofold: it presents RAG as an alternative to fine-tuning for incorporating philosophical knowledge and uses the engineering arrangement to ask whether externally retrieved resources can count as part of an extended artificial cognitive system. The evidence is the system design and philosophical comparison between internal model memory, external retrieval, and the conditions under which information becomes functionally available to the agent. Limitations include the gap between an interpretive philosophical proposal and a demonstrated cognitive status, the dependence of the result on how “accessibility,” memory, and agency are operationalized, and the absence of evidence that RAG produces human-like understanding. The paper matters for philosophical context engines because it directly precedes claims about external philosophical memory: retrieval utility and extended-mind status must be evaluated separately, with explicit criteria for causal dependence, accessibility, reliability, and user/system agency. Source: institutional repository (https://eprints.soton.ac.uk/502110)

8. Galatolo, Rappuoli, Winkle & Beloucif — “Beyond Ethical Alignment: Evaluating LLMs as Artificial Moral Assistants” (2025)

This paper formalizes what an Artificial Moral Assistant should do and introduces AMAeval, a benchmark designed to test explicit moral reasoning rather than only agreement with a target ethical verdict. Drawing on philosophical literature, the framework distinguishes deductive and abductive moral reasoning and evaluates open LLMs both on generating and assessing reasoning chains; the reported results show substantial variation across models, persistent weakness in abductive reasoning, and a separation between evaluation and generation abilities. The contribution is therefore a bridge from normative theory to operational evaluation, with moral reasoning treated as structured deliberation involving conflicting values rather than mere alignment. Limitations include dependence on the selected AMA theory, the difficulty of validating generated reasoning chains independently, possible rubric and dataset biases, and the risk that verbalized chains rationalize rather than reveal the causal basis of decisions. It is a key baseline for philosophical moral assistants: any new system should compare against AMAeval and report premise-level validity, abductive adequacy, uncertainty, plural-tradition robustness, and independent human or expert judgments—not just verdict agreement. Source: arXiv 2508.12754 (https://arxiv.org/html/2508.12754v1)

9. Asghari & Bia■y — “Triangles, Justice, and AI: Testing Large Language Models’ Comprehension of Political Ideologies” (Philosophy & Digitality, 2025)

This study tests whether LLMs can apply Rawlsian and Nozickian theories of justice to a novel thought experiment involving sentient geometric shapes whose number of angles affects political status. Seven models are evaluated at three levels inspired by Bloom’s taxonomy—recall, application, and reflection—so the task distinguishes remembering theory from transferring it to an unfamiliar scenario and critically reflecting on its implications. The evidence shows large variation across models: one system gives sophisticated theory-consistent responses while others produce confused or generic answers, suggesting that some models can map ideological concepts onto novel cases without implying human-like understanding. Limitations include a small theory set, a single stylized scenario, possible sensitivity to prompt wording and model familiarity with Rawls and Nozick, and scoring that may conflate doctrinal fluency with philosophical comprehension. The paper is a direct precedent for evaluating philosophical-theory transfer; future benchmarks should add paraphrased and adversarial scenarios, opposing traditions, counterexamples, cross-theory consistency, premise-level argument analysis, and blinded expert scoring. Source: Philosophy & Digitality (https://journals.ub.uni-koeln.de/index.php/phidi/article/view/11663)

10. Chen et al. — “Leveraging large language models to assist philosophical counseling: prospective techniques, value, and challenges” (Humanities and Social Sciences Communications, 2025)

This article surveys prospective uses of prompt engineering, fine-tuning, and RAG in philosophical counseling, including counselor recommendation, pre-counseling assessment, in-session guidance, session summarization, outcome evaluation, and cultural adaptation. Its contribution is an application-oriented account of how LLMs could expand access and reduce administrative burden while insisting that systems should augment rather than replace professional counselors. The evidence is primarily conceptual and illustrative: the article synthesizes counseling requirements, proposes technical workflows, and discusses generated summaries and recommendation functions alongside risks involving hallucination, privacy, explainability, trust, and the lack of genuine understanding or empathy. Its limitations are the absence of controlled counseling-outcome experiments, limited evidence that proposed methods improve philosophical rather than generic conversational support, and the high-stakes difficulty of mental-health triage and personalized advice. It is relevant to philosophical dialogue generation because it shows that useful assistance, safe deployment, and legitimate philosophical agency are distinct claims; a new assistant needs bounded tasks, human oversight, privacy protections, factual and interpretive faithfulness tests, and failure analysis rather than a generic chatbot demonstration. Source: Nature (https://www.nature.com/articles/s41599-025-04657-7)

11. Millière & Buckner — “The Philosophy of Language Models” (Philosophy Compass, 2026)

This broad survey synthesizes philosophical disagreements about LLMs across syntactic competence, compositionality, meaning, representation, attitudes, reasoning, agency, and consciousness. Its central contribution is methodological: it argues that progress requires both explicit philosophical commitments and close engagement with empirical evidence, including interpretability and causal evidence where behavioral performance underdetermines the underlying mechanism. The article’s evidence is a structured review of classic arguments and contemporary empirical work, framed around the conflict between human-like-capacity interpretations and shallow pattern-matching interpretations. As a survey, it does not provide a new experimental adjudication, and its conclusions remain sensitive to unresolved disputes over what counts as meaning, reasoning, agency, or consciousness. For philosophical LLM generation and evaluation, it is the most current high-level warning against unqualified claims: research should maintain separate evidence tracks for output quality, doctrinal or argumentative competence, internal representation, causal mechanism, and philosophical status, and should state clearly which track its results support. Source: Philosophy Compass/Wiley (https://compass.onlinelibrary.wiley.com/doi/10.1111/phc3.70095)

12. Abdali et al. — “Self-reflecting Large Language Models: A Hegelian Dialectical Approach” (2025; arXiv version updated 2026)

This paper builds a self-reflection framework around a Hegelian dialectical pattern: an initial proposition is challenged by an opposition and reconciled through a more comprehensive synthesis. It applies the iterative procedure to scientific-idea generation and error correction, compares dynamic temperature annealing with constant temperature, and introduces Multi-Agent Majority Voting (MAMV) so multiple LLMs assess validity and novelty when domain experts are unavailable; the paper reports gains over baselines on GSM8K, GSM-hard, GSM-Symbolic, and MMLU Pro, with qualitative results for open-ended ideation. Its limitations are especially important for philosophical evaluation: the Hegelian terminology may function mainly as prompt scaffolding rather than an operational account of dialectic, MAMV is circular when LLMs judge LLM-generated ideas, and model agreement is a weak proxy for validity, novelty, or philosophical insight. The paper is direct precedent for dialectical prompting, multi-agent critique, and philosophical idea generation; a stronger system should use independently authored rubrics, expert adjudication on a frozen subset, source and argument verification, and novelty tests that do not rely solely on LLM consensus. Source: arXiv 2501.14917 (https://arxiv.org/abs/2501.14917)

Cross-paper implication

Across the 12 papers, the literature already covers philosopher-style imitation, philosophical-text RAG and intertextuality, classical AI-philosophy benchmarks, awareness probes, extended-mind interpretations, moral reasoning, political-theory transfer, philosophical counseling, and dialectical self-critique. The defensible remaining gap is narrower: an argument-form-aware philosophical context engine that identifies premises, conclusions, inference type, source tradition, and uncertainty; retrieves or compresses proof-relevant passages; and evaluates generated outputs with independent philosophical rubrics, contamination controls, faithfulness measures, and human/expert validation rather than style, verdict agreement, or circular LLM judging alone.

16 Historical Logics

Paragraph summaries

**1. Aristotle,* Prior Analytics*. Aristotle's* Prior Analytics*, especially Books I–II, is the primary ancient source for categorical syllogistic as a theory of terms, premises, figures, moods, conversion, reduction, and demonstration. Its historical contribution is not merely a list of valid patterns: it presents a vocabulary and proof practice in which some syllogisms are "perfect," others are derived by reduction, and demonstration is tied to necessary consequence. For this project, it should be treated as a source layer from which argument templates and deduction traces are extracted, with Greek wording, translation, quantifier scope, existential assumptions, modality, and textual uncertainty preserved. A formalizer must distinguish Aristotle's own procedures from later predicate-logical recodings; success in a modern encoding establishes a property of that reconstruction, not proof that Aristotle endorsed the modern system. Source: Perseus Digital Library (https://www.perseus.tufts.edu/hopper/text?doc=Perseus:text:1999.01.0053).

2. John Corcoran, "Aristotle's Prior Analytics and the Theory of Syllogism" (1974). Corcoran's contribution is to present Aristotelian syllogistic as an internally structured deductive system rather than as an unfinished anticipation of modern predicate logic. The key historical/formal distinction is between a deductional reconstruction—where rules, derivations, and proof structure matter—and a semantic or model-theoretic reconstruction that may redescribe the system from outside. This supports a project architecture in which historical calculi retain their own primitives, derivation steps, and proof-theoretic behavior while remaining comparable to modern formalisms through explicit translations. Corcoran's Aristotelian focus is a strength for the ancient module but also a boundary: medieval semantic theories, Arabic relational developments, Indian inference, and Mohist disputation require separate schemas rather than being treated as extensions of one syllogistic core. Source route: WorldCat record for *Ancient Logic and Its Modern Interpretations* (https://search.worldcat.org/title/ancient-logic-and-its-modern-interpretations/oclc/1020378).

3. Klaus Glashoff, "Aristotelian Syntax from a Computational–Combinatorial Point of View" (2005). Glashoff provides the most direct computational precedent in this set. He formalizes Aristotle's logical system through rule-based reduction relations on a defined basic set, applies structural tools such as the Newman lemma, reproduces Aristotle's derivational method, and analyzes the resulting rule system with a computer program. Historically, the paper reinforces the view that Aristotelian syllogistic is a formal system in its own right, not merely a fragment awaiting translation into predicate logic; formally, it demonstrates how typed terms, reduction rules, normalization behavior, and generated derivation traces can be made executable. The project can directly adapt this pattern for a provenance-preserving Aristotelian module, while keeping syntax, semantics, and later interpretive additions separate. Source: Oxford Academic (https://academic.oup.com/logcom/article-abstract/15/6/949/1086849) and author PDF (https://glashoff.eu/Texte/reduction2.pdf).

4. Catarina Dutilh Novaes,* Formalizing Medieval Logical Theories: Suppositio, Consequentiae and Obligationes *(2007). Dutilh Novaes formalizes three major medieval areas—supposition theory, Buridan's theory of consequence, and obligationes—and then examines what formalization itself does to historical material. The historical contribution is to show that medieval logic includes semantic-pragmatic and dialogical practices that cannot be reduced to a small set of Aristotelian syllogistic moods; the formal contribution is an "algorithmic hermeneutics" that makes interpretive choices explicit rather than pretending that formal notation is neutral transcription. For this project, the book supplies a methodology and several candidate executable schemas: context-sensitive term interpretation, consequence relations, stipulated cases, commitment and retraction rules, and logical games. Its Latin-medieval focus means that Arabic, Indian, and Mohist materials should be formalized with their own technical vocabularies and ontological commitments. Source: Springer (https://link.springer.com/book/10.1007/978-1-4020-5853-0).

5. Alexander Broadie,* Introduction to Medieval Logic*, 2nd ed. (1993). Broadie's synthesis maps the breadth of late medieval logical analysis through figures including Walter Burley, William Ockham, John Buridan, Albert of Saxony, and Paul of Venice. It covers truth, categorical and molecular propositions, validity conditions, supposition—including intentional contexts—and medieval syllogistics, thereby documenting how far medieval logicians developed beyond the Aristotelian core. Its historical value for the project is curatorial: it identifies families of phenomena that a corpus must not collapse into "syllogism," while its formal relevance lies in the distinctions that need annotation before executable rules can be written. Because it is a scholarly overview rather than a machine-readable specification, implementation would still require edition-level alignment, terminology normalization, competing analyses, and explicit treatment of context-sensitive reference. Source: Oxford Academic (https://academic.oup.com/book/6319).

6. Catarina Dutilh Novaes and Stephen Read (eds.), The Cambridge Companion to Medieval Logic *(2016).** This companion presents medieval logic as a connected field spanning ancient inheritance, Arabic logic before and after Avicenna, Latin traditions, and thematic developments in supposition, propositions, sophisms, syllogism, consequence, modality, and obligationes. Its historical contribution is corrective: it resists narrow national or chronological boundaries and foregrounds transmission, adaptation, and internal innovation across Greek, Syriac/Arabic, and Latin contexts. For the project, it motivates a provenance graph in which a formal rule can be linked to a source passage, commentator, translation lineage, and later adaptation, while structural similarity is kept distinct from documented historical transmission. The volume does not itself establish a common annotation language, so the missing computational layer is a shared alignment and uncertainty model rather than a universal medieval calculus. Source: Cambridge University Press (https://www.cambridge.org/core/books/abs/cambridge-companion-to-medieval-logic/frontmatter/90B806867E65585809C37DBE09390BC7).

7. Paolo Mancosu and Massimo Mugnai,* Syllogistic Logic and Mathematical Proof *(2023). Mancosu and Mugnai trace the long debate over whether syllogistic can capture mathematical proof, from Aristotle through medieval authors and later figures such as Kant and De Morgan. The relevant historical/formal contribution is their treatment of medieval extensions of syllogistic, including oblique terms, relational sentences, and expository syllogism, within a broader account of the relationship between syllogistic reasoning and mathematical practice. For this project, the book supplies concrete rule families that should be represented separately from categorical syllogism: relational and oblique propositions, expository reasoning, and historically situated claims about what counts as proof. It also warns against evaluating historical systems solely by modern adequacy criteria; the computational task is to model the resources and limits of each reconstruction, not to declare one tradition universally sufficient. Source: Oxford Academic (https://academic.oup.com/book/46132).

8. Khaled El-Rouayheb,* Relational Syllogisms and the History of Arabic Logic, 900–1900 *(2010). El-Rouayheb reconstructs a millennium of Arabic logical work on relational and “unfamiliar” syllogisms, drawing on manuscripts and rare books across regions and periods from the tenth century through the nineteenth. Its historical contribution is substantial: it challenges the idea that Islamic logic merely preserved Greek logic or stagnated after the medieval period, showing sustained debate and original development, including Ottoman analyses of relational inference as a distinct kind of syllogism. For the project, this is the central Arabic module and a strong reason to represent relation-sensitive inference as historically primary rather than as an exception to Aristotelian logic. A computational implementation should preserve Arabic technical terms, manuscript and translation provenance, chronology, and the distinction between inherited Aristotelian forms and later developments; the book itself is historical analysis, not an executable calculus. Source: RMIT bibliographic record (https://researchrepository.rmit.edu.au/discovery/fulldisplay?context=L&docid=alma9921478072501341&vid=61RMIT_INST%3ARMITU).

**9. Wilfrid Hodges, “Avicenna’s Logics with Sentence-Type Terms: 1. Wholly* mutta■il *Syllogisms” (published online 2024; issue 2025). Hodges analyzes an early Avicennian work, the* Short Epitome in Logic*, in which hypothetical logic is built from Aristotelian-looking syllogistic patterns but uses sentences rather than nouns as terms. The historical contribution is to identify a neglected change in the term domain and to show how Avicenna’s system anticipates ideas later associated with adverbs of quantification and Boole’s secondary propositions. Formally, it is a valuable test of typed historical representation: identical surface labels such as “syllogism” can conceal different syntactic categories, quantificational behavior, and semantics. The project should therefore encode the term ontology and sentence-type structure explicitly, rather than importing a noun-term Aristotelian schema; the article covers one Avicennian work and one family of syllogisms, so it cannot stand for Islamic logic as a whole. Source: Oxford Academic (https://academic.oup.com/logcom/article-abstract/35/7/exae032/7721282).

**10. Bimal Krishna Matilal,* The Character of Logic in India *(1998). Matilal presents Indian logic as a diverse philosophical and linguistic tradition centered not only on formal deduction but also on debate, proof, inference, epistemic sources, defeaters, and the relation between language and reality. The Ny■ya five-member inference—thesis, reason, example, application, and conclusion—is tied to* vy■pti* (pervasion or invariable concomitance), while the book also treats Buddhist, Jain, and later Navya-Ny■ya developments. Its historical contribution is therefore anti-reductive: Indian logical practices should not be measured only by whether they instantiate Aristotelian or Fregean forms. For this project, Matilal supports a Ny■ya argument record that stores warrant, example, application, epistemic status, and defeater conditions alongside the conclusion, while retaining Sanskrit technical terms and distinguishing modern analytic paraphrase from historical doctrine. Source: Google Books (https://books.google.com/books/about/The_Character_of_Logic_in_India.html?hl=en&id=NzZRu12ngLAC&output=html_text).

**11. Jonardon Ganeri (ed.),* Indian Logic: A Reader *(2001). Ganeri’s reader assembles landmark primary and secondary materials from the modern study of Indian logic, including work on Ny■ya, the Indian syllogism, research methodology, the concept of* pak■a*, Navya-Ny■ya inference, and the history of interpretation. Its historical contribution is historiographical as well as substantive: it exposes how modern scholarship has framed Indian reasoning, including the danger of treating “Indian

logic” as a single homogeneous system or judging it only by comparison with Western logic. For the project, the reader is useful as a corpus-construction guide: passages need tradition labels, source dates, original-language text, translation alternatives, and metadata about scholarly disagreement. It does not provide a canonical formal calculus, so computational work should preserve competing readings and distinguish Ny■ya, Buddhist, Jain, and Navya-Ny■ya materials rather than normalizing them into one Indian schema. Source: Routledge (https://www.routledge.com/Indian-Logic-A-Reader/Ganeri/p/book/9780203037119).

12. Mrityunjoy Panday and Sudipta Ghosh, “Cubical Type Theoretic Navya-Ny■ya” (2026). Panday and Ghosh propose a formalization of Navya-Ny■ya’s technical language in CCHM De Morgan cubical type theory. The paper’s formal contribution is unusually ambitious: it encodes relation, delimitor (*avacchedaka*), absence (*abh■va*), pervasion (*vy■pti*), identity, higher relations, pary■pti, and qualifier–qualificand structure; it also reports internal signature theorems, metatheoretic results, worked encodings of *Tattvacint■ma■i* passages, and a Cubical Agda implementation sketch. Historically, its importance is methodological because it treats dependent delimitation, typed absence, non-extensional identity, and unbounded relational depth as load-bearing features rather than noise to be flattened into first-order predicates. The project can use it as the clearest precedent for a typed relational intermediate representation, while treating its claims as frontier preprint evidence requiring independent validation against Sanskrit commentaries and broader Ny■ya history. Source: arXiv (https://arxiv.org/abs/2605.12548).

13. A. C. Graham,* Later Mohist Logic, Ethics and Science *(1978). Graham’s translation and analysis reconstructs the Later Mohist Canons and related writings as a rare surviving body of ancient Chinese work on logic, ethics, optics, mechanics, geometry, names, descriptions, and methods of disputation. Its historical contribution is to make a technically rich but textually difficult tradition available for systematic study while emphasizing that Mohist reasoning is not simply an uncompleted Aristotelian syllogistic. For this project, the book suggests a different computational vocabulary: lexical distinctions, definitions, classification, analogy, counterexample, disputational moves, and empirical or technical reasoning should be modeled as a family of operations linked to original Chinese text, translation choices, emendations, and commentary. The major limitation is philological instability: terse Canons and damaged transmission make Graham’s reconstruction an interpretive layer that must not be silently encoded as uncontested primary meaning. Source route: WorldCat (https://search.worldcat.org/title/later-mohist-logic-ethics-and-science/oclc/4495009) and bibliographic/contents record (https://obnb.uk/p6734657-later-mohist-logic-ethics-and-science).

14. Catarina Dutilh Novaes,* Formal Languages in Logic: A Philosophical and Cognitive Analysis *(2012). Dutilh Novaes broadens the notion of formal language beyond mathematical symbol systems and analyzes the historical, philosophical, and cognitive roles that formalisms play in reasoning. Its contribution to the history of logic is a methodological guardrail: formalization is an intervention that makes some distinctions explicit, changes cognitive access to inferential structure, and can suppress or distort other distinctions; therefore, historical practices need not use modern formal languages to be logically sophisticated. For this project, the book supports evaluating an encoding by what it preserves, exposes, and loses, and by documenting the intended level of abstraction rather than claiming tradition-neutrality. It does not prescribe a data model, theorem prover, or cross-tradition benchmark, so its role is to govern the design of the layered representation: source text, translation, tradition-specific schema, executable encoding, proof trace, and disagreement record. Source: Cambridge University Press (https://www.cambridge.org/core/books/formal-languages-in-logic/7D1DD805F7B3C70A0C32F2F6A66BE3DD).

Synthesis for the project

Together, these sources support a provenance-preserving argument record rather than a universal historical logic. Each record should retain source passage and edition, language and translation, tradition and date range, premises and claims, inferential operation, semantic commitments, defeaters or exceptions, confidence, formal encoding, proof/check trace, and unresolved scholarly alternatives. The strongest shared computational object is a historically typed transition from linguistic material to conclusion; comparison should occur at multiple levels—surface structure, rule behavior, semantic commitments, and proof-theoretic properties—without treating successful modern verification as evidence that the historical author accepted the modern encoding.

17 Cross Cultural Logic

The retained literature is best read as a body of work on cross-cultural reasoning, commonsense, values, and multilingual evaluation rather than as evidence for discrete “non-Western logics.” The summaries below were checked against the canonical arXiv, ACL Anthology, DOI, or publisher records listed for each paper.

1. **LogiQA: A Challenge Dataset for Machine Reading Comprehension with Logical Reasoning — Liu et al. (IJCAI 2020).** LogiQA contributes 8,678 expert-written multiple-choice reading-comprehension instances designed to test several deductive reasoning types, and its experiments show that contemporary neural models remained far below the human ceiling. Its central evidence is benchmark performance on questions derived from Chinese civil-service examination materials, making it a concrete example of institutionally situated reasoning data rather than a claim that Chinese people use a distinct logic. The main limitation is construct validity: exam genre, translation, educational conventions, and domain familiarity are entangled with logical ability, and the dataset should not be generalized to “Chinese reasoning” as a whole. Culturally, it demonstrates why source context and authorship matter; formally, it is the strongest item in this set for testing explicit deductive reasoning, but its cultural relevance is contextual rather than evidence of an alternative formal system. (arXiv (https://arxiv.org/abs/2007.08124); IJCAI record/DOI (https://doi.org/10.5555/3491440.3491941))

2. **LogiQA 2.0—An Improved Dataset for Logical Reasoning in Natural Language Understanding — Liu et al. (IEEE/ACM TASLP 2023).** LogiQA 2.0 contributes a larger, reannotated version of the original benchmark, combining logical reasoning in machine reading comprehension and natural-language inference, with refined texts and professional manual translation. Reported dataset changes include increased scale, improved annotation quality, and removal or reduction of items whose distinctive Chinese idioms make cross-lingual comparison difficult; these design choices make it useful for separating reasoning from translation and culture-specific wording. Its limitation is that the resulting English-facing benchmark may remove precisely the contextual features that made the original material culturally situated, while the exam-derived task remains a narrow slice of reasoning. The paper is therefore culturally relevant as a study of how Chinese-source materials are transported across languages, and formally relevant as a more controlled deductive benchmark—not as evidence for a uniquely Chinese logic. (DOI (https://doi.org/10.1109/TASLP.2023.3293046); dataset record (https://huggingface.co/datasets/datatune/LogiQA2.0))

3. **Common Sense Beyond English: Evaluating and Improving Multilingual Language Models for Commonsense Reasoning — Lin, Lee, Qiao, and Ren (ACL-IJCNLP 2021).** This paper contributes the 561,000-sentence Mickey corpus in 11 languages, the language-general Mickey Probe, and cross-lingual versions of CSQA and CODAH in 16 languages, together with multilingual contrastive pretraining (MCP) for improving performance. Its evidence comes from zero-shot probing and cross-lingual commonsense experiments showing substantial discrepancies across languages and measurable gains from MCP, including a reported improvement on X-CSQA. The principal limitation is that the translated X-CSQA/X-CODAH tasks inherit English benchmark assumptions and translation artifacts, so language coverage is not the same as culturally native task construction; the paper also primarily tests multilingual transfer rather than deep cultural variation. Its cultural relevance is methodological—showing that commonsense performance beyond English cannot be inferred from English scores—while its formal relevance is indirect because the tasks concern plausibility and everyday inference rather than validity in a formal logic. (ACL Anthology (https://aclanthology.org/2021.acl-long.102/))

4. **GeoMLAMA: Geo-Diverse Commonsense Probing on Multilingual Pre-Trained Language Models — Yin et al. (EMNLP 2022).** GeoMLAMA contributes 3,125 prompts in English, Chinese, Hindi, Persian, and Swahili covering geographically differentiated concepts associated with American, Chinese, Indian, Iranian, and Kenyan contexts. Experiments probing 11 multilingual pretrained models find that larger models do not necessarily encode geo-diverse concepts better, that models are not simply Western-biased in every probe, and that a country’s native language is not reliably the best language for eliciting its cultural knowledge. The limitation is that cloze-style relational probes measure stored associations and accessibility, not full reasoning or lived cultural competence; country labels also simplify within-country diversity. Culturally, the paper directly operationalizes geographically variable commonsense; formally, it shows why language, knowledge retrieval, and reasoning must be treated as separate axes rather than treating a wrong culturally situated answer as a logical error. (ACL Anthology (https://aclanthology.org/2022.emnlp-main.132/); arXiv (https://arxiv.org/abs/2205.12247))

5. **Cultural Bias and Cultural Alignment of Large Language Models — Tao, Viberg, Baker, and Kizilcec (PNAS Nexus 2024).** The paper contributes a disaggregated comparison of outputs from five GPT-family models against nationally representative survey data and places model value profiles on the Inglehart–Welzel cultural map. Its evidence indicates that default outputs exhibit culturally uneven value alignment, with the publisher record describing a tendency toward self-expression values associated with English-speaking and Protestant-European countries, while cultural prompting can

shift outputs. Limitations include dependence on English prompts and phrasing, the use of survey-derived value constructs as a proxy for culture, and the inability of observational output comparisons to establish why a model exhibits a given bias. The cultural relevance is high for value-conditioned judgment and alignment; the formal relevance is low because the study does not test deductive validity, but it warns that apparently neutral reasoning or recommendations may encode culturally uneven priors. (PNAS Nexus (https://doi.org/10.1093/pnasnexus/pgae346); arXiv (https://arxiv.org/abs/2311.14096))

6. **Understanding the Capabilities and Limitations of Large Language Models for Cultural Commonsense — Shen et al. (NAACL 2024).** This study contributes a systematic evaluation of LLMs across general and culture-specific commonsense benchmarks, examining multiple countries, languages, and prompting conditions. Its evidence shows significant performance differences across cultures, that cultural context can alter performance even on general commonsense, and that query language affects culture-related results; the ACL record confirms these as the paper's central findings. The limitation is that benchmark and prompt design still mediate what counts as cultural knowledge, and translated or country-indexed items cannot fully represent intra-cultural variation or pragmatic competence. Culturally, it supplies direct evidence that models do not possess a single uniform commonsense; formally, it reinforces the need to distinguish knowledge access and contextual interpretation from logical validity. (ACL Anthology (https://aclanthology.org/2024.naacl-long.316/); arXiv (https://arxiv.org/abs/2405.04655))

7. **CDEval: A Benchmark for Measuring the Cultural Dimensions of Large Language Models — Wang et al. (C3NLP Workshop 2024).** CDEval contributes a benchmark built with GPT-4-assisted generation followed by human verification, covering six cultural dimensions across seven domains to examine consistency and variation in LLM outputs. Its evidence is comparative benchmark evaluation showing that mainstream models exhibit both stable patterns and meaningful differences across cultural dimensions and domains. The main limitation is reliance on automated item generation and model-based cultural framing, which can reproduce the assumptions of the generation model or reduce complex cultures to predefined dimensions; the benchmark measures cultural behavior more than reasoning process. Its cultural relevance lies in making plural cultural values an explicit alignment and evaluation target; its formal relevance is indirect, useful for testing whether explanations or decisions vary with cultural framing but not for judging logical entailment. (ACL Anthology PDF (https://aclanthology.org/2024.c3nlp-1.1.pdf); arXiv (https://arxiv.org/abs/2311.16421))

8. **WorldValuesBench: A Large-Scale Benchmark Dataset for Multi-Cultural Value Awareness of Language Models — Zhao et al. (LREC-COLING 2024).** WorldValuesBench contributes more than 20 million demographic-context/value-question examples derived from 94,728 World Values Survey participants and frames evaluation as predicting human response distributions from demographic context. Its evidence shows that even strong open and closed models find the task difficult under Wasserstein-distance evaluation, with large differences in the fraction of questions on which models approximate human distributions closely. Limitations follow from the survey itself: demographic attributes are imperfect proxies for culture, response distributions can conceal minority disagreement, and predicting aggregate values is not the same as understanding reasons or applying them in context. Culturally, it offers unusually broad empirical coverage for value-conditioned behavior; formally, it is not a logic benchmark, but it supports a separate taxonomy axis for value-sensitive judgment rather than conflating value disagreement with invalid inference. (ACL PDF (https://aclanthology.org/anthology-files/anthology-files/pdf/lrec/2024.lrec-main.1539.pdf); arXiv (https://arxiv.org/abs/2404.16308))

9. **SeaEval for Multilingual Foundation Models: From Cross-Lingual Alignment to Cultural Reasoning — Wang et al. (NAACL 2024).** SeaEval contributes a broad benchmark spanning standard NLP, reasoning, multilingual semantics, and cultural practices, nuances, and values. Its reported evidence is especially relevant to robustness: models vary under paraphrased instructions, show exposure and positional/majority-label biases, give inconsistent answers to semantically equivalent multilingual questions, and lack balanced multilingual capability. The limitation is breadth over depth: the benchmark mixes language proficiency, factual knowledge, reasoning, and cultural comprehension, so an observed gap does not by itself identify the causal failure mode; cultural categories also remain benchmark-defined. Culturally, it establishes that multilingual equivalence does not guarantee culturally or semantically stable behavior; formally, it supports evaluating reasoning invariance under language changes while keeping the task's inferential content fixed. (ACL Anthology (https://aclanthology.org/2024.naacl-long.22/))

10. **LLM-GLOBE: A Benchmark Evaluating the Cultural Values Embedded in LLM Output — Karinshak et al. (arXiv 2024).** LLM-GLOBE contributes a benchmark grounded in the empirically validated GLOBE cultural-psychology framework and an "LLMs-as-a-Jury" pipeline for large-scale analysis of open-ended outputs, including comparisons between Chinese and US LLMs. Its evidence suggests that open-generation tasks reveal interpretable similarities and differences in the cultural value systems embedded in model outputs and may be more informative than narrow fixed-answer tests. Limitations include

the validity of using LLM judges to score cultural values, the risk that GLOBE dimensions impose a particular social-science ontology, and the difficulty of separating training-data effects from model or deployment effects. The cultural relevance is direct for latent value structure and cross-cultural alignment; formal relevance is limited because open-ended value expression is not proof-theoretic reasoning, though it matters for culturally situated explanations and decisions. (arXiv (https://arxiv.org/abs/2411.06032))

11. **Global MMLU: Understanding and Addressing Cultural and Linguistic Biases in Multilingual Evaluation — Singh et al. (ACL 2025).** Global MMLU contributes an expanded 42-language evaluation set with professional and community translation, human verification, and separate culturally sensitive and culturally agnostic subsets. Its evidence shows that 28% of original MMLU questions require culturally sensitive knowledge, that 84.9% of geography questions focus on North American or European regions, that translation introduces artifacts, and that model rankings change when culturally sensitive subsets are isolated. The limitation is that the framework still begins from MMLU's inherited task and domain distribution, while cultural-sensitivity labels are themselves annotation judgments and may not capture all target-language effects. Culturally, it is strong evidence that translation alone cannot globalize a benchmark; formally, it helps disentangle reasoning performance from culturally specific premises and language artifacts. (arXiv (https://arxiv.org/abs/2412.03304); ACL record (https://aclanthology.org/2025.acl-long.919/))

12. **Can LLMs Grasp Implicit Cultural Values? Benchmarking LLMs' Metacognitive Cultural Intelligence with CQ-Bench — Liu et al. (arXiv/OpenReview 2025).** CQ-Bench contributes conversational story-based tests of implicit cultural-value inference, built from World Values Survey and GlobalOpinions values and organized into attitude detection, value selection, and value extraction tasks. Its evidence shows a capability split: some reasoning-focused models approach human-level performance on value selection, yet performance remains weaker on nuanced attitude detection and open-ended value extraction; the paper also reports gains from fine-tuning smaller models on a small culturally rich sample. Limitations include synthetic multi-character conversations, GPT-4o-assisted validation, and the possibility that agreement with a model judge reflects annotation pipeline consistency rather than genuine cultural understanding. Culturally, it advances from explicit factual knowledge to pragmatic and socially embedded interpretation; formally, it concerns defeasible context-sensitive inference rather than deductive validity and should be classified accordingly. (arXiv (https://arxiv.org/abs/2504.01127))

13. **Commonsense Reasoning in Arab Culture — Sadallah et al. (ACL 2025).** This paper contributes ArabCulture, a Modern Standard Arabic commonsense benchmark authored and validated by native speakers across 13 countries, four broad regions, 12 daily-life domains, and 54 subtopics. Its evidence comes from zero-shot evaluation of open-weight models up to 32B parameters, which struggle across Arab cultures and show regional performance variation; the ACL record explicitly contrasts this native-authored design with machine-translated English datasets. The limitation is that MSA and country-level coverage cannot represent all dialects, communities, or intra-country disagreement, and multiple-choice commonsense still compresses context-sensitive reasoning into a single gold answer. Culturally, it is a strong correction to Anglocentric Arabic evaluation; formally, it measures situated everyday inference rather than formal logic, making it valuable for a separate cultural-commonsense axis. (ACL Anthology (https://aclanthology.org/2025.acl-long.380/))

14. **DRISHTIKON: A Multimodal Multilingual Benchmark for Testing Language Models' Understanding on Indian Culture — Maji et al. (EMNLP 2025).** DRISHTIKON contributes a multimodal, multilingual Indian-culture benchmark covering 15 languages, all Indian states and union territories, and more than 64,000 aligned text-image pairs across cultural themes such as festivals, attire, cuisine, arts, and heritage. Its evidence is evaluation of vision-language models in zero-shot and chain-of-thought settings, with pronounced limitations on low-resource languages and less-documented traditions and on reasoning over culturally grounded multimodal inputs. Limitations include the difficulty of separating visual recognition, language proficiency, cultural knowledge, and reasoning, plus the risk that broad national coverage still underrepresents local disagreement and living traditions. Culturally, it greatly expands Indian representation beyond translated text; formally, it is primarily a cultural-understanding and multimodal reasoning benchmark, not evidence for a unitary Indic logic. (ACL Anthology (https://aclanthology.org/2025.emnlp-main.68/); arXiv (https://arxiv.org/abs/2509.19274))

15. **SANSKRITI: A Comprehensive Benchmark for Evaluating Language Models' Knowledge of Indian Culture — Maji et al. (2025).** SANSKRITI contributes 21,853 curated question-answer pairs spanning 28 states, eight union territories, and 16 attributes of Indian culture, including rituals, history, cuisine, language, religion, medicine, sports, and personalities. Its evidence is comparative evaluation of large, Indic, and small language models showing substantial disparities and persistent difficulty with region-specific cultural queries. The limitation is that a large knowledge benchmark can reward memorization and retrieval without demonstrating contextual inference, pragmatic sensitivity, or logical validity; state and topic taxonomies also risk flattening intra-regional diversity. Culturally, it is a valuable India-centered resource for exposing gaps in regional coverage; formally, it should be treated as cultural knowledge/context evaluation adjacent to reasoning, not as evidence for

“Indic logic.” (arXiv (https://arxiv.org/abs/2506.15355); ACL Findings PDF (https://aclanthology.org/2025.findings-acl.228.pdf))

Cross-paper conclusion

Across all 15 retained papers, the verified evidence supports a mature research direction on culturally situated reasoning and evaluation, but not yet a comparative science of distinct non-Western formal logics. The strongest formal-reasoning precedent is LogiQA and LogiQA 2.0; the strongest culturally differentiated reasoning precedents are GeoMLAMA, ArabCulture, Global MMLU, SeaEval, and CQ-Bench. The recurring confound is that model failure may reflect translation, language accessibility, missing cultural knowledge, annotation conventions, or social-norm disagreement rather than invalid inference. A defensible taxonomy should therefore keep separate axes for formal logical validity, culturally situated commonsense, value-conditioned judgment, pragmatic/implicit-value inference, multimodal cultural understanding, and multilingual/translation robustness.

18 Toulmin Defeasible

The following summaries cover the 15 papers retained in the subtopic-18 review. Each linked source was checked against a primary paper, publisher, proceedings, ACL, arXiv, or author/repository record; where a page exposed only metadata or an abstract, the summary is limited to claims supported there.

1. **Simari & Loui (1992), "A Mathematical Treatment of Defeasible Reasoning and Its Implementation."** Primary PDF (https://cs.uns.edu.ar/~grs/Publications/1992-SimariLoui-AIJ.pdf) presents an argument-based defeasible reasoning system that combines Poole-style specificity with Pollock-style warrant and defeat. Its computational representation consists of a consistent evidential context, Horn-like defeasible rules, minimal argument structures, subarguments, counterarguments, and a specificity ordering used to determine which conclusions survive; the paper also proves termination and reduces the search space for specificity checks. Reasoning is therefore dialectical rather than monotonic: a conclusion is tentatively supported until competing arguments and rebuttals are assessed. The main limitation is deliberate formal restriction to Horn-style contexts and rules, with no natural-language extraction, rich Toulmin role inventory, provenance modeling, or modern learned component prediction. For this project, it is a foundational substrate for representing claims, warrants, exceptions, and defeat, and a useful benchmark for distinguishing a genuine defeasible proof object from a merely Toulmin-labeled text span.

2. **Nute (1994), "Defeasible Logic."** Oxford Academic record (https://academic.oup.com/book/53007/chapter-abstract/421965116) develops a formal account of defeasible logic organized around strict rules, defeasible rules, presumptions, defeaters, and tentative conclusions. The computational representation is rule-based and proof-procedural: conclusions can be supported provisionally while explicit defeaters block or undercut them, allowing nonmonotonic reasoning without treating every inference as classical entailment. Its central use is formal knowledge representation and reasoning under exceptions, rather than argument extraction or natural-language explanation. The chapter's limitation for a Toulmin-centered system is that it captures rule status and defeat more directly than explanatory roles such as data, backing, qualifier, or rebuttal, and it does not address NLP or LLM context selection. It is relevant as the formal contrast case: a project that calls a warrant "defeasible" should specify whether it has Nute-like rule semantics and proof conditions, or only an informal explanatory label.

3. **Verheij (2003), "DefLog: On the Logical Interpretation of Prima Facie Justified Assumptions."** Maastricht University record (https://cris.maastrichtuniversity.nl/en/publications/deflog-on-the-logical-interpretation-of-prima-facie-justified-ass/) treats assumptions as prima facie justified rather than definitely true and introduces dialectical interpretations that distinguish actually justified assumptions from defeated ones. The representation is sentence-based rather than argument-graph-based: assumptions, dialectical negations, partial stages, and stable-semantics-like interpretations determine whether an assumption remains justified. Reasoning use centers on formalizing defeat and the multiplicity or existence of dialectical interpretations, including prima-facie conditionals. The limitation is that DefLog does not provide the full six-part Toulmin decomposition, document-level evidence extraction, or learned language processing; its abstraction also leaves provenance and rhetorical explanation outside the core formalism. It matters here because it supplies a precise semantics for rebuttal and belief revision that could underwrite the "current status" of a Toulmin claim graph instead of treating a predicted warrant as automatically valid.

4. **Garcia & Simari (2004), "Defeasible Logic Programming: An Argumentative Approach."** arXiv record (https://arxiv.org/abs/cs/0302029) introduces DeLP, combining logic programming with defeasible argumentation. Facts and weak rules form arguments for queries, while a dialectical warrant procedure evaluates whether an argument is undefeated in the presence of counterarguments; the system explicitly supports contradictory goals, incomplete information, and dynamic domains. Computationally, DeLP turns logic-program clauses into argument structures and uses dialectical analysis to determine warranted conclusions rather than relying on a single derivation. Its main limitation is the gap between symbolic predicates/rules and natural-language argumentative structure: it does not extract Toulmin roles, infer implicit warrants from documents, or optimize external context for an LLM. The project relevance is direct: DeLP offers an implementable reference engine for testing whether a provenance-bearing claim/warrant/rebuttal graph yields a warranted conclusion under conflict, and it clarifies that "support" and "acceptance" must be evaluated separately.

5. **Verheij (2003), "Dialectical Argumentation with Argumentation Schemes: An Approach to Legal Logic."** Author/paper record (https://www.ai.rug.nl/~verheij/publications/ai&l2003.htm) formalizes argumentation schemes as generalized inference patterns whose conclusions are defeasible, concrete, and context-dependent rather than necessarily valid. The computational representation embeds schemes and their premises in a formal dialectical argumentation model, making it possible to analyze attacks and counterarguments while preserving the practical, domain-sensitive character of legal reasoning. The principal use is legal logic: schemes structure recurring reasoning patterns and dialectical procedures

determine whether their conclusions withstand challenge. A limitation is that the paper does not solve scheme discovery from raw text or provide a broad NLP/LLM evaluation methodology; it also warns implicitly against reducing all legal reasoning to one universal schema. For this project, it is the key bridge from Toulmin-like explanatory patterns to formal defeat: a context engine should retrieve not only evidence for a claim but also the premises and critical conditions that make the selected scheme applicable.

6. **Gordon, Prakken & Walton (2007), "The Carneades Model of Argument and Burden of Proof."** Publisher record (https://www.sciencedirect.com/science/article/pii/S0004370207000677) presents Carneades as a computational model built around argument graphs, ordinary premises, assumptions, exceptions, audiences, proof standards, burdens of proof, and critical questions associated with argumentation schemes. Arguments are represented separately from their acceptance status, allowing the same graph to be evaluated under different audiences or standards such as preponderance of evidence. Reasoning uses include informal-logic analysis, legal argument, and explicit handling of pro/con conflict and burden shifts. Its limitation for an NLP-context project is that the model presupposes structured arguments and does not itself extract them from open-domain language, rank retrieved passages under a token budget, or evaluate LLM answers. Carneades is nevertheless the strongest representational precedent: a new system should explain whether its proposed Toulmin graph adds capabilities beyond Carneades or simply provides an extraction and context-selection layer compatible with it.

7. **Walton & Gordon (2011), "Formalizing Informal Logic."** Journal page (https://ojs.uwindsor.ca/index.php/informal_logic/article/view/4335) explains how the Carneades Argumentation System formalizes core concerns of informal logic through argument graphs, audiences, proof standards, and argumentation schemes. The computational representation captures premise acceptability, relevance and sufficiency, pro/con conflict, and scheme appropriateness, while proof standards determine whether a position is accepted for a given audience. Its reasoning use is not to turn every informal argument into deductive logic, but to make practical argumentative evaluation explicit and inspectable. The paper's limitation is scope: it is a formal modeling contribution rather than a natural-language extraction, retrieval, compression, or LLM study, and the linked record does not establish performance on contemporary benchmarks. It is relevant as a design baseline for separating graph construction, audience-relative acceptance, and proof standards—three functions that a context-engineering system should not collapse into a single confidence score.

8. **Verheij (2009), "The Toulmin Argument Model in Artificial Intelligence: Or, How Semi-Formal, Defeasible Argumentation Schemes Creep into Logic."** Author page/PDF (https://www.ai.rug.nl/~verheij/publications/toulmin2009.htm) traces Toulmin's six-part model—data, claim, warrant, backing, qualifier, and rebuttal—into default logic, defeaters, dialectical argumentation, argumentation schemes, and AI/legal reasoning. The paper's computational contribution is primarily conceptual synthesis: Toulmin roles become increasingly formal when implemented as rules, argument structures, attack relations, and proof procedures, but the model itself is not a complete inference calculus. Its reasoning use is therefore explanatory and architectural, helping connect semi-formal practical justification to nonmonotonic logic without confusing rhetorical adequacy with deductive validity. The limitation is that it does not present an end-to-end NLP system or an empirical context-selection evaluation. For this project, it is the essential boundary-setting paper: a useful computational object should preserve Toulmin's explanatory distinctions while adding explicit defeaters, provenance, and acceptance semantics rather than claiming that six labels alone constitute reasoning.

9. **Reisert, Inoue, Okazaki & Inui (2015), "A Computational Approach for Generating Toulmin Model Argumentation."** ACL Anthology record and PDF (https://aclanthology.org/W15-0507/) is the most direct NLP implementation in the set. Given a target claim, the system aggregates statements from multiple documents and constructs Toulmin-style claim, data, and warrant relations, using causal PROMOTE/SUPPRESS relations, sentiment, and contextual similarity to identify supporting material and generate implicit warrants or backing. The computational object is therefore a document-grounded argument structure rather than a purely symbolic proof; its use is automatic argument generation from heterogeneous evidence. Limitations include dependence on earlier feature-based/document-processing assumptions, incomplete coverage of all Toulmin components, and no token-budget-aware compression, explicit defeasible proof calculus, or LLM downstream evaluation. It is directly relevant and a serious novelty constraint: any project proposing Toulmin-aware evidence assembly must distinguish itself through retrieval control, proof-preserving compression, contradiction handling, or measurable reasoning gains rather than evidence aggregation alone.

10. **Reisert, Inoue, Okazaki & Inui (2017), "A Corpus of Deep Argumentative Structures as an Explanation to Argumentative Relations."** arXiv record (https://arxiv.org/abs/1712.02480) defines a task for explaining argumentative relations with a small set of deeper patterns instead of only labeling surface support or attack. The representation uses structured patterns such as value judgment and bipolar causality to explain why one segment supports another; an annotation study covers 357 relations, reports 74.6% coverage on a test set, and finds 85.9% inter-annotator agreement for the proposed

pattern set. Its reasoning use is to expose compact intermediate links that make argumentative relations interpretable and computationally analyzable. Limitations are the constrained corpus/domain, pattern coverage gaps, and the absence of a retrieval or compression policy that uses these structures operationally. The project relevance is strong for proof-preserving context selection: deep relation patterns can serve as compact warrant edges or missing-premise cues, but they must be integrated with provenance and defeasible attack status rather than treated as complete proofs.

11. **Singh, Inoue, Mim, Naitoh & Inui (2021), "Annotating Implicit Reasoning in Arguments with Causal Links."** arXiv record (https://arxiv.org/abs/2110.13692) focuses on implicit argumentation knowledge connecting premises and claims, rather than only commonsense or factual gaps. Inspired by the Argument from Consequences scheme, it proposes a semi-structured causal template that makes missing reasoning links explicit, then develops a two-phase annotation process and releases guidelines and collected implicit reasonings. Computationally, the template represents an unspoken causal warrant that can be attached to argumentative components; its reasoning use is explanatory reconstruction of enthymematic links. The study reports substantial expert agreement for quality evaluation but also raises concerns about whether crowdsourcing can reliably collect high-quality semi-structured reasoning. It does not perform retrieval, token compression, formal defeasible inference, or LLM evaluation. For this project, the causal links are useful compact proof scaffolds and retrieval targets: a system could search for evidence that supports, attacks, or fills a missing causal warrant while preserving uncertainty about whether the reconstructed link is actually warranted.

12. **Hagen et al. (2021), "Developing a Conversational Agent's Capability to Identify Structural Wrongness in Arguments Based on Toulmin's Model of Arguments."** Frontiers article (https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2021.645516/full) operationalizes Toulmin components in an educational conversational agent that identifies structural wrongness and uses conditional dialogue behavior to provide feedback. The representation maps argument elements and their relationships into dialogue-state conditions, allowing the agent to react differently when a claim lacks appropriate data, warrant, or other structural support. Its reasoning use is pedagogical diagnosis and interactive repair, not formal warrant computation or dialectical proof. The limitation is consequently important: structural wrongness detection can flag missing or misplaced components but does not establish truth, defeasible acceptance, source reliability, or resistance to rebuttal, and the work is not a context-retrieval benchmark. It is relevant as a conversational overlap and as evidence that Toulmin structure can control next-turn behavior; a new context manager should add provenance-aware evidence selection, belief revision, or measurable contradiction handling.

13. **Atkinson & Bench-Capon (2021), "Argumentation Schemes in AI and Law."** Publisher record (https://journals.sagepub.com/doi/10.3233/AAC-200543) reviews the role of argumentation schemes in AI and law and argues that a single generic Toulmin-like scheme is too coarse to represent the diversity of legal reasoning. The computational perspective favors multiple domain-sensitive schemes, associated critical questions, and richer state-transition or dialectical machinery for evaluating arguments. Schemes thus function as structured, defeasible reasoning templates whose applicability depends on premises, context, and challenges, rather than as universal slots filled identically in every domain. The limitation is that this is a review/design analysis rather than an end-to-end extraction or LLM experiment, so it does not supply a ready benchmark for open-domain context selection. Its project relevance is a design warning: use Toulmin roles as an interoperable core or explanatory view, but preserve domain-specific schemes, critical questions, and exception conditions instead of forcing all reasoning into one flat ontology.

14. **Lawrence & Reed (2019), "Argument Mining: A Survey."** ACL Anthology record and PDF (https://aclanthology.org/J19-4006/) surveys automatic identification and extraction of argumentative structure from natural language, covering components, relations, schemes, philosophical foundations, datasets, methods, and open challenges. Its computational representation spans argument segments and links, with attention to the difficulty of mapping linguistic evidence onto competing argumentation ontologies, including Toulmin-inspired data, claim, and warrant categories. The reasoning use is landscape-level: it clarifies what argument mining can extract and why deeper reasoning remains difficult, rather than proposing one defeasible proof engine. Limitations include the survey's date relative to current LLM methods and the persistent gap between component/relation classification and valid nonmonotonic inference. For this project, it supplies the evaluation vocabulary and a crucial methodological caution: component F1 or relation accuracy alone cannot establish defeasible reasoning quality; extraction, warrant reconstruction, defeat handling, proof-supporting context recall, answer quality, and cost should be measured separately.

15. **Kim & Yang (2026), "TRACE: Toulmin-based Reasoning Assessment through Constructive Elements for LLM CoT Evaluation."** PMLR record (https://proceedings.mlr.press/v306/kim26bw.html) introduces TRACE, a metric that analyzes chain-of-thought using Toulmin constructive elements together with Flavell's metacognitive framework rather than judging only final-answer accuracy. The representation treats generated reasoning as a structured set of argumentative elements and

uses that structure to assess reasoning quality; experiments reportedly cover 26.3K QA samples across seven reasoning models, find correlation with benchmark accuracy, and show usefulness as a reinforcement-learning reward signal. Its reasoning use is evaluation and training feedback for LLM traces, not external evidence retrieval, formal defeasible entailment, or hidden-context selection. Limitations therefore include dependence on observable reasoning traces and the fact that structural quality does not by itself guarantee source-grounded truth or correct defeat behavior. TRACE is the strongest current LLM-era overlap and should be a comparison point for any project claiming Toulmin-based reasoning evaluation, while a context-engineering contribution should target retrieval control, proof verification, contradiction robustness, or token/latency trade-offs instead.

Cross-paper implication for the project

Across the set, Toulmin provides an interpretable decomposition of justification, while defeasible logic and argumentation systems provide the semantics for exceptions, attack, burden, and acceptance. NLP work contributes extraction and reconstruction of implicit warrants; newer LLM work contributes structure-aware trace evaluation. The remaining integrated opportunity is a provenance-preserving defeasible graph that links claim, data, warrant, backing, qualifier, rebuttal/defeater, critical questions, and current dialectical status, then uses that graph to select or compress external context and evaluates proof-supporting recall, contradiction handling, answer quality, validity, token cost, and latency separately.

19 Abduction Analogy

Verified against the publisher or author-hosted record on 2026-10-12. The source list contains 15 retained papers (not 17); the summaries below cover all 15 numbered entries in the agent brief. Two metadata corrections are applied: the first paper is by Paul and Frank, and the Occam paper's verified title is **Do Language Models Follow Occam's Razor? An Evaluation of Parsimony in Inductive and Abductive Reasoning**.

Abduction and hypothesis generation

1. **Paul & Frank (2021), "Generating Hypothetical Events for Abductive Inference" (STARSEM 2021).** The paper studies Abductive NLI as explanation selection and adds a specialized language model that generates plausible next events from hypothetical scenarios; a multitask model then compares event consequences associated with candidate hypotheses against the observed outcome. It reports improvement over vanilla pretrained language models fine-tuned on Abductive NLI, with manual analysis suggesting that modeling possible future events helps select explanations. The main limitation is task scope: this is constrained binary/choice-style abductive inference rather than open-ended hypothesis generation, and its notion of plausibility remains tied to the dataset's event knowledge. For structured-context engineering, it establishes a useful observation → candidate explanation → consequence simulation pattern, showing why explicit temporal and causal bridges can improve abductive selection. Source: https://aclanthology.org/2021.starsem-1.6/

2. **Bai et al. (2024), "Advancing Abductive Reasoning in Knowledge Graphs through Complex Logical Hypothesis Generation" (ACL 2024).** The authors formulate complex logical hypothesis generation over knowledge graphs: a generated hypothesis must explain a set of observations through graph-supported relations. A supervised generator produces hypotheses structurally closer to references, but that objective does not reliably generalize to unseen observations; the proposed Reinforcement Learning from Knowledge Graph (RLF-KG) instead rewards agreement between observations and conclusions entailed by the graph, improving explanation quality and achieving state-of-the-art results on three knowledge graphs. The limitation is that structural closeness and graph entailment are proxies for explanatory adequacy, while performance depends on the completeness and correctness of the underlying KG. The paper directly supports relation-explicit context engineering: constraints and feedback tied to the knowledge structure are more useful than unconstrained generation. Source: https://aclanthology.org/2024.acl-long.72/

3. **Zhou et al. (2024), "Hypothesis Generation with Large Language Models" (NLP4Science 2024).** This work generates initial scientific or data-driven hypotheses from a few labeled examples and iteratively updates them using predictive feedback and a multi-armed-bandit-inspired reward strategy, evaluating on synthetic and real datasets. The reported method improves over few-shot prompting and shows gains over supervised learning in some settings, indicating that LLM hypothesis generation benefits from an explicit propose–test–refine loop rather than one-shot completion. Its limitation is conceptual: predictive hypotheses are not necessarily explanatory, novel, or scientifically valid, so improvement in predictive reward does not establish genuine abduction. In relation to structured context, the paper motivates feeding models compact examples, evaluation feedback, and a controlled refinement objective, while warning that task reward can substitute for explanation unless the latter is measured separately. Source: https://aclanthology.org/2024.nlp4science-1.10/

4. **Sun & Saparov (2025), "Do Language Models Follow Occam's Razor? An Evaluation of Parsimony in Inductive and Abductive Reasoning" (arXiv:2509.03345).** The paper introduces InAbHyD, a programmable synthetic benchmark with incomplete world models and observations, and evaluates hypotheses not only for correctness but also for parsimony under an Occam-style metric. It finds that language models can handle simple cases but struggle to produce high-quality, sufficiently simple hypotheses as worlds become more complex; in-context learning and reinforcement learning with verifiable rewards do not eliminate the difficulty. The synthetic setup improves control and diagnostic resolution but limits ecological validity because it abstracts away noisy observations, open-world uncertainty, and domain-specific knowledge. The arXiv record checked on October 12, 2026 reports that the paper was accepted at the EMNLP 2026 main conference, although the accessible source remains the arXiv record. Its relation to context engineering is direct: abductive prompts should expose constraints, coverage, and simplicity trade-offs, not merely ask for a plausible explanation. Source: https://arxiv.org/abs/2509.03345

5. **Zheng et al. (2025), "LogiDynamics: Unraveling the Dynamics of Inductive, Abductive and Deductive Logical Inferences in LLM Reasoning" (EMNLP 2025).** LogiDynamics evaluates inductive, abductive, and deductive inference in controlled textual, visual, and symbolic environments, varying difficulty and response format while examining hypothesis selection and iterative refinement. The study reports that explicit selection and refinement can substantially improve reasoning as task demands scale, and it places abduction within a broader family of logical inference dynamics rather than treating it as

a single completion behavior. The controlled environments and analogy-oriented task framing limit direct transfer to unconstrained real-world explanations, and system-style labels should not be interpreted as evidence about human cognition. For structured contexts, the key implication is architectural: separate candidate generation, evidence-based selection, and refinement, then test each stage rather than treating fluent final text as proof of reasoning. Source: https://aclanthology.org/2025.emnlp-main.1045/

Analogy identification, generation, and relational transfer

6. **Chen et al. (2022), "E-KAR: A Benchmark for Rationalizing Natural Language Analogical Reasoning" (Findings of ACL 2022).** E-KAR introduces a knowledge-intensive benchmark drawn from Chinese and English civil-service examinations, with 1,655 Chinese and 1,251 English problems, requiring both analogical question answering and free-text rationalization; candidate answers and explanations receive manual annotations. The authors find the benchmark challenging for contemporary models in both answer selection and explanation generation, making it difficult to be correct for the correct relational reason. Because the questions require substantial background knowledge, performance confounds analogy with retrieval and world knowledge, and generated rationales need not be faithful accounts of the model's internal process. The benchmark is nevertheless highly relevant to context engineering because it requires relation-grounded explanations rather than surface-level answer matching. Source: https://aclanthology.org/2022.findings-acl.311/

7. **Bhavya, Xiong & Zhai (2022), "Analogy Generation by Prompting Large Language Models: A Case Study of InstructGPT" (INLG 2022).** This study prompts InstructGPT to generate analogous concepts and explanations, comparing prompt wording, decoding temperature, and model scale. Precise imperative prompts and low temperature perform best; model size strongly affects quality, with the largest evaluated InstructGPT reaching human-level performance for concept generation while remaining weaker on explanation generation. The limitation is that judged fluency and appropriateness of generated analogies do not demonstrate relational transfer, causal analogy, or faithful reasoning, and the experiments reflect narrow tasks and an early instruction-tuned model family. Its contribution to structured-context design is controlled evidence that prompt form and decoding settings materially change analogy outputs, so these variables must be reported and held constant in comparisons. Source: https://aclanthology.org/2022.inlg-main.25/

8. **Hu et al. (2023), "In-Context Analogical Reasoning with Pre-Trained Language Models" (ACL 2023).** The authors convert visual Raven's Progressive Matrices into language-based abstractions and use in-context prompts to select the correct completion, analyzing model complexity, demonstrations, and prior knowledge. The method shows how making perceptual relations explicit in text can enable pretrained language models to solve a visual analogy-style task, but textualization may measure language-mediated reconstruction rather than native visual analogy, and results depend heavily on the quality of the abstraction and available prior knowledge. The paper therefore provides a direct structured-context intervention: relational descriptions can help, but the representation itself becomes part of the task and may carry much of the difficulty or answer signal. Source: https://aclanthology.org/2023.acl-long.109/

9. **Wijesiriwardene et al. (2023), "ANALOGICAL: A Novel Benchmark for Long Text Analogy Evaluation in Large Language Models" (Findings of ACL 2023).** ANALOGICAL benchmarks whether models can identify analogous situations across extended narratives, targeting analogy beyond short word or sentence pairs. Its method stresses long-context reading and relational retrieval, making it useful for separating short-range matching from story-scale analogy identification. The central limitation is construct overlap: success requires memory, retrieval, reading comprehension, and narrative abstraction in addition to analogy, so poor performance cannot be attributed to relational mapping alone. For context engineering, the benchmark underscores that adding more context does not guarantee better transfer; systems need mechanisms that retrieve and align the relevant relational structure within long inputs. Source: https://aclanthology.org/2023.findings-acl.218/

10. **Yuan et al. (2023), "StoryAnalogy: Deriving Story-level Analogies from Large Language Models to Unlock Analogical Understanding" (EMNLP 2023).** StoryAnalogy constructs a 24,000 story-pair corpus with human annotations based on an extended Structure-Mapping Theory and evaluates both story-level analogy identification and generation. The reported ChatGPT result is about 30% on a multiple-choice identification task, compared with more than 85% for humans, revealing a large gap in mapping extended narrative relations. The limitation is that story analogies combine long-context retrieval, commonsense interpretation, linguistic variation, and annotation complexity, so the gap is not a pure measure of analogy failure. The paper contributes a relationally annotated resource for testing whether models align event roles and structures rather than merely matching topical vocabulary, making it especially relevant to structured narrative context. Source: https://aclanthology.org/2023.emnlp-main.706/

11. **Petersen & van der Plas (2023), "Can Language Models Learn Analogical Reasoning? Investigating Training Objectives and Comparisons to Human Performance" (EMNLP 2023).** The study tests several training objectives on basic

analogies designed to resemble human analogical-reasoning evaluations rather than standard word-embedding analogy tasks, and compares trained models with a human-baseline dataset. Models learn the evaluated behavior from relatively small amounts of data and approach human performance after training. The limitation is that small, basic, supervised tasks show learnability under task-specific training but do not establish zero-shot, cross-domain, or compositional relational transfer. Its relevance to context engineering is diagnostic: explicit training or examples can teach an analogy task, so evaluations must distinguish memorized task behavior from general reasoning and should include held-out relations and domains. Source: https://aclanthology.org/2023.emnlp-main.1022/

12. **Ye et al. (2024), "AnaloBench: Benchmarking the Identification of Abstract and Long-context Analogies" (EMNLP 2024).** AnaloBench collects 340 human-written analogies and evaluates 12 open-source and three proprietary models on two human-relevant abilities: recalling related experiences from a large pool and applying analogy to complex, lengthy scenarios. Scaling produces some improvements, but yields minimal gains for lengthy scenarios and for retrieving relevant analogues from large information pools. The benchmark is modest in size and its human-written scenarios impose a particular domain and difficulty distribution, which limits broad generalization claims. It directly informs retrieval-aware context engineering by separating analogue recall from mapping and transfer, showing that larger models alone do not solve the long-context bottleneck. Source: https://aclanthology.org/2024.emnlp-main.725/

13. **Qin et al. (2025), "Relevant or Random: Can LLMs Truly Perform Analogical Reasoning?" (Findings of ACL 2025).** This paper compares self-generated relevant examples with irrelevant or random examples before solving reasoning tasks, testing whether improvements arise from genuinely useful analogical context rather than from adding any extra text. The experimental design treats generated exemplars as candidate analogical supports and finds that context relevance matters, supporting a relevance-sensitive account of analogical prompting. A limitation is that relevant examples can still provide semantic priming, lexical cues, or task hints, and operational relevance in the benchmark is not a complete theory of relational analogy. The practical relation is strong: context selection should be evaluated against random and irrelevant controls, and gains should not be attributed to analogy unless mapping and transfer are separately demonstrated. Source: https://aclanthology.org/2025.findings-acl.1230/

14. **Wijesiriwardene et al. (2025), "KnowledgePrompts: Exploring the Abilities of Large Language Models to Solve Proportional Analogies via Knowledge-Enhanced Prompting" (COLING 2025).** KnowledgePrompts introduces a 15,000-item proportional-analogy multiple-choice dataset and compares exemplar prompts with structured and targeted knowledge prompts. The reported best model reaches 55%, and targeted knowledge performs better than the other knowledge formats, indicating that the content and granularity of context matter more than simply adding demonstrations. The limitation is that multiple-choice proportional analogies may not measure open-ended relational transfer, while targeted knowledge can encode much of the relation needed to select the answer. For context engineering, the paper provides direct evidence for relation-explicit, targeted context over undifferentiated example dumps, while also motivating controls for answer leakage and knowledge sufficiency. Source: https://aclanthology.org/2025.coling-main.268/

15. **Das & Balke (2025), "From Prototypical to Relational: How LLMs Navigate Complex Analogies" (INLG 2025).** The authors build a four-way complex-analogy benchmark in which all answer choices are semantically plausible, using relations from Wikidata and AnalogyKB and evaluating relational overlap alongside embedding similarity and prototypicality. Models including GPT-4 perform better on embedding- and prototype-based measures than on fine-grained relational mappings, indicating that plausible surface similarity can mask weak structure alignment. The benchmark's relation mining and GPT-4-assisted salience ranking introduce pipeline and model dependence, while making every option plausible reduces the evaluation to a less sharply defined ground truth than ordinary multiple choice. Its contribution is a clean diagnostic separation between prototypical similarity and relational correspondence, which is essential when testing whether structured context actually supports analogy rather than merely improving topical matching. Source: https://aclanthology.org/2025.inlg-main.28/

Cross-paper implications

Across these 15 studies, relation-explicit context consistently appears more useful than undifferentiated examples, but context can help through retrieval, priming, representation, or answer cues without proving relational reasoning. Abductive systems should therefore score coverage/consistency, parsimony, and unseen-case generalization; analogy systems should separately measure analogue retrieval, structure mapping, and transfer. Fluent explanations are insufficient evidence of faithful reasoning, so robust evaluations should include irrelevant-context controls, distractors, alternative plausible hypotheses, long-context tests, and cross-domain or unseen-relation transfer.

20 Ai Academic Writing

1. Gao et al. (2023), “Comparing scientific abstracts generated by ChatGPT to real abstracts with detectors and blinded human reviewers”

Gao et al. generated 50 medical abstracts from article titles and journal identities using an early ChatGPT version, then compared them with the originals using plagiarism/similarity tools, the GPT-2 Output Detector, journal-format checks, and blinded human reviewers. The generated texts were superficially formatted as scientific abstracts, but only 8/50 correctly followed the target journal’s heading/structure conventions; the detector achieved AUROC 0.94, reviewers correctly identified 68% of generated abstracts, and incorrectly labeled 14% of original abstracts as AI-generated. The design tests plausibility and provenance detection rather than scientific truth, because the model did not receive the underlying articles, methods, or data, and the corpus and model were from late 2022/early 2023. For this project, the paper supports separating fluency, format compliance, factuality, and authorship detection into distinct evaluation dimensions rather than treating detectability or surface quality as evidence of research quality. (DOI: https://doi.org/10.1038/s41746-023-00819-6)

2. Herbold et al. (2023), “A large-scale comparison of human-written versus ChatGPT-generated essays”

Herbold et al. compared a large corpus of human argumentative student essays with essays generated by ChatGPT-3.5 and ChatGPT-4 from matched prompts, using teacher ratings on standard quality criteria and computational linguistic analysis of logical structure, language complexity, vocabulary richness, and linking. The AI essays received higher average quality ratings, with GPT-4 outperforming GPT-3.5 and humans on several structural and linguistic dimensions; the models also showed recognizable regularities such as formulaic conclusions, more nominalizations, and fewer discourse/epistemic markers. The study concerns student essays rather than evidence-heavy scholarly research, and its prompts were generally not designed to test citation accuracy, empirical validity, or original research reasoning, so the result cannot be generalized to reliable academic scholarship. It is directly useful here as a decomposition of writing quality into argument structure, topical coverage, language, complexity, vocabulary, and cohesion, while demonstrating why those dimensions must be scored separately from truth and evidential support. (DOI: https://doi.org/10.1038/s41598-023-45644-9)

3. Yeadon et al. (2024), “Evaluating AI and human authorship quality in academic writing through physics essays”

Yeadon et al. evaluated 300 short physics-essay submissions, equally divided between pre-ChatGPT student work and GPT-4 outputs, within a course covering history and philosophy of physics, science communication, and ethics in academia; five independent markers graded blinded essay sets and then judged likely authorship, while five commercial detectors were also compared. Human and AI essays did not differ significantly in grades ($p = 0.107$), markers were only marginally better than chance at authorship identification, and ZeroGPT performed best in the study’s binary detector comparison, reporting 98% accuracy under that specific setup. The sample came from one module, used short essays and one model/version, and detector results are corpus-dependent; the detector headline should not be treated as evidence of general authorship reliability. The study is valuable for this project because it cleanly separates academic-writing quality from provenance classification and provides a philosophy-adjacent test context in which content quality, reasoning, and authorship should be evaluated as different endpoints. (DOI: https://doi.org/10.1088/1361-6404/ad669d)

4. Kong and Liu (2024), “A comparative genre analysis of AI-generated and scholar-written abstracts for English review articles in international journals”

Kong and Liu constructed two corpora of 160 scholar-written and 160 ChatGPT-generated review-article abstracts spanning four hard- and soft-science disciplines, then compared their rhetorical moves, move order, obligatory and optional elements, and move embedding. They found significant differences in the frequency of three of five rhetorical moves, preferred sequencing, and embedding frequency, although the two groups shared some embedding combinations and broad structural patterns. The evaluation is about genre realization and disciplinary organization, not whether the abstracts’ claims are true, whether evidence supports them, or whether the underlying reviews are novel or methodologically sound; it also uses abstracts rather than full papers. For this project, the paper offers an operational alternative to vague judgments that text “sounds academic”: disciplinary genre knowledge, rhetorical-move coverage, ordering, and embedding can be scored separately from grammar, readability, factuality, and argument validity. (DOI: https://doi.org/10.1016/j.jeap.2024.101432)

5. Geng and Trotta (2024), “Is ChatGPT Transforming Academics’ Writing Style?”

Geng and Trotta analyzed approximately one million arXiv abstracts submitted from May 2018 through January 2024 and calibrated a statistical style model using real abstracts and simulated ChatGPT-modified abstracts produced with a simple “revise the following sentences” prompt. They estimated increasing ChatGPT-like stylistic influence, especially in computer science, with roughly 35% of abstracts classified as exhibiting the modeled style under their baseline assumptions. The estimate measures resemblance to a simulated style distribution, not actual ChatGPT use, authorship, factuality, quality, or research merit; it is also sensitive to prompt choice, calibration data, field differences, and temporal shifts in academic language. This paper matters for the project as a warning that human/AI comparisons and detector performance may drift as people adopt model-like phrasing, making style-based provenance a moving target rather than a stable proxy for authorship or scholarly quality. (arXiv: https://arxiv.org/abs/2404.08627)

6. Thelwall (2024), “Can ChatGPT evaluate research quality?”

Thelwall configured GPT-4 to apply UK Research Excellence Framework-style criteria for significance, rigor, and originality to 51 of the author’s own journal articles, repeating the evaluation 15 times and comparing the outputs with the author’s self-evaluations. Single-run scores correlated weakly with the self-ratings (mean r = 0.281), while averaging repeated runs improved the correlation to 0.509; after removing the weakest articles, the average-score correlation fell to r = 0.200 and was no longer statistically significant, indicating difficulty with fine-grained judgments. The study uses a convenience sample from one author and one field, relies on self-evaluation rather than independent expert consensus, and is prompt/model dependent, so it is not a general validation of LLM peer review. It directly supports this project’s requirement to test evaluator stability, calibration, human agreement, and criterion-level validity rather than accepting plausible rationales or one-shot scores as evidence that an AI judge can assess philosophical or academic research quality. (DOI: https://doi.org/10.2478/jdis-2024-0013)

7. Levin et al. (2024), “Association of reviewer experience with discriminating human-written versus ChatGPT-written abstracts”

Levin et al. created 10 fabricated gynecologic-oncology abstracts and matched ChatGPT-3.5 abstracts from the same titles and results, then asked 30 reviewers—10 residents, 10 junior reviewers, and 10 senior reviewers—to classify 20 blinded abstracts, yielding 600 judgments. Reviewers correctly classified 50% overall, including only 46.3% of ChatGPT abstracts; seniority, publication experience, and AI familiarity were associated with better identification, while human-written abstracts were identified more accurately than generated ones. Because both sets were fabricated and matched, the study tests provenance discrimination rather than medical correctness, clinical usefulness, or research quality; its small, domain-specific sample also limits generalization. For this project, it shows that evaluator expertise can affect authorship judgments and should be recorded, but that even experienced reviewers’ ability to identify AI text does not establish that they can assess argument validity, factuality, or scholarly contribution. (DOI: https://doi.org/10.1136/ijgc-2023-005162)

8. Bohlmann and Berger (2024), “ChatGPT and the Writing of Philosophical Essays: An Experimental Study with Prospective Teachers on How the Turing Test Inverted”

Bohlmann and Berger conducted a philosophy-specific experiment in which prospective teachers judged philosophical essays written by students and by ChatGPT, comparing human discrimination performance with an AI classifier. The indexed scholarly record reports approximately 78.7% classification accuracy, making the study unusually direct evidence about whether philosophical prose can be identified by its provenance. The available record exposes less methodological detail than the full methods needed for assessing philosophical competence, and authorship classification is not the same as evaluating interpretation, premise quality, validity, counterargument handling, or originality; the result therefore should not be read as a measure of philosophical writing quality. Its main value for this project is domain specificity: philosophy requires a separate evaluation layer for reasoning and textual engagement even when a classifier or reader can distinguish likely AI authorship. (DOI: https://doi.org/10.5840/teachphil202451200)

9. Schwitzgebel, Schwitzgebel, and Strasser (2023/2024), “Creating a Large Language Model of a Philosopher”

Schwitzgebel and collaborators fine-tuned GPT-3 on Daniel Dennett’s writings, asked Dennett and the model the same 10 philosophical questions, collected four model responses per question without cherry-picking, and recruited 425 participants consisting of Dennett experts, philosophy-blog readers, and ordinary participants to identify the human philosopher’s answer. Experts achieved 51% accuracy against a 20% chance baseline, philosophy-blog readers performed similarly, and ordinary participants were near chance; for two questions, an AI response was selected more often than Dennett’s actual answer. The task is short-response style and voice imitation, not full academic papers, source-grounded exegesis, valid argument construction, or research originality, and the arXiv version should be distinguished from the later journal publication. For this project, it demonstrates that convincing philosophical voice and human-like reception are independent of argument validity, factuality, attribution, and scholarly accountability, reinforcing the need for construct-specific evaluation. (arXiv: https://arxiv.org/abs/2302.01339)

10. Alkaissi and McFarlane (2023), “Artificial Hallucinations in ChatGPT: Implications in Scientific Writing”

Alkaissi and McFarlane used prompted short paragraphs on medical and non-medical topics as an editorial case study of ChatGPT’s scientific-writing behavior, checking whether fluent outputs contained plausible but incorrect claims and references. They report that the model could produce convincing prose while mixing accurate material with fabricated or erroneous facts and citations, illustrating how apparent expertise can conceal unsupported content. This is not a systematic benchmark: the prompt set is small, the design is editorial rather than independently rated, and it does not estimate hallucination rates across models, domains, or prompt conditions. For this project, the paper supplies a foundational factuality warning and motivates explicit claim-level checks for source existence, bibliographic correctness, entailment, uncertainty, and evidence support instead of using coherence or style as a proxy for truth. (DOI: https://doi.org/10.7759/cureus.35179)

11. Bhattacharyya et al. (2023), “High Rates of Fabricated and Inaccurate References in ChatGPT-Generated Medical Content”

Bhattacharyya et al. prompted ChatGPT-3.5 to generate 30 short medical papers with at least three references each, then verified reference authenticity and the accuracy of seven bibliographic elements using Medline, Google Scholar, and the Directory of Open Access Journals. Of 115 generated references, 47% were fabricated, 46% were authentic but inaccurate, and only 7% were both authentic and accurate; incorrect PMID, volume, page, and year fields were common, with an average of 4.3 inaccurate elements per reference. The study is limited to early ChatGPT-3.5, medical prompts, and short generated papers, so rates should not be transferred directly to philosophy or current models, and it evaluates references rather than complete argument quality. It directly supports a paper-level reference audit in this project covering existence, metadata match, relevance, and whether each citation actually supports the claim for which it is supplied. (DOI: https://doi.org/10.7759/cureus.39238)

12. Ide, Nakagawa, and collaborators (2023), “Can ChatGPT Be Considered an Author of a Medical Article?”

Ide et al. applied the four International Committee of Medical Journal Editors authorship criteria to ChatGPT, distinguishing contribution to conception/design or analysis, drafting/revision, final approval, and accountability for accuracy and integrity. Their analysis concluded that ChatGPT can assist with contribution and drafting but cannot give final approval or accept responsibility for the article, so it should not be listed as an author under ICMJE criteria even though its use should be disclosed. This is a normative/accountability analysis rather than an empirical test of generated prose, scientific validity, or model behavior across tasks, and its conclusion is tied to a particular medical-publishing framework. For this project, it provides a useful multidimensional authorship taxonomy: generation assistance, intellectual contribution, approval, attribution, and responsibility should not be collapsed into a single “AI-authored” label. (DOI: https://doi.org/10.2188/jea.JE20230030)

13. Aylsworth and Castro (2024), “Should I Use ChatGPT to Write My Papers?”

Aylsworth and Castro present a philosophical normative analysis of whether students should use chatbots to write humanities papers, examining competing arguments about cheating, education, capacity loss, agency, and technological assistance before defending a Kantian account centered on the duty to cultivate and safeguard one’s autonomy. Their conclusion is that students have moral reasons to write certain humanities papers themselves because the writing process cultivates the capacity to set and pursue one’s own ends, while permissible assistance may remain available for less educationally central tasks. The paper does not empirically measure ChatGPT’s quality, factuality, reasoning, or student behavior, and its claims depend on a normative account of education and autonomy rather than experimental results. It is nevertheless important to this project because it defines a target construct that technical metrics can miss: a text may be fluent, accurate, and logically organized yet fail an assignment’s agency, authorship, or educational-purpose requirements. (DOI: https://doi.org/10.1007/s13347-024-00809-w)

Verification note

All 13 retained records were checked against a publisher, repository, PubMed/PMC, arXiv, or institutional scholarly record. The verification also corrected the source attribution for the reference-integrity study: the paper listed in the input inventory as “Sadasivan et al.” is published as Bhattacharyya, Miller, Bhattacharyya, and Miller (2023), DOI 10.7759/cureus.39238. The set contains empirical evaluations, style/provenance studies, a citation-integrity study, an authorship-accountability analysis, and normative philosophy; these constructs should not be pooled into one undifferentiated quality score.

21 Scholastic Method Ai Direct

Verification date: 2026-10-12

Paper count: 10 retained papers

Scope note: The retained set contains no verified paper that directly implements the medieval scholastic *quaestio/disputatio* sequence as an AI or LLM architecture. The papers below are therefore summarized by their exact or adjacent relation to the project: structured argument extraction, objection/contestability handling, formal argument semantics, debate dialogue, or question-driven critical reasoning.

1. Hong et al. (2022), “Scholastic: Graphical Human-AI Collaboration for Inductive and Interpretive Text Analysis”

Hong et al. present *Scholastic*, a human–AI text-analysis interface that combines machine-in-the-loop clustering with researcher-applied codes, structured metadata, hierarchical document and word clusters, and interactive visualizations for inductive and interpretive analysis.[1] The paper is an important verification of a likely false positive: “Scholastic” is the name of an HCI/ML system, not a reference to medieval scholastic method, and the system does not represent a *quaestio*, objections, distinctions, a *respondeo*, or replies. Its exact relevance is consequently low, although it offers a useful precedent for human-guided construction of structured context over a text corpus rather than treating documents as undifferentiated input.[1]

2. Freedman et al. (2025), “Argumentative Large Language Models for Explainable and Contestable Claim Verification”

Freedman et al. develop Argumentative Large Language Models (ArgLLMs), which augment language models with an explicit argumentation layer for explainable and contestable claim verification.[2][11] The system constructs argumentation frameworks and applies formal reasoning to evaluate claims, making contestability and explanation first-class properties rather than relying only on an unstructured generated answer. This is a strong adjacent precedent for a scholastic-AI project because objections can be treated as contesting reasons that must be represented and answered; however, the paper does not implement the ordered medieval workflow of question, objection list, distinctions, determination, and targeted replies, nor does it define a scholastic *respondeo* stage.[2][11]

3. Castagna, Sassoon, and Parsons (2024), “Can formal argumentative reasoning enhance LLMs performances?”

Castagna, Sassoon, and Parsons introduce MQArgEng, a proof-of-concept pipeline that inserts computational-argumentation semantics as an argumentation-engine plug-in for an LLM.[3] The reported preliminary MT-Bench experiments indicate moderate gains across most topical categories, supporting the premise that an external formal argument layer can improve LLM reasoning or conversation without retraining the base model. The paper is therefore a high-value comparator for compact philosophical context built around explicit argument relations, but its argumentation engine remains generic: it does not encode scholastic objections, distinctions, authorities, determination, or an ordered reply structure, and it should not be treated as a direct implementation of the scholastic method.[3]

4. Sanayei et al. (2025), “Can LLMs Judge Debates? Evaluating Non-Linear Reasoning via Argumentation Theory Semantics”

Sanayei et al. test whether LLMs can approximate Computational Argumentation Theory semantics when judging dialogue-formatted debates.[4] Using QuAD-style argumentative structures, the study asks models to rank or assess arguments without directly receiving the underlying graph and finds that performance worsens as inputs become longer or discourse flow is disrupted, while stronger prompting can partly reduce positional and length biases. Its direct contribution to a scholastic-AI design is methodological: it shows why extracting a compact argument graph or other explicit structure may be preferable to asking an LLM to recover non-linear relations from a long debate transcript. The work evaluates argument acceptability and ranking, not scholastic determination or the ordered resolution of objections.[4]

5. Alfano et al. (2026), “LLM-based Argument Mining meets Argumentation and Description Logics: a Unified Framework for Reasoning about Debates”

Alfano et al. propose a unified pipeline that uses LLM-assisted argument mining to turn raw debate text into a fuzzy argumentative knowledge base, representing arguments as entities with support or attack relations and initial strengths, applying quantitative argumentation semantics, and embedding the resulting structures into fuzzy description logic for query answering.[5] The full framework separates tasks such as argument tagging, entity extraction, quantitative bipolar argumentation framework extraction, and fuzzy argument-box construction, making it the closest recent technical comparator in this set for span classification, explicit relational context, formal propagation, and queryable reasoning. Its novelty threat is nevertheless adjacent rather than direct: the framework models debates and formal support/attack networks, not the scholastic *quaestio*, enumerated objections, distinctions, determination, or reply ordering.[5]

6. Mitsuda, Higashinaka, and Saito (2022), “Combining Argumentation Structure and Language Model for Generating Natural Argumentative Dialogue”

Mitsuda, Higashinaka, and Saito combine a hand-crafted argumentation structure with a language model that rewrites propositions—using keywords and stance information—into the next natural-language utterance while preserving argumentative context.[6] Their manual evaluation reports improved naturalness without loss of argumentation consistency, establishing a clear separation between a structured argumentative skeleton and surface-language realization. That separation parallels a scholastic-AI architecture in which a system first extracts or constructs a deductive/disputational skeleton and only then generates polished philosophical prose. The paper remains an adjacent dialogue-generation precedent: it does not model objections, distinctions, determination, or scholastic replies as explicit protocol stages.[6]

7. Sazid et al. (2022), “A Unified Representation and a Decoupled Deep Learning Architecture for Argumentation Mining of Students’ Persuasive Essays”

Sazid et al. propose a unified representation and decoupled deep-learning architecture for identifying non-argumentative units, premises, claims, major claims, and relations such as support, attack, for, and against in students’ persuasive essays.[7] The paper is relevant because it demonstrates how text spans can be assigned argument roles and converted into a compact relational representation rather than being processed as undifferentiated prose. That capability could support extraction of premises, claims, and counter-considerations from scholastic texts, but the source domain is monologic essay argument mining: it has no dialogue moves, explicit objection/reply pairs, distinction phase, authority handling, or scholastic resolution stage. It is therefore a medium-strength structural precedent rather than a direct novelty threat.[7]

8. Macagno (2021), “Argumentation Schemes in AI: A Literature Review. Introduction to the Special Issue”

Macagno reviews AI research using Walton-style argumentation schemes and characterizes schemes as dialectical instruments with premises, conclusions, and critical questions.[8] The review explains how critical questions can challenge a scheme’s applicability or shift the burden of proof back to a proponent, and it surveys applications spanning argument mining, representation, multi-agent systems, law, policy, healthcare, economics, and education. This is the strongest conceptual bridge in the set between scholastic-style dialectical organization and computational implementation: critical questions, defeasible presumptions, burden shifting, nested arguments, and dialogue protocols function as partial computational analogues of objections and replies. The limitation is decisive for novelty: the article is a literature review of argumentation schemes, not an LLM system and not a full implementation of the medieval *quaestio/disputatio* sequence.[8]

9. Walton (2016), “Some Artificial Intelligence Tools for Argument Evaluation: An Introduction”

Walton surveys AI tools for evaluating defeasible arguments in legal and everyday conversational settings, emphasizing approaches that go beyond deductive logic and Bayesian rules through argument graphs, argumentation schemes, and systems such as Carneades.[9] The paper supplies foundational prior art for using an intermediate argument representation between natural-language discourse and formal evaluation, which is directly relevant to extracting and checking the reasoning skeleton of scholastic texts. It predates current LLMs and does not address automatic extraction of scholastic components, medieval textual organization, or an explicit question–objection–determination–reply protocol. Its relevance is thus foundational and conceptual rather than a direct system-level overlap.[9]

10. Shi et al. (2026), “Tackling the Root of Misinformation by Teaching Laypeople about Logical Fallacies via Socratic Questioning and Critical Argumentation”

Shi et al. introduce LFTutor, an LLM-based tutoring system that combines intent-driven Socratic questioning with critical argumentation to teach users to identify logical fallacies.[10] The study reports automatic and human-evaluation gains over baseline LLMs that lack the pedagogical strategies, showing that question-driven challenge and reflective response can be operationalized as a controlled interaction pattern. This is relevant to a scholastic-AI project as a nearby design precedent for generating objections, eliciting commitments, and prompting replies instead of producing a single unsupported answer. Its method is Socratic pedagogy rather than medieval scholastic disputation, and the opened record does not define a *quaestio*, formal objection list, distinction stage, *respondeo*, or determination data structure.[10]

Synthesis

Across all ten verified papers, the direct duplicate risk remains low: none operationalizes the complete medieval scholastic sequence as an AI/LLM reasoning architecture. The high-risk adjacent convergence is clear, however. Existing work already covers explicit argument representations, support/attack relations, formal argumentation semantics, contestability, debate judging, argument mining, structured dialogue generation, and question-driven critical reasoning. A defensible novelty claim should therefore center on making the scholastic sequence itself a first-class computational object: identify and normalize the question, enumerate objections, distinguish senses or cases, locate authorities and reasons, produce a determination, and attach targeted replies to each objection with provenance.

Sources

[1] https://arxiv.org/abs/2208.06133

[2] https://ojs.aaai.org/index.php/AAAI/article/view/33637

[3] https://arxiv.org/abs/2405.13036

[4] https://arxiv.org/abs/2509.15739

[5] https://arxiv.org/abs/2603.02858

[6] https://aclanthology.org/2022.aacl-short.9

[7] https://aclanthology.org/2022.argmining-1.6

[8] https://philpapers.org/archive/MACASI-11.pdf

[9] https://philpapers.org/rec/WALSAI-4

[10] https://aclanthology.org/2026.acl-long.2209

[11] https://arxiv.org/abs/2405.02079

22 Scholastic Digital Humanities Ai

1. Maliszewski, “It takes a village to write a book: Mapping anonymous contributions in Stephen Langton’s *Quaestiones Theologiae” (2025)

Maliszewski presents a proposed computational study of Stephen Langton’s *Quaestiones Theologiae*, a collection derived from reportationes of oral teaching, to investigate editorial layers and anonymous contributions. The workflow combines transformer-based handwritten-text recognition and transcription alignment with stylometric features such as frequent words, part-of-speech patterns, and pseudo-affixes, and it explicitly compares manually composed material with automatically extracted transcriptions. Its scholarly target is authorship attribution and reconstruction of the collection’s formation, not the recovery or validation of scholastic inferences; HTR and stylometry operate as evidence about textual production below the level of argument reasoning. The paper is therefore a direct precedent for computational analysis of a scholastic reportatio corpus, while remaining a study design whose results depend on transcription quality, corpus size, and the instability of stylometric signals in short questions. Verified record: arXiv:2508.12830 (https://arxiv.org/abs/2508.12830); publisher DOI listed in the source record: 10.1017/chr.2025.10004 (https://doi.org/10.1017/chr.2025.10004).

2. Clérice, Vlachou-Efstathiou, and Chagué, “CREMMA Medii Aevi: Literary Manuscript Text Recognition in Latin” (2023)

CREMMA Medii Aevi contributes a segmentation and handwritten-text-recognition dataset for Medieval Latin literary manuscripts dating from the eleventh through the sixteenth centuries. The paper adds approximately 263,000 characters under guidelines designed to interoperate with earlier Latin and Medieval French HTR resources, with particular attention to abbreviations, and reports models that improve recognition on unknown Latin manuscripts. This is corpus digitization and AI-assisted transcription: the models make manuscript text more accessible, but they do not normalize the text, identify scholastic genre structure, reconstruct a quaestio, or reason over an argument. Its relevance to scholastic studies is infrastructural, since manuscript witnesses can become machine-readable inputs for later linguistic, transmission, or argument-level work; the dataset itself is broader than scholastic disputations. Verified records: DOI 10.5334/johd.97 (https://doi.org/10.5334/johd.97); article PDF (https://malamatenia.github.io/pdfs/cremma-medii-aevi.pdf); Zenodo dataset record (https://zenodo.org/records/7510411).

3. Koch et al., “A Tailored Handwritten-Text-Recognition System for Medieval Latin” (2023)

Koch and colleagues build an end-to-end HTR pipeline for handwritten lemma entries on the Bavarian Medieval Latin Dictionary’s record cards. The system uses image-segmentation models to locate relevant regions and transformer-based vision-encoder/GPT-2-decoder combinations to transcribe them, with extensive data augmentation; the reported best setup achieves a character error rate of 0.015 and is compared with Google Cloud Vision. The contribution is clearly AI transcription for a low-resource historical language rather than AI reasoning: it extracts lexical evidence from images but does not interpret scholastic propositions, represent disputational roles, or test inferences. Because the source material consists of dictionary cards rather than quaestiones or commentaries, its direct relevance is methodological—showing how domain-specific HTR can precede downstream Medieval Latin analysis—and it does not constitute a scholastic argument resource. Verified record: ACL Anthology (https://aclanthology.org/2023.alp-1.12/); paper PDF (https://aclanthology.org/2023.alp-1.12.pdf); arXiv record (https://arxiv.org/abs/2308.09368).

4. McGillivray, Passarotti, and Ruffolo, “The Index Thomisticus Treebank Project: Annotation, Parsing and Valency Lexicon” (2009)

This paper introduces the Index Thomisticus Treebank as a dependency-annotated resource for Medieval Latin texts by Thomas Aquinas and describes its annotation, parsing experiments, and valency lexicon. The project turns a large Thomistic corpus into machine-readable morphosyntactic structures that support supervised parsing and lexical investigation. It is corpus infrastructure and NLP parsing, not AI reasoning: dependency trees identify syntactic relations and valency patterns, but they do not distinguish objections, authorities, *sed contra*, *respondeo*, replies, or the inferential warrants of a scholastic disputation. The treebank is nevertheless a foundational resource for computational work on Aquinas, including texts adjacent to *Sentences* commentary and disputed-question traditions, and it establishes that the relevant bottleneck is not merely access to Latin text but the addition of genre-aware and argument-level annotation. The source file’s author attribution should be corrected: the ACL record lists Barbara McGillivray, Marco Passarotti, and Paolo Ruffolo. Verified record: ACL Anthology (https://aclanthology.org/2009.tal-2.5/).

5. Cecchini, Passarotti, Marongiu, and Zeman, “Challenges in Converting the Index Thomisticus Treebank into Universal Dependencies” (2018)

This paper revises the conversion of the Index Thomisticus Treebank into Universal Dependencies by harmonizing part-of-speech tags, lemmatization, and dependency relations with other Latin treebanks while correcting errors and inconsistencies in the earlier conversion. The authors evaluate the revised process against a manually checked gold standard and report improved conversion quality, especially for labeled attachment. The work concerns annotation normalization and interoperability, not reasoning: UD makes scholastic Latin accessible to broader NLP tools, but a standardized dependency representation is not a historically neutral account of medieval semantics and does not encode the architecture of a *quaestio* or assess argument validity. Its importance is methodological because it demonstrates that even apparently basic preprocessing decisions can materially affect downstream comparisons, retrieval, and parsing. Verified record: ACL Anthology (https://aclanthology.org/W18-6004/); paper PDF (https://aclanthology.org/W18-6004.pdf); Zenodo repository record (https://zenodo.org/records/2552203).

6. Gamba, Passarotti, and Ruffolo, “More Data and New Tools: Advances in Parsing the Index Thomisticus Treebank” (2021)

Gamba, Passarotti, and Ruffolo evaluate how larger training data and newer parser combinations affect automatic dependency parsing of the Index Thomisticus Treebank. The experiments compare configurations including DeSR and neural parsers and examine ensemble behavior on Medieval Latin Aquinas data, showing that parser performance depends on language-specific modeling and that different systems reproduce characteristic errors. This is supervised AI/NLP at the syntactic level, not reasoning over scholastic arguments: improved attachment scores do not show that a model has reconstructed premises, objections, authorities, replies, or inferential force. The paper is useful as a benchmark precedent because it tests modern parsing assumptions against a historically and linguistically distinctive corpus, while leaving open the much harder problem of connecting parses to disputational structure and historically adequate argument interpretation. Verified record: CEUR-WS paper PDF (https://ceur-ws.org/Vol-2989/long_paper20.pdf); Zenodo DOI record (https://doi.org/10.5281/zenodo.5599095).

7. Mambrini, Passarotti, Moretti, and Pellegrini, “The Index Thomisticus Treebank as Linked Data in the LiLa Knowledge Base” (2022)

This LREC paper publishes annotation layers from the Index Thomisticus Treebank as Linguistic Linked Open Data within the LiLa Knowledge Base, linking Medieval Latin corpus tokens and dependency structures to interoperable lexical resources. The contribution is knowledge representation and corpus interoperability: persistent identifiers and explicit links make it possible to query text, morphology, syntax, and lexical information across resources. That infrastructure should not be mistaken for an argument graph or reasoning engine, because linked tokens and lexical relations do not by themselves encode premises, conclusions, objection/reply roles, or validity conditions. The work matters for scholastic AI because provenance-preserving links are a prerequisite for assembling contexts that retain the connection between a model’s input and its textual, lexical, and annotation sources. The source file’s author list should be corrected to the LREC record: Francesco Mambrini, Marco Passarotti, Giovanni Moretti, and Matteo Pellegrini. Verified record: institutional publication record (https://publicatt.unicatt.it/handle/10807/210702); LREC PDF (https://www.lrec-conf.org/proceedings/lrec2022/pdf/2022.lrec-1.428.pdf).

8. Nowak, Zi■bura, Wróbel, and Smywi■ski-Pohl, “eFontes: Part of Speech Tagging and Lemmatization of Medieval Latin Texts: A Cross-Genre Survey” (2024)

The eFontes study develops transformer-based models for Medieval Latin lemmatization, part-of-speech tagging, and morphological-feature prediction using Universal Dependencies resources and a newly developed corpus of Polish Medieval Latin. It evaluates performance across genres and addresses orthographic variation and Latinized vernacular vocabulary, reporting strong but task-specific accuracy. This is AI/NLP annotation rather than reasoning: the models assign linguistic labels to tokens and do not infer the interpretation or argumentative role of a passage in a scholastic question. The paper is relevant because variation in spelling, vocabulary, and genre can propagate errors into search, quotation alignment, semantic modeling, and any later attempt at argument mining. Its results support treating linguistic annotation as an uncertainty-bearing layer rather than as a transparent substitute for historical reading. Verified record: arXiv:2407.00418 (https://arxiv.org/abs/2407.00418).

9. *Corpus Thomisticum* official research infrastructure

The *Corpus Thomisticum* is a long-standing digital research infrastructure providing searchable editions of Aquinas's works, bibliography, research tools, manuscript material, and database functions for words, phrases, quotations, similarities, correlations, and statistics. It is a corpus and retrieval platform rather than a scholarly paper or AI-reasoning system: its search and concordance capabilities enable discovery and comparison, but they do not perform argument-aware retrieval, reconstruct disputational roles, or validate the reasoning of a passage. For scholastic computational research, its significance is foundational because it supplies broad access to Latin texts and manuscript-oriented materials, while also making clear that searchable text is only the first layer of a reasoning-oriented pipeline. Any downstream model must still distinguish editions, witnesses, editorial choices, and passage-level evidence. Verified record: official introduction (https://corpusthomisticum.org/wintroen.html).

10. Rockwell and Passarotti, "The Index Thomisticus as a Digital Humanities Big Data Project" (2019)

Rockwell and Passarotti analyze the Index Thomisticus as a major Digital Humanities infrastructure project, emphasizing its history, scale, data representation, concordance functions, and the project-management and media-archaeological conditions that made large-scale computational work on Aquinas possible. The paper is a history and analysis of digital infrastructure, not an AI model and not a formal account of scholastic reasoning. Its principal contribution to the present topic is to prevent anachronism: computational engagement with Aquinas predates current machine learning and demonstrates that digitization, indexing, and corpus-scale retrieval have their own long history. The article therefore weakens any novelty claim framed as simply "using computation on Aquinas" and redirects novelty toward provenance-preserving, genre-aware argument representation or evaluation. Verified record: DOI 10.6092/issn.2532-8816/8575 (https://doi.org/10.6092/issn.2532-8816/8575); open article PDF (https://umanisticadigitale.unibo.it/article/download/8575/9248).

11. ERC-DEBATE, "Debate: Innovation as Performance in Late-Medieval Universities"

The ERC-DEBATE project studies late-medieval university *principia* as public academic performances and builds a corpus-oriented research program around authorship, transmission, neglected texts, and the intellectual dynamics of debate. The project's CORDIS reporting describes *principia* as structured events combining oral performance, argument, authority, competition, and institutional setting, while the project corpus page frames the recovery and organization of texts across university contexts. This is corpus construction, digital editing, and manuscript/transmission research, not an evaluated AI-reasoning benchmark: identifying or attributing debate texts is different from extracting premises, modeling historically specific argumentative moves, or checking validity. Its importance is that it supplies a debate-centered corpus domain in which future argument-level annotation could be meaningful and provenance-sensitive. Verified records: DEBATE project site (https://debate-erc.com/); corpus page (https://debate-erc.com/corpus/); CORDIS project record (https://cordis.europa.eu/project/id/771589).

12. University of Würzburg, *Arabic and Latin Corpus*

The University of Würzburg's *Arabic and Latin Corpus* catalogs and digitizes Arabic–Latin translations from roughly the tenth through the fourteenth centuries, providing citable digital materials concerning texts, authors, translators, and witnesses. Its primary achievement is transmission infrastructure: it supports access, comparison, and historical study of cross-language movement rather than automatic reasoning about whether an Arabic argument and its Latin translation preserve the same premises, modality, quantification, or warrant. For scholastic computational work, the corpus is especially important because Arabic–Latin translation networks shaped the availability and vocabulary of medieval philosophy and science, yet bilingual alignment alone does not establish inferential equivalence. The resource therefore belongs in the prerequisite corpus layer, with argument-level translation evaluation remaining an open research problem. Verified records: University of Würzburg project page (https://www.uni-wuerzburg.de/en/zpd/projects/); Arabic and Latin Corpus site (https://www.arabic-latin-corpus.philosophie.uni-wuerzburg.de/).

13. UniQ: database of the Prague debate on universals

UniQ is a specialized database developed by the Institute of Philosophy of the Czech Academy of Sciences for the medieval debate on universals at the University of Prague between 1348 and 1500. It interlinks philosophical texts, authors, manuscripts, bibliographic sources, dating, authorship information, digital editions, and manuscript witnesses, and provides separate tools for manuscript and text records using linked-data principles. This is a debate-centered corpus and bibliographic graph, not AI reasoning: its relations describe scholarly and transmission entities rather than extracted premises, objections, replies, or formal validity judgments. UniQ is nevertheless a strong model for provenance-aware organization of a specific scholastic controversy and could serve as an upstream layer for argument annotation or model evaluation. Verified record: Digital Humanities project page (https://digitalhumanities.cz/en/db/uniq-2/).

14. Dutilh Novaes, *Formalizing Medieval Logical Theories: Suppositio, Consequentiae and Obligationes* (2007)

Dutilh Novaes develops historically informed formal reconstructions of medieval theories of supposition, consequence, and obligationes, presenting supposition theory as algorithmic hermeneutics, analyzing consequence with tools related to model theory and two-dimensional semantics, and treating obligationes as logical games. Unlike the other retained items, this book directly addresses formal representations of historical reasoning, but it is a philosophical and interpretive formalization rather than a trained AI system or corpus-scale argument miner. Its central methodological value is its warning that formalization is not neutral transcription: the choice of representation must preserve the conceptual and logical practices of the historical tradition instead of silently replacing them with modern predicate logic. The book supplies the strongest retained precedent for actual reasoning-oriented modeling, while leaving open the alignment of formal structures with manuscript passages, textual variants, competing interpretations, and executable evaluation over large scholastic corpora. Verified record: Springer book page (https://link.springer.com/book/10.1007/978-1-4020-5853-0).

15. Betti and van den Berg (eds.), *Data-Driven History of Ideas* and the e-Ideas program

The data-driven history-of-ideas program represented by Betti and van den Berg combines qualitative scholarship with quantitative and computational methods for tracing the origins, development, and spread of concepts, while emphasizing the need to represent concepts rather than merely count words and to build high-quality multilingual corpora. The related e-Ideas project combines ontology design, NLP-based machine learning, and distributional semantics under the principle that domain expertise must guide bottom-up corpus exploration. This is computational intellectual history and concept analysis, not direct argument reasoning over scholastic quaestiones: concept trajectories, topic distributions, and semantic proximity can reveal patterns but do not by themselves identify the argumentative function of a term inside an objection, authority citation, or *respondeo*. The methodological precedent is valuable for corpus-scale conceptual comparison, but transfer to medieval scholastic texts would require genre-aware segmentation, historical validation, and explicit separation of concept modeling from inference checking. Verified records: Data-Driven History of Ideas collection (https://openhumanitiesdata.metajnl.com/collections/historyofideas); e-Ideas project (https://conceptsinmotion.org/e-ideas/).

Overall finding

Across the fifteen retained sources, the dominant contribution is infrastructure: digitized or searchable corpora, HTR, linguistic annotation, dependency parsing, linked data, authorship/transmission analysis, and debate or manuscript metadata. These systems are essential prerequisites for scholastic AI but should not be described as reasoning systems merely because they use transformers, machine learning, knowledge graphs, or statistical text analysis. Dutilh Novaes provides the clearest retained precedent for formalized historical reasoning, while Maliszewski provides the closest direct computational study of scholastic reportationes; neither supplies an evaluated end-to-end system that takes a medieval quaestio or disputation as input and produces a provenance-preserving, historically validated argument graph with checked inference. The defensible open space is therefore the missing middle between corpus infrastructure and formal reasoning: genre-aware argument-role annotation, explicit provenance and uncertainty, alignment between source passages and alternative formalizations, and evaluation that separates textual fidelity, historical adequacy, argument-form recognition, and logical validity.

23 Disputatio Dialectical Ai

1. Uckelman (2013), “Medieval Disputationes de obligationibus as Formal Dialogue Systems”

Structure: The paper treats a medieval *obligatio* as an alternating dialogue between an Opponent and a Respondent, beginning from a positum and governed by rules for admitting, denying, or responding to propositions. Method: Uckelman reconstructs one medieval obligation theory in Prakken’s formal dialogue-system framework and gives it a Dynamic Epistemic Logic semantics, including dialogue states and conditions for consistency or successful response. Finding: The analysis shows that medieval obligational disputation can be represented as an executable formal dialogue system rather than merely described historically. Limitation: The implementation covers a particular family of obligationes, leaves some proof-system and knowledge assumptions under-specified, and identifies complexity questions for future work; it is not a full Thomistic *quaestio*. Scholastic overlap: This is the closest protocol-level analogue in the set because it directly models regulated opposition, constrained reply, and terminal evaluation, although it lacks the named *sed contra*, *respondeo*, and reply-to-objection stages. Source: https://link.springer.com/article/10.1007/s10503-012-9266-7

2. Maudet, Parsons & Rahwan (2007), “Argumentation in Multi-Agent Systems: Context and Recent Developments”

Structure: This survey organizes computational argumentation around conflicting arguments, their interaction, and dialogue protocols across reasoning and communication tasks. Method: It reviews argumentation-based belief revision, deliberation, planning, persuasion, inquiry, negotiation, and dialogue games, including locutions and commencement, commitment, combination, and termination rules. Finding: The survey argues that argumentation supplies a principled way for agents to reconcile information and reach consistent conclusions, while argument schemes and critical questions provide structured ways to challenge claims. Limitation: It is a selective, explicitly “idiosyncratic” survey rather than a single implemented system, and its general dialogue-game treatment does not prescribe a scholastic answer schema. Scholastic overlap: Its rule-governed proponent/opponent interaction and critical-question machinery map naturally onto objection generation and reply control, while its warning against overly rigid proof dialogues supports flexible handling of alternative scholastic replies. Source: https://web.mit.edu/irahwan/www/docs/argmas2006survey.pdf

3. Castagna, Kökciyan, Sassoon, Parsons & Sklar (2024), “Computational Argumentation-based Chatbots: a Survey”

Structure: The paper classifies argumentation-based chatbots by four functions—argument extraction, reply structuring, argumentation-based reasoning, and reply delivery—and surveys dialogue types such as inquiry, persuasion, deliberation, negotiation, verification, and education. Method: The authors define computational-argumentation chatbot criteria, review systems against abstract and structured argumentation frameworks, argument schemes, critical questions, semantics, and evaluation practices, then synthesize benefits and open problems for generative-LLM chatbots. Finding: Computational argumentation offers a route to more coherent, inspectable, and explainable chatbot responses, but the literature contains relatively few mature systems and evaluations are uneven. Limitation: The survey depends on heterogeneous prior systems, many of which are domain-limited, knowledge-base dependent, or unable to support broad conversational interaction; it does not itself implement or validate a complete disputation engine. Scholastic overlap: Its argument graphs, typed attacks, critical questions, and protocol-based reply structures provide a broad substrate for a machine-readable objection-to-reply pipeline, but the scholastic stages and distinction between objection types remain to be specified. Source: https://arxiv.org/html/2401.03454v1

4. Bolton, Calderwood, Christensen, Kafrouni & Drori (2020), “High Quality Real-Time Structured Debate Generation”

Structure: The system represents a debate as a tree of arguments and debate paths, with stance-based branching and a high-level grammar for multi-turn generation. Method: The authors train and evaluate a framework over a corpus of tree-structured debates annotated with argument metadata; generation is designed to be embedding-model agnostic and is assessed using competitive-debate criteria for style, content, and strategy. Finding: The system produces complex debates in real time at quality judged close to human performance on those metrics, and releases data, models, and code for reproducibility. Limitation: The evaluation does not test whether a debate is complete or covers the most important points, and generated responses can contain unknown-token artifacts that may bias human judgments; the structure is stance-oriented rather than semantically typed. Scholastic overlap: Debate trees and paths are a strong structural analogue for a root question with nested objections and replies, but the system lacks explicit *sed contra*, *respondeo*, role semantics, and per-objection resolution. Source: https://arxiv.org/pdf/2012.00209

5. Kargupta, Agarwal, August & Han (2025), “Tree-of-Debate: Multi-Persona Debate Trees Elicit Critical Thinking for Scientific Comparative Analysis”

Structure: Tree-of-Debate turns papers into specialized LLM personas and expands a dynamically branching debate tree whose nodes compare contributions, similarities, and differences before synthesis. Method: The authors combine segment-level retrieval, persona construction, self-deliberation, adaptive debate-depth expansion, and debate synthesis, then compare against baselines on scientific comparative-analysis datasets using domain-expert and human evaluations. Finding: Structured debate substantially improves contextualization—reported at an average 95.21% versus 75.57% for the strongest baseline—and supports richer comparative summaries, though gains vary by domain and depth. Limitation: Factuality can decline in domains with less representation in model pretraining, and the approach remains dependent on LLM retrieval, persona quality, and debate-depth choices; its target is literature comparison rather than proposition-level truth adjudication. Scholastic overlap: Specialized disputant roles and dynamic branching resemble assigned participants and expanding objection/counter-objection trees, but the system does not impose a canonical *quaestio* protocol or a principled *respondeo*. Source: https://aclanthology.org/2025.acl-long.1422/ ; https://arxiv.org/html/2502.14767v2

6. Vasileiou, Kumar, Yeoh, Son & Toni (2024), “Dialectical Reconciliation via Structured Argumentative Dialogues”

Structure: DR-Arg models an explainer agent and an explainee whose knowledge differs, using structured arguments and counterarguments in a dialogue that terminates when the explainee understands the relevant decision model. Method: The paper gives operational semantics and theoretical guarantees for termination and success, then evaluates the framework through computational experiments and human-subject studies. Finding: Argumentative dialogue is presented as a viable way to reconcile knowledge discrepancies and improve human understanding of AI decisions, extending model-reconciliation approaches beyond single-shot explanations and assumed human models. Limitation: The framework is aimed at explainability and convergence on an AI decision, not at open-ended truth-seeking or historical scholastic disputation; its understanding metric and domain assumptions constrain generalization. Scholastic overlap: It is close to an explanatory *respondeo*: the explainer gives reasons, addresses counterarguments, and seeks a satisfactory reconciliation, but it does not explicitly separate objections, *sed contra*, main determination, and individual replies. Source: https://proceedings.kr.org/2024/73/kr2024-0073-vasileiou-et-al.pdf

7. Faugier, Armetta, Bonifati & Yun (2024), “Assisted Debate Builder with Large Language Models”

Structure: ADBL2 represents debates as bipolar argumentation trees whose edges encode support or attack, while preserving an editable, inspectable tree artifact. Method: The authors build an open-source debate-builder interface, use relation-based argument mining to verify existing relations and propose new arguments, and fine-tune a Mistral-7B model for relation classification across domains. Finding: The reported fine-tuned model reaches 90.59% overall F1 and is described as outperforming prior approaches, making it useful for semi-automated construction and correction of debate graphs. Limitation: The paper is a demonstration and relation-mining system rather than a complete dialogue manager; support/attack labels do not by themselves capture distinctions such as undercutting, qualification, distinction, or reply-to-reply, and performance depends on the available datasets and domains. Scholastic overlap: ADBL2 supplies the graph-construction layer needed to turn textual objections and replies into linked, reviewable structures, but it does not decide the scholastic role or discharge status of a move. Source: https://arxiv.org/abs/2405.13015

8. Du, Li, Torralba, Tenenbaum & Mordatch (2023), “Improving Factuality and Reasoning in Language Models through Multiagent Debate”

Structure: Multiple instances of a language model independently propose answers and reasoning, debate them over several rounds, and converge on a common final answer. Method: The authors apply a shared debate procedure and prompts to black-box models across mathematical, strategic, and factuality-oriented tasks, comparing multiagent debate with single-model baselines. Finding: The paper reports improved mathematical and strategic reasoning and better factual validity, including fewer fallacious answers and hallucinations. Limitation: The protocol is primarily prompt-defined, and the work does not provide explicit argument types, provenance links, formal attack semantics, or a proof that the final consensus correctly resolves every objection; “debate” can therefore function as persuasive aggregation rather than inspectable disputation. Scholastic overlap: Its rounds of opposition followed by synthesis resemble objection, response, and judgment procedurally, but it lacks a stable question-centered schema and a reasoned distinction between main answer and replies. Source: https://arxiv.org/abs/2305.14325

9. Khan et al. (2024), “Debating with More Persuasive LLMs Leads to More Truthful Answers”

Structure: Two stronger LLM debaters argue for opposing answers while a weaker model or human judge selects between them. Method: The study evaluates debate as a way for non-experts to assess expert-level answers and additionally optimizes debaters for persuasiveness without direct ground-truth labels. Finding: Debate improves answer selection, with the abstract reporting 76% accuracy for non-expert models and 88% for humans versus naive baselines of 48% and 60%; unsupervised persuasion optimization further improves truth identification. Limitation: The setup fixes two opposing sides and delegates resolution to a judge, so it does not require the Respondent to integrate objections into a principled determination; performance also depends on judge competence and persuasive asymmetries. Scholastic overlap: The paper operationalizes adversarial objection as a truth-seeking aid before judgment, but its winner-selection format is only a procedural near-analogue to disputation and lacks formal replies, distinctions, and authority handling. Source: https://arxiv.org/abs/2402.06782

10. Hadoux & Hunter (2017), “Strategic Sequences of Arguments for Persuasion Using Decision Trees”

Structure: A persuasion dialogue is represented as a decision tree containing possible moves, chance or uncertainty, belief updates, and terminal valuations. Method: The authors transform persuasion into a decision problem, define update and valuation functions, compare established decision rules for selecting argument sequences, and analyze their behavior under uncertainty about the persuadee’s model. Finding: The framework shows how decision rules can prioritize moves strategically; in the reported experiments, MaxiMax performs best when the persuadee model is accurate, while rule performance changes as model error grows. Limitation: The objective is persuasive utility rather than correctness or principled discharge of objections, and practical use requires a suitable valuation function, update function, and model of the interlocutor. Scholastic overlap: It offers a planning mechanism for choosing which objection, critical question, or reply to present next, but its optimization target must be replaced or constrained by scholastic relevance and argumentative validity. Source: https://ojs.aaai.org/index.php/AAAI/article/view/10673/10532

11. Gorur, Rago & Toni (2024), “Can Large Language Models perform Relation-based Argument Mining?”

Structure: The paper treats relation-based argument mining as classification of relations among argument components, principally support and attack, as a prerequisite for building argument structures. Method: It evaluates prompted general-purpose and quantized open-source LLMs on ten datasets and compares them with a RoBERTa baseline. Finding: The study reports that Llama 70B-4bit and Mixtral 8x7B-4bit surpass the RoBERTa baseline, with Llama 70B-4bit performing better at the cost of slower inference and greater GPU requirements. Limitation: The main experiments use a binary support/attack task even though practical applications often require a ternary no-relation label; datasets are English-only, models are constrained by hardware, and the method uses prompting rather than fine-tuning. Scholastic overlap: This is enabling infrastructure for a disputation parser because a system must identify whether a passage supports, attacks, or relates to another, but the coarse relation inventory does not yet distinguish objection, rebuttal, undercutter, qualification, or reply. Source: https://arxiv.org/html/2402.11243

12. Peldszus & Stede (2019), “Determining Relative Argument Specificity and Stance for Complex Argumentative Structures”

Structure: The paper studies manually curated argument trees for 741 controversial topics, with 95,312 claims and generally two-to-six levels of depth, focusing on stance and relative specificity within complex structures. Method: It compares feature-based logistic regression, bag-of-words models, and fine-tuned BERT under controls for path distance, parent-child relation, and stance similarity. Finding: Claim length is strongly predictive of specificity; fine-tuned BERT performs best overall, while stance is easier for parent-child pairs and specificity becomes harder when claims are linguistically similar or evaluated under tighter controls. Limitation: The dataset and task address classification rather than dialogue generation or reply validation, and the authors leave integration into coherent argument generation for future work. Scholastic overlap: Specificity and stance modeling can help a scholastic system determine whether an objection targets the whole conclusion, a particular premise, or a narrower inference, supporting typed replies and distinctions rather than flat pro/con labeling. Source: https://aclanthology.org/P19-1456.pdf

24 Islamic Comparative Scholastic Ai

1. Hodges, “Ibn S■n■ on Analysis: 1. Proof Search. Or: Abstract State Machines as a Tool for History of Logic”

Hodges studies Avicenna’s discussion of analysis in *Qiy■s* as a historical proof-search procedure for syllogisms, representing its states and transitions with an Abstract State Machine. The evidence is a close reading, translation, and formal reconstruction of the Arabic logical passage rather than a machine-learning experiment or benchmark. Its main limitation is scope: it formalizes one logical procedure and does not model the semantics, rhetoric, or social context of broader kal■m or scholastic debate. For this project, it is the strongest direct precedent for converting a medieval Arabic reasoning procedure into an executable, inspectable computational representation, especially a target representation for premises, transformations, and conclusions. [1]

2. Hodges, “A Correctness Proof for al-Barak■t’s Logical Diagrams”

This article reconstructs Ab■ al-Barak■t’s twelfth-century diagrammatic method for finding conclusions of premise-pairs and identifying pairs with no conclusion, then gives a rigorous proof that the procedure is correct. The computational method is symbolic reconstruction and correctness proof, not NLP: the author reverse-engineers a procedure that the medieval source presents without instructions or proof. The evidence is the logical behavior of the reconstructed diagrams and the proof that the resulting outputs match the intended syllogistic consequences. The limitation is that the system remains a narrow formal treatment of syllogistic inference and does not address historical language processing, argument identification, or theological disagreement. It matters here as a model for translating Arabic logical artifacts into verifiable symbolic-AI components rather than treating them as opaque historical prose. [2]

3. Abdelwahab, Daghbouche, and Shannan, “The Algorithm of Islamic Jurisprudence (Fiqh) with Validation of an Entscheidungsproblem”

This paper proposes a formal account of *u■■l al-fiqh* by expressing selected jurisprudential questions through mathematical logic, a logic-query language, and a generic algorithm intended to decide legal rulings. Its method is conceptual formalization using subsets of first-order logic, with discussion of completeness, decidability, complexity, and an Islamic-law Entscheidungsproblem. The evidence is the paper’s definitions, worked methodological distinctions, and proposed formal architecture, not a released implementation, corpus, or empirical evaluation. Its limitation is substantial: broad claims about complete algorithmic coverage and decidability are programmatic and should not be treated as validated performance results; it also concerns jurisprudence rather than kal■m proper. It is relevant as an adjacent example of rule-based knowledge representation for Islamic scholastic reasoning and as a caution against confusing formal specification with demonstrated historical or legal competence. [3]

4. Abdelwahab et al., “Computational Extraction of Legal Causes via al-■abr wa al-Taqs■m: A Set-Theoretic Formalization for Closed Fiqh Chapters”

This paper formalizes the classical u■■l al-fiqh technique of *al-■abr wa al-taqs■m*—examination and division—as a set-theoretic procedure for extracting candidate legal causes within closed jurisprudential chapters. The computational method maps the traditional elimination and classification process into sets, predicates, and operations that can in principle support rule-based extraction. The evidence is the explicit mathematical reconstruction of the method and its application to legal-cause analysis, rather than a trained NLP model or benchmarked information-extraction system. Its limitation is the closed-domain and highly formal setting: results do not establish robustness on varied Arabic legal prose, contested interpretations, or theological argument. For this project, it demonstrates how a historically situated scholastic technique can become an executable intermediate layer between textual evidence and an argument graph, while confirming that current formalization work is concentrated in fiqh rather than comparative kal■m. [4]

5. Maliszewski, “It Takes a Village to Write a Book: Mapping Anonymous Contributions in Stephen Langton’s *Quaestiones Theologiae*”

Maliszewski presents a registered-report protocol for analyzing collaborative production and editorial layers in Stephen Langton’s medieval Christian theological *quaestiones*, which derive from *reportationes* of oral teaching. The method combines handwritten-text recognition, automated transcription alignment, stylometry based on frequent words, part-of-speech sequences, pseudo-affixes, and unsupervised clustering. The evidence includes exploratory principal-component analyses separating Langton’s corpus from other scholastic collections and revealing internal stylistic diversity, while the proposed study compares manually edited with HTR-extracted data. Its chief limitation is that short *quaestiones* provide weak stylometric samples and the work primarily identifies authorship and transmission layers, not argumentative structure or doctrinal validity. It is directly useful here as a Christian scholastic preprocessing and corpus-method precedent: reliable modeling of disputation requires first separating reportators, editors, witnesses, and textual layers. [5][6]

6. Maliszewski, published Cambridge version of “It Takes a Village to Write a Book”

The Cambridge-published version confirms the same registered-report study and supplies the bibliographic and publication record for the Langton investigation: the corpus contains multiple versions of many *quaestiones*, and the proposed workflow compares stylometry on manually edited text with transformer-based HTR output. Its computational method remains function-word and other stylometric features, POS sequences, pseudo-affixes, clustering, and transcription alignment, while the evidence is the article’s exploratory PCA and corpus statistics rather than a completed argument-mining evaluation. The limitation is unchanged: short texts weaken authorial-style inference, and authorship or editorial-layer detection does not recover the semantic structure of scholastic objections and replies. The published record is relevant to this project as a verified, reusable computational-humanities protocol for corpus formation and transmission-aware preprocessing; it should be treated as the same study as the open preprint, not as independent corroboration. [6]

7. Lange et al., “Text Mining Islamic Law”

This article surveys and demonstrates the use of digital-humanities and text-mining methods in Islamic legal studies, including corpus construction, computational comparison, and analysis of Qur’anic and legal materials across schools of law. Its method is empirical text mining combined with historical and legal interpretation, using machine-readable corpora to test or refine claims in the secondary literature. The evidence is the article’s comparative computational analyses, which in some cases confirm scholarly claims and in others correct them. Its limitation is disciplinary and representational: text mining can reveal distributional and textual patterns, but it does not by itself recover the full force of legal reasoning, authority, intent, or dialectical exchange. The relation to this project is infrastructural and methodological: it shows that computational Islamic studies can be historically meaningful when corpus design and domain interpretation remain explicit, while underscoring the lack of mature argument-mining resources for kal■m. [7]

8. OpenITI, Open Islamicate Texts Initiative

OpenITI is a machine-actionable corpus initiative for premodern Arabic and other Islamicate texts rather than a single research paper or AI model. Its computational contribution is corpus encoding, normalization, metadata, versioning, and machine-readable text organization designed to support search, quantitative analysis, authorship study, terminology extraction, and future NLP. The evidence is the initiative’s description of its corpus goals, scholarly data model, and coverage of the written Arabic and Islamicate heritage. Its limitations include uneven edition alignment, OCR and transcription noise, incomplete genre and author metadata, and the need for expert validation before using texts as gold-standard training data. For this project, OpenITI is the principal enabling infrastructure for assembling comparable corpora across kal■m, falsafa, fiqh, Jewish-Arabic philosophy, and related traditions; it is a prerequisite resource, not evidence that comparative scholastic argument mining has already been solved. [8]

9. Yousef et al., “Enhancing State-of-the-Art NLP Models for Classical Arabic”

Yousef and colleagues develop Classical-Arabic NLP resources and fine-tune deep-learning models for named-entity recognition, person-relationship classification, toponym classification, onomastic-section detection, onomastic-element classification, and date recognition. The method uses a large OpenITI-derived subcorpus, active learning, expert annotation, iterative correction, and fine-tuning of Classical-Arabic transformer models. The evidence includes reported improvements such as fine-tuned NER F1 scores around 94% for persons and toponyms and an onomastic-boundary F1 around 88%, alongside a corpus of roughly 71 million tokens and hundreds of thousands of biographies. Limitations include the in-progress status, task-specific biography domain, sparse expert labels, weaker performance for underrepresented classes and toponym relations, and no direct modeling of premises or rebuttals. It supplies the language-processing layer needed before extracting authorities, participants, and textual relations from historical Islamic scholastic works. [9]

10. Hawwari, Diab, and Zaghouani, “A Pilot PropBank Annotation for Quranic Arabic”

This paper creates a pilot Quranic Arabic PropBank by assigning predicate-argument structures and semantic roles to the 50 most frequent verbs in the Quranic Arabic Dependency Treebank. The computational method combines dependency syntax, manually created verb frames, Arabic root-meaning references, double annotation, and adjudication; the pilot covers 7,227 verbal occurrences and is intended eventually to extend to all 1,466 Quranic verb types. The evidence is the feasibility of constructing frame files and the observed sense differences between Quranic Arabic and modern newswire Arabic, including greater ambiguity in the Quranic resource. Its limitation is coverage: the pilot annotates only 50 verbs and semantic roles are not equivalent to discourse-level argumentation, doctrinal inference, or intertextual authority. For this project, it is a foundational semantic layer for distinguishing linguistic predicate structure from later theological interpretation and for building Arabic resources that can feed argument-mining pipelines. [10]

11. Altammami and Atwell, “Challenging the Transformer-based Models with a Classical Arabic Dataset: Quran and Hadith”

This study evaluates monolingual, bilingual, and multilingual transformer models on semantic relatedness between Quran verses and Hadith teachings, using a deliberately difficult Classical-Arabic dataset assembled with expert consultation. The method trains or fine-tunes models on Quran–Quran relatedness data and tests generalization on Quran–Hadith pairs, reducing the risk that random splits merely measure memorization. The evidence is the dataset-construction methodology, cross-domain evaluation, and the reported finding that current models struggle with the complex semantics, orthography, rhetoric, and implicit meanings of Quranic and Hadith texts. Its limitation is task scope: binary semantic relatedness is not structured argument mining, and performance depends on expert pair construction rather than a complete model of religious reasoning. It provides a useful evaluation warning for this project: strong results on modern Arabic or in-domain splits do not establish competence with historically situated scholastic language or reasoning. [11]

12. Alnefaie, Atwell, and Alsalka, “HAQA and QUQA: Constructing Two Arabic Question-Answering Corpora for the Quran and Hadith”

This paper constructs two reusable Arabic QA corpora: HAQA for Hadith and QUQA for Quran questions, containing 1,598 and 3,382 question–answer pairs respectively. The method integrates existing datasets with expert-authored books, OCR, manual review, automated cleaning, metadata construction, and source/evidence fields distinguishing expert commentary from answer-bearing Quranic or Hadith passages. The evidence is the released corpus design, source-selection process, explicit credibility metadata, and stated coverage and size. Its limitations are that QA pairs are not annotated for argumentative roles, legal-school disagreement, or dialectical relations, and the mixture of source text and expert explanation can complicate faithfulness evaluation. It is relevant as a nearby benchmark and retrieval resource for Islamic textual systems, especially for testing citation grounding and answer extraction before attempting historically faithful modeling of kal■m debates. [12]

13. “Computational Linguistic Analysis of the Biblical Text”

This book chapter surveys computational-linguistic approaches to biblical languages and textual analysis, including corpus methods, linguistic theories, databases, and computational representations of Biblical Hebrew and related scriptural languages. Its method is synthetic and methodological rather than a single new model: it compares how different linguistic frameworks shape computational analysis of the Biblical text. The evidence is the chapter’s discussion of existing computational resources, theories, and applications across biblical-language scholarship. Its limitation is that it provides enabling infrastructure and methodological orientation, not an annotated corpus or system for medieval Jewish scholastic argument, Christian *quaestiones*, or comparative theological debate. For this project, it supplies the comparative-tradition bridge: Hebrew and biblical NLP can support historical textual normalization and semantic analysis, but those capabilities must not be mistaken for computational modeling of Jewish scholastic reasoning. [13]

14. Tsarfaty et al., “What’s Wrong with Hebrew NLP? And How to Make it Right”

This paper presents ONLP/YAP, a joint morpho-syntactic parsing framework for Modern Hebrew designed to avoid error propagation in morphologically ambiguous Semitic text. The computational method jointly searches over morphological segmentation, disambiguation, part-of-speech and feature assignment, lemmas, and dependency trees using structured prediction and beam search rather than a brittle pipeline. The evidence is the system architecture, open implementation and output layers, and reported improvements over competing Hebrew models, with support for lattices and out-of-domain analysis. Its limitation is domain and task mismatch: it targets Modern Hebrew language processing and does not analyze medieval Jewish philosophy, rabbinic disputation, or arguments; performance also depends on lexicon coverage and annotated data. It is relevant as transferable infrastructure for Jewish scholastic corpora, where historically sensitive segmentation and syntactic analysis are prerequisites for reliable semantic and argument-level extraction. [14]

15. Mirarab and DarestaniFarahani, “Explainable Large Language Model for Islamic and Humanities Sciences Based on Knowledge Graphs”

This article proposes a conceptual framework integrating knowledge graphs with large language models for Islamic studies and the humanities, emphasizing explainability, conceptual inference, and representation of complex relations. The method is an analytical study based on a systematic literature review and purposeful selection of existing knowledge-graph and LLM resources; its stated output is a framework rather than a trained system. The evidence is the article’s explicit architecture, metadata, and claims that graph representations can organize concepts while LLMs extract and process natural-language information. Its central limitation is evidentiary: no publicly reported benchmark, corpus, ablation, expert agreement study, or historical kal■m evaluation demonstrates that the proposed framework works in practice. It is nevertheless directly relevant as a design hypothesis for an ontology of doctrines, authorities, objections, distinctions, and inferential relations, provided the project treats it as a proposal rather than validated evidence. [15]

16. Mouhoub, “Islamic Large Language Models: From Knowledge Acquisition to Trustworthy and Hallucination-Resistant AI”

Mouhoub surveys Islamic LLMs and trustworthy Islamic AI across Arabic NLP, Islamic corpora, Quranic QA, knowledge benchmarks, retrieval-augmented generation, legal reasoning, inheritance reasoning, hallucination evaluation, and citation-aware generation. The method is a structured literature survey and research agenda organized around source hierarchy, exact citation verification, Classical-Arabic variation, madhhab-aware disagreement, retrieval, and expert evaluation. The evidence is the synthesis of existing models, resources, and benchmarks, together with the argument that fluency is insufficient without grounded sources and verifiable reasoning. Its limitation is that it is a survey rather than a new empirical benchmark or system, and its broad field map may include emerging resources whose comparative validity remains unsettled. For this project, it provides the clearest current specification of evaluation requirements: source-grounded extraction, explicit uncertainty and disagreement, citation accuracy, and tests against fabricated authorities and anachronistic reasoning. [16]

17. Lawrence and Reed, “Argument Mining: A Survey”

Lawrence and Reed define argument mining as the automatic identification and extraction of inferential structure expressed in natural language and review methods for detecting argumentative components and relations. The computational method surveyed includes discourse analysis, machine learning, structured prediction, component classification, relation identification, and increasingly deeper modeling of reasoning structure. The evidence is a broad synthesis of argument-mining tasks, datasets, techniques, and unresolved challenges across domains rather than a domain-specific Islamic or scholastic experiment. Its limitation is transfer: generic argument-mining categories and benchmarks do not automatically capture scholastic distinctions, authority relations, objections, replies, modality, genre conventions, or multilingual historical variation. It is the methodological anchor for this project because it defines the target capability, but it must be combined with expert-annotated corpora and historically grounded schemas for kal■m, Jewish philosophy, and Christian *quaestiones*. [17]

18. Verification and project-level finding

The 17 retained items were checked against the supplied URLs and accessible publisher, repository, ACL, arXiv, Cambridge, Brill, OpenITI, and author-page records; the Hodges proof-search paper was additionally cross-checked against an accessible mirror because the retained author PDF did not download in the local manifest, and the chapter/PDF records with OCR restrictions were verified through bibliographic or publisher metadata. The set contains four symbolic/formalization studies, one computational Christian-scholastic protocol, Islamic-law text-mining and corpus infrastructure, Classical-Arabic and Hebrew NLP resources, Quran/Hadith QA and semantic datasets, one Islamic knowledge-graph/LLM proposal, a current Islamic-LLM survey, and a general argument-mining survey. Collectively, the evidence supports a sparse-and-fragmented literature finding: the strongest computational precedents formalize procedures or build enabling corpora, while no retained source provides a mature, openly benchmarked, multilingual AI system for historically faithful comparative modeling of kal■m, Jewish scholastic argument, and Christian scholastic *quaestiones*.

Sources

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[16] https://arxiv.org/abs/2606.16629

[17] https://aclanthology.org/J19-4006.pdf

25 Ai Philosophy Dialectical Methods

1. **Murray Shanahan, “Talking About Large Language Models” (2022/2023).** Contribution: this conceptual paper argues that LLMs should be understood as simulators of language-conditioned behavior rather than casually anthropomorphized as agents with stable beliefs, knowledge, or intentions; its central evidence is a technical-philosophical account of how fluent language invites misleading human interpretations. Limitation: it offers no generation protocol, philosophical benchmark, or empirical test of scholastic reasoning. Direct novelty threat: it weakens any claim that a system has newly created an “AI philosopher” merely because it produces convincing philosophical discourse, but it does not pre-empt a typed disputation engine. Evaluation relevance: separate role or voice consistency from argument validity and test prompt paraphrase, persona reversal, and adversarial framing controls. Source: arXiv:2212.03551 (https://arxiv.org/abs/2212.03551).

2. **Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo, “Are Emergent Abilities of Large Language Models a Mirage?” (2023).** Contribution: the paper shows mathematically and empirically that apparently discontinuous scaling abilities can be produced by nonlinear metrics, thresholding, and reporting choices rather than by a genuine capability transition; it supports this with analyses of InstructGPT/GPT-3 tasks, BIG-Bench, and vision models. Limitation: it studies general scaling claims, not philosophical reasoning or structured dialogue, and does not prescribe a scholastic evaluation rubric. Direct novelty threat: it directly challenges broad claims that a prompt protocol “creates reasoning” based only on a jump in final-answer accuracy or debate win rate. Evaluation relevance: use continuous obligation-level scores, confidence intervals, calibration, metric-sensitivity analyses, and budget-matched comparisons rather than relying on coarse aggregate success. Source: arXiv:2304.15004 (https://arxiv.org/abs/2304.15004).

3. **Eric Schwitzgebel, David Schwitzgebel, and Anna Strasser, “Creating a Large Language Model of a Philosopher” (2023).** Contribution: the authors fine-tune GPT-3 on Daniel Dennett’s writings, ask Dennett and the model the same ten questions, and collect 425 judgments about which answers were human-authored; experts identified Dennett’s responses at 51%, above the 20% chance level but below the preregistered 80% expectation, while ordinary participants were near chance and some model answers were preferred. Limitation: the experiment measures stylistic or positional indistinguishability in a small question set, not validity, provenance, objection coverage, or dialectical structure. Direct novelty threat: it pre-empts broad claims that generating recognizable philosophical positions or passing philosopher-authorship discrimination is itself novel. Evaluation relevance: add blinded expert discrimination, position fidelity, question-level variance, historical attribution checks, and separate style imitation from argument quality. Source: arXiv:2302.01339 (https://arxiv.org/abs/2302.01339).

4. **Hangfeng He, Hongming Zhang, and Dan Roth, “SocREval: Large Language Models with the Socratic Method for Reference-free Reasoning Evaluation” (Findings of NAACL 2024).** Contribution: SocREval uses a Socratic-method-inspired prompt to have GPT-4 evaluate reasoning chains without requiring a gold reasoning chain, and reports improved performance over reference-free and reference-based metrics on four human-annotated datasets, with analyses of cost, prompt robustness, and example selection. Limitation: the method evaluates reasoning-chain quality rather than generating a philosophical disputation, and its questions do not encode *objection*, *sed contra*, determination, distinction, or reply roles. Direct novelty threat: it is a direct precedent for sequential interrogation of intermediate reasoning, so novelty cannot rest on Socratic questioning alone. Evaluation relevance: adapt its staged questioning idea to audit each scholastic obligation, but validate the evaluator against expert annotations and test judge bias and self-preference. Source: ACL Anthology (https://aclanthology.org/2024.findings-naacl.175/).

5. **Shunyu Yao et al., “Tree of Thoughts: Deliberate Problem Solving with Large Language Models” (2023).** Contribution: Tree of Thoughts generalizes chain-of-thought into branching units of thought with self-evaluation, lookahead, pruning, and backtracking; on Game of 24, creative writing, and mini-crosswords, it demonstrates that explicit search over intermediate states can substantially improve task performance, including a reported GPT-4 increase from 4% to 74% on Game of 24. Limitation: the tasks are puzzles and planning problems, and the framework has no philosophical ontology, authority handling, or expert evaluation of distinctions and replies. Direct novelty threat: it pre-empts claims that branching intermediate reasoning or search is novel. Evaluation relevance: compare semantic scholastic branches against generic ToT under equal model-call, token, latency, and retrieval budgets, and measure whether objections are distinct, relevant, and resolved rather than merely alternative thoughts. Source: arXiv:2305.10601 (https://arxiv.org/abs/2305.10601).

6. **Yilun Du et al., “Improving Factuality and Reasoning in Language Models through Multiagent Debate” (2023).** Contribution: the paper has multiple instances independently propose answers, debate their reasoning over several rounds, and converge on a final answer; it reports gains in mathematical and strategic reasoning and reduced factual errors and hallucinations across evaluated tasks. Limitation: agents are not assigned scholastic semantic types, and the method does not require a fixed question, explicit contrary authority, determination, or one reply mapped to every objection. Direct novelty

threat: it is the closest established threat to claims that iterative objection/rebuttal exchange itself improves reasoning. Evaluation relevance: include it as a primary baseline and measure objection coverage, correction rate, reply alignment, final accuracy, and marginal quality per model call. Source: arXiv:2305.14325 (https://arxiv.org/abs/2305.14325).

7. **Geoffrey Irving, Paul Christiano, and Dario Amodei, "AI Safety via Debate" (2018).** Contribution: this foundational proposal formulates two-agent, turn-taking debate with a human or overseer judging which side provides more truthful and useful information, gives a complexity-theoretic motivation, and reports an initial MNIST experiment in which debate improved classifier accuracy. Limitation: its objective is scalable oversight and winner selection, not clarification through a fixed philosophical question structure, and its initial empirical setting is narrow. Direct novelty threat: it pre-empts generic claims about adversarial two-sided debate, human judging, or exposing flaws through contestation. Evaluation relevance: borrow judge-reliability, hidden-information, and adversarial-robustness tests while explicitly checking that debate victory is not being conflated with truth, validity, or scholastic compliance. Source: arXiv:1805.00899 (https://arxiv.org/abs/1805.00899).

8. **Chi-Min Chan et al., "ChatEval: Towards Better LLM-based Evaluators through Multi-Agent Debate" (ICLR 2024).** Contribution: ChatEval builds a multi-agent referee team with role-diverse prompts that debates candidate texts and aggregates the judgments; experiments on two benchmarks report improved accuracy and correlation with human assessment, and show that diverse roles matter more than repeating one role. Limitation: the target is preference or quality agreement, so the system may reward fluency, persuasion, or style without verifying individual premises and inferences. Direct novelty threat: it directly challenges novelty claims about role-specialized evaluator debate or multi-agent LLM judging. Evaluation relevance: compare against role-diverse judging, report criterion-level human agreement, and add adversarial cases such as fluent invalid arguments, verbosity bias, duplicated objections, and fabricated authorities. Source: OpenReview (https://openreview.net/forum?id=FQepisCUWu).

9. **Tian Liang et al., "Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate" (2023; ICLR-era work).** Contribution: this paper identifies a degeneration-of-thought problem in iterative self-reflection and proposes multi-agent debate in which agents exchange opposing arguments under a judge, reporting improvements on commonsense machine translation and counter-intuitive arithmetic; it also finds that adaptive stopping and moderate debate depth matter and that mixed-model judging can be unfair. Limitation: the method seeks diversity and better answers but does not define scholastic roles or require authority, distinctions, determination, or objection-specific replies. Direct novelty threat: it pre-empts claims that a system's novelty lies simply in generating divergent objections through multiple agents. Evaluation relevance: score objection diversity, non-redundancy, relevance, and adversarial strength separately, and test judge fairness when agents or judges differ by model family. Source: arXiv:2305.19118 (https://arxiv.org/abs/2305.19118).

10. **Yunxuan Li et al., "Improving Multi-Agent Debate with Sparse Communication Topology" (Findings of EMNLP 2024).** Contribution: the authors study how communication connectivity affects multi-agent debate and report that sparse topologies can match or outperform fully connected debate while reducing computational cost on GPT and Mistral models; they also extend the setup to multimodal reasoning and alignment labeling. Limitation: topology is treated as a systems variable, not as a semantic representation of objection, authority, distinction, or reply, and the benchmarks are not philosophical. Direct novelty threat: it weakens claims that specialized communication channels or agent organization are novel without a semantic protocol. Evaluation relevance: use sparse-topology debate as an efficiency baseline and compare channels by proof-relevant role, coverage, latency, token cost, and quality per call. Source: ACL Anthology (https://aclanthology.org/2024.findings-emnlp.427/).

11. **Reza Sanayei, Srdjan Vesic, Eduardo Blanco, and Mihai Surdeanu, "Can LLMs Judge Debates? Evaluating Non-Linear Reasoning via Argumentation Theory Semantics" (Findings of EMNLP 2025).** Contribution: the paper converts natural-language debates into argument-graph problems and tests whether LLMs can approximate Quantitative Argumentation Debate semantics, which assigns acceptability from attack and support relations; models show moderate alignment but degrade with longer inputs or disrupted discourse flow, while advanced prompting reduces length and position biases. Limitation: the graph lacks scholastic distinctions among authority, determination, and replies, and the models receive dialogue rather than the underlying graph. Direct novelty threat: it is a direct precedent for graph-level, non-linear debate evaluation and therefore limits claims that argument-relation auditing is unprecedented. Evaluation relevance: add attack/support consistency, graph-level acceptability, length and order perturbations, and graph-aware versus dialogue-only ablations to objection–reply evaluation. Source: ACL Anthology (https://aclanthology.org/2025.findings-emnlp.1159/).

12. **Weiqi Wang and Yangqiu Song, "MARS: Benchmarking the Metaphysical Reasoning Abilities of Language Models with a Multi-task Evaluation Dataset" (2024).** Contribution: MARS formulates metaphysical reasoning as a three-step discriminative process concerning action changes, resulting state changes, and situational transitions, then evaluates 20 language models and finds all three tasks difficult even for strong systems; conceptualization-taxonomy

pretraining is reported as a possible aid. Limitation: MARS is a domain benchmark rather than a dialectical generator, and its tasks do not evaluate objections, authorities, distinctions, determinations, or replies. Direct novelty threat: it pre-empts claims that philosophy-specific reasoning evaluation is novel merely because generic benchmarks are replaced with philosophical content. Evaluation relevance: use its task-family design as a model for philosophy-specific datasets, then add expert-labeled disputation stages, counterfactuals, and proof obligations. Source: arXiv:2406.02106 (https://arxiv.org/abs/2406.02106).

13. **Ramon Ruiz-Dolz, Zlata Kikteva, and John Lawrence, "Mining Complex Patterns of Argumentative Reasoning in Natural Language Dialogue" (ACL 2025).** Contribution: the paper introduces QT-Schemes, a corpus of 441 dialogue arguments annotated with 24 argumentation schemes, and evaluates Transformer and LLM approaches for mining scheme structure from textbook-like and real-world argumentation. Limitation: scheme mining identifies stereotyped inference patterns but does not establish proof validity, implement a disputation workflow, or guarantee that all argumentative units receive a scheme label. Direct novelty threat: it is the closest representation precedent for a typed argumentative ontology and directly threatens claims that structured argument-form extraction is new. Evaluation relevance: adapt its annotation strategy for scholastic roles and report macro-F1, premise/conclusion relations, provenance, and transfer from natural dialogue to expert-authored disputations. Source: ACL Anthology (https://aclanthology.org/2025.acl-long.368/).

14. **Elfia Bezou-Vrakatseli, Oana Cocarascu, and Sanjay Modgil, "Can Large Language Models Understand Argument Schemes?" (Findings of ACL 2025).** Contribution: this systematic study evaluates seven LLMs on Walton-style argument-scheme classification under zero-shot, few-shot, and chain-of-thought prompting, including manually annotated and generated enthymemes, and compares formal definitions with model-generated descriptions; larger models perform satisfactorily in some settings while smaller models remain challenged. Limitation: classification of argument form is not end-to-end reasoning, proof validity, or successful reply generation, and performance varies by scheme and prompting regime. Direct novelty threat: it directly pre-empts claims that LLM recognition of argumentative form is itself novel. Evaluation relevance: include scheme/form classification as an upstream metric, test macro-F1 and cross-domain robustness, and measure whether correct scheme recognition predicts objection quality or downstream answer success. Source: ACL Anthology (https://aclanthology.org/2025.findings-acl.702/).

15. **Yisong Miao et al., "Discursive Socratic Questioning: Evaluating the Faithfulness of Language Models' Understanding of Discourse Relations" (ACL 2024).** Contribution: DiSQ evaluates discourse understanding through sequences of targeted, counterfactual, and consistency questions rather than a single classification score; on PDTB, the best GPT model reaches only 41% of ideal performance, while the method remains robust to domain shifts and paraphrase variations. Limitation: it tests discourse-relation faithfulness, not philosophical validity, authority fidelity, or compliance with scholastic roles. Direct novelty threat: it is a direct precedent for multi-question faithfulness auditing and weakens claims that sequential cross-question consistency tests are new. Evaluation relevance: apply targeted, counterfactual, and consistency probes to ensure a determination remains stable across replies, detect equivocation, and distinguish superficial agreement from preserved reasoning relations. Source: ACL Anthology (https://aclanthology.org/2024.acl-long.341/).

16. **Chuyi Kong et al., "PlatoLM: Teaching LLMs in Multi-Round Dialogue via a User Simulator" (ACL 2024).** Contribution: PlatoLM introduces a Socratic user simulator trained from questions extracted from genuine human–machine conversations and reports state-of-the-art MT-Bench performance among LLaMA-based 7B models, with more human-like questioning patterns and richer topic structures than prior simulated-dialogue approaches. Limitation: it targets realistic multi-turn dialogue training and response quality, not philosophical argument validity, typed objections, authorities, distinctions, or objection coverage. Direct novelty threat: it pre-empts broad claims about using a Socratic simulator or multi-round questioning as such, but not claims about executable scholastic disputation semantics. Evaluation relevance: treat dialogue quality and questioning realism as separate baselines, then evaluate whether a scholastic protocol adds auditable intermediate obligations beyond conversational naturalness. Source: ACL Anthology (https://aclanthology.org/2024.acl-long.424/).

Cross-paper conclusion

Across the verified corpus, the strongest threats cover generic debate, multi-agent judging, branching search, Socratic interrogation, argument-scheme recognition, graph-based debate evaluation, and philosophy-specific benchmarking. None of these inspected papers combines a typed and auditable sequence of question, objections, *sed contra* authority, determination, distinctions, and one-to-one replies with expert-annotated evaluation of those intermediate obligations. The defensible novelty boundary is therefore the integrated scholastic protocol and its proof-relevant evaluation—not LLM debate, philosophical text generation, Socratic questioning, structured reasoning, or argument mining in isolation. This remains a plausible but unproven claim until budget-matched ablations, expert annotations, provenance checks, and judge-bias controls are run.